

n8n Forensic Reconstruction

Current public surface, source history, architectural forms, branches, documentation, APIs, schemas, artifacts, contradictions, and unknowns

Target https://n8n.io/	Classification TYPE 3 — HYBRID	Research period 14–18 August 2026 Asia/Tehran	Evidence sprints 6 of 6 completed
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Evidence labels: [OBSERVED] · [STATED] · [INFERRED] · [UNKNOWN]
Status boundaries: CURRENT · HISTORICAL · REMOVED · REVERTED · BRANCH-ONLY · FUTURE · UNKNOWN

Markdown SHA-256 at render: ea027b368fe122194798f76c486aeda8b1b05097138fa4b8c9110f0cf54cef64

Passive, read-only, public-surface research. No product evaluation, recommendation, redesign, competitor comparison, vulnerability test, or legal conclusion.

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n8n forensic reconstruction — working evidence record

Sprint metadata

- **Target supplied:** <https://n8n.io/>
- **Research dates:** Sprint 1 on 2026-08-14 Asia/Tehran; Sprints 2–4 on 2026-08-15 Asia/Tehran; Sprint 5 on 2026-08-17 Asia/Tehran; Sprint 6 on 2026-08-18 Asia/Tehran. HTTP/Git timestamps are recorded in UTC.
- **Cumulative evidence through:** Sprint 6 of 6 (final consolidation)
- **Sprint 1 scope:** reconnaissance, repository topology, origin/current-scale baseline, and early-history orientation.
- **Sprint 2 scope:** sampled branch reconstruction, consequential commits, and deleted/reverted/branch-only material.
- **Sprint 3 scope:** pinned current-source architecture: workspace/package graph, bootstrap and process roles, internal/Public APIs, local/queued execution, lifecycle persistence, DB/config/module systems, task runners and Code-node integration, multi-main/Redis orchestration, Engine 2.0 artifacts, tests, and deployment forms.
- **Sprint 4 scope:** target-authored documentation and GitBook information architecture; complete documentation-wiki history; deterministic documentation-branch sampling; VuePress/MkDocs/GitBook transitions; migration, repairs, removals and reverts; releases/changelog; target terminology; current OpenAPI/schema/artifact cross-checks; contradictions and unknowns.
- **Sprint 5 scope:** current marketing-site routing and infrastructure signals; exact sitemap growth/duplicates/generated route graph; deterministic rendered-route sample; Nuxt/Cloudflare/cache/header forms; public template/catalog APIs and workflow artifacts; machine-readable surfaces; source/document/live drift and unknowns.
- **Sprint 6 scope:** cross-source audit and final consolidation; unified evidence topology, timeline, architecture forms, interface/status matrices, attempt/removal registry, contradictions, unknowns, provenance audit, and final PDF production.
- **Final synthesis navigation:** Sections 55–64 consolidate the evidence from all six sprints; Sections 1–54 retain the underlying chronological evidence record.
- **Target classification: TYPE 3 — HYBRID.**
 - [OBSERVED] The supplied target is a live public URL, not a repository URL.
 - [OBSERVED] The rendered homepage and raw HTML both link directly to <https://github.com/n8n-io/n8n>.
 - [OBSERVED] GitHub and its API identify that repository as public and owned by [n8n-io](#).
- **Access boundary:** passive and read-only. No credentials were supplied. No authentication, workflow execution, mutation, login-wall access, vulnerability testing, or secret collection was attempted. HTTP probes used GET. Repository acquisition used read-only Git smart HTTP; it did not mutate the remote.

Evidence-label discipline

- **[OBSERVED]** directly inspected response, source object, tree, diff summary, Git metadata, or target-authored artifact.
- **[STATED]** wording in target-authored pages, READMEs, commit messages, or documentation.
- **[INFERRED]** minimal connective reconstruction.

- **[UNKNOWN]** unresolved by evidence inspected in this sprint.

No architectural-quality judgment or recommendation is made.

1. Reconnaissance decision and sprint plan

1.1 Scale decision

[OBSERVED] The public repository exposed 1,636 branch refs, 2,112 tags, and 23,011 commits reachable from `master` at collection time. The live sitemap index exposed 12 child sitemaps containing 334,603 URL entries in aggregate. Both exceed the mandatory sampling thresholds.

[INFERRED] A one-session investigation would risk omitting branch, historical, current-code, documentation, and live-surface evidence. Six evidence sprints were therefore selected:

1. Reconnaissance, repository topology, and origin/early-history baseline (this sprint).
2. Sampled branch investigation, consequential commits, and deleted/abandoned material.
3. Current source architecture: packages, runtime/processes, APIs, persistence, schemas, tests, and deployment.
4. Target-authored documentation, the hosting-platform wiki and in-repository/external documentation systems, releases/changelog, terminology, and contradictions.
5. Live public-surface architecture/infrastructure, routes, public APIs, integrations/templates, and source/deployment drift.
6. Cross-source audit, unified evidence matrix, timeline/diagrams/unknowns, and final PDF.

1.2 Sampling rule fixed at reconnaissance

[OBSERVED] Repository size triggers branch sampling. The Sprint 2 sample frame will include:

- default branch `master`;
- the three oldest current non-default branch tips by committer date;
- the three newest current non-default branch tips by committer date;
- branch names matching target terms `arch`, `doc`, `spec`, `prototype`, `experiment`, `v2`, `rewrite`, `core`, or `memory` (metadata triage of every match; deeper object inspection where unique evidence exists);
- release lines and any sampled branch found to be more than 10 commits ahead of `master` after its ancestry is fetched;
- any additional branch implicated by an important deletion, revert, or partial merge.

[OBSERVED] Live-surface size also triggers sampling. All top-level public route families will be mapped in Sprint 5, while the 11,452 workflow pages and very large cross-integration families will be stratified rather than exhaustively fetched.

[UNKNOWN] Branch creation dates are not native Git data. “Oldest/newest branch” in this report means oldest/newest **tip committer timestamp among refs still advertised**, not branch creation time.

2. Resource and method ledger

2.1 Disk/file budget

Stage	Observed disk	Observed files	Notes
Baseline	0	0	Empty workspace.
Initial blobless bare clone, depth 1	1.5 MB	24	No checkout.
1,636 branch tips, depth 1, tree:0	4.3 MB	2,131	All advertised heads represented as refs.
Full master commit graph, tree:0, plus all tags	25 MB	3,789	23,011 reachable commits; 2,112 tag refs.
Current/root tree and selected blobs	34 MB	~3,801	Historical files read with git show; no checkout.
Broad early-history name/deletion scan	167 MB	not retained	[OBSERVED] Promisor-object hydration unexpectedly fetched historical objects and exceeded the 80 MB working target. The scan was stopped; the clone was immediately deleted.
Post-cleanup before writing this record	0	0	Temporary Git repository and caches removed.

[OBSERVED] The temporary excursion to 167 MB violated the requested sub-80 MB operating target even though no checkout or binary export was made. This is recorded rather than concealed. All hydrated objects were removed immediately. Later sprints must use bounded commit/path queries rather than repository-wide historical summary scans.

2.2 Git acquisition

[OBSERVED] Commands were read-only and no checkout was performed:

```
git clone --filter=blob:none --bare --depth=1 https://github.com/n8n-io/n8n.git n8n.git
git fetch --filter=tree:0 --depth=1 origin '+refs/heads/*:refs/remotes/origin/*'
git fetch --filter=tree:0 --deepen=50000 origin master:refs/remotes/origin/master
git fetch --filter=tree:0 --depth=1 origin '+refs/tags/*:refs/tags/*'
```

Historical files were streamed from objects with `git show <ref>:<path>` and trees with `git ls-tree`; no branch/commit was checked out.

3. Live HTTP evidence log

3.1 Direct target reconnaissance

Each excerpt is at most 500 characters. TLS `ssl_verify_result=0` means curl's certificate verification succeeded.

ID	UTC timestamp	Method and URL	Status / latency / relevant headers	Body excerpt or structural result
L-001	2026-08-13T23:06:29Z	GET <code>https://n8n.io/</code>	200; 0.085724s; content-type: text/html; charset=utf-8; server: cloudflare; strict-transport-security: max-age=0; includeSubDomains; x-frame-options: DENY; TLS verify 0; 350,806 bytes	<!DOCTYPE html><html lang="en"><head>...<title>AI Workflow Automation Platform - n8n</title>...
L-002	2026-08-13T23:06:29Z	GET <code>https://n8n.io/robots.txt</code>	200; 0.054841s; text/plain; Cloudflare; TLS verify 0; 76 bytes	User-agent: * Disallow: /cdn-cgi/ Sitemap: <code>https://n8n.io/sitemap_index.xml</code>
L-003	2026-08-13T23:06:30Z	GET <code>https://n8n.io/sitemap.xml</code>	307 to <code>/sitemap_index.xml</code> , then 200; 0.119403s; text/xml; x-sitemap-prerendered: 2026-08-13T17:04:39.343Z; TLS verify 0; 1,053 bytes	XML sitemap index beginning with <code>sitemap-pages.xml</code> , <code>sitemap-integrations.xml</code> , <code>sitemap-workflows.xml</code> , and <code>sitemap-creators.xml</code> .
L-004	same collection window	GET <code>https://n8n.io/</code> (streamed link extraction; repeated)	curl success; response not persisted	Direct links included <code>https://github.com/n8n-io</code> and <code>https://github.com/n8n-io/n8n</code> .
L-005	2026-08-13T23:07:33Z	GET <code>https://n8n.io/sitemap_index.xml</code>	200 (independently established by L-003); body streamed only	12 child sitemap locations.
L-006	same collection window	Tool-mediated GET <code>https://n8n.io/</code>	success; transport latency/headers not exposed by tool	Rendered extraction called it "AI Workflow Automation Platform - n8n", linked <code>llms.txt</code> , and linked the source repository while stating "Top 50 Github" and "200.5k stars".

[OBSERVED] The first shell attempt to collect headers had a command-construction error before a URL was supplied to the header request; its separate excerpt GETs did succeed. L-001–L-003 are corrected, complete requests and supersede that malformed attempt.

3.2 Sitemap child requests

All were read-only GETs at 2026-08-13T23:07:33–23:07:37Z, TLS verify result 0, and status 200. Bodies were held in memory only long enough to count `<url>` entries and retain a generic XML excerpt.

ID	URL	Latency	Bytes	<code><url></code> entries
L-010	<code>https://n8n.io/sitemap-pages.xml</code>	0.130747s	15,311	233
L-011	<code>https://n8n.io/sitemap-integrations.xml</code>	0.154790s	127,006	2,024
L-012	<code>https://n8n.io/sitemap-workflows.xml</code>	0.244623s	1,387,438	11,452
L-013	<code>https://n8n.io/sitemap-creators.xml</code>	0.138731s	25,255	442
L-014	<code>https://n8n.io/sitemap-cross-integrations.xml</code>	0.304659s	3,524,575	46,963

ID	URL	Latency	Bytes	<url> entries
L-015	https://n8n.io/sitemap-cross-integrations-non-native-1.xml	0.204400s	3,724,069	49,999
L-016	https://n8n.io/sitemap-cross-integrations-non-native-2.xml	0.237529s	3,713,184	49,999
L-017	https://n8n.io/sitemap-cross-integrations-non-native-3.xml	0.176548s	3,771,703	49,999
L-018	https://n8n.io/sitemap-cross-integrations-non-native-4.xml	0.327539s	3,738,242	49,999
L-019	https://n8n.io/sitemap-cross-integrations-non-native-5.xml	0.263208s	3,723,654	49,999
L-020	https://n8n.io/sitemap-cross-integrations-non-native-6.xml	0.224710s	1,730,338	23,434
L-021	https://n8n.io/sitemap-ai-benchmark.xml	0.081465s	4,332	60
	Aggregate			334,603

Common excerpt (under 300 characters): <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" ...

3.3 GitHub API and page requests

ID	UTC timestamp	Method and URL	Status / latency	Excerpt / parsed fields
H-001	2026-08-13T23:07:32Z	GET https://api.github.com/repos/n8n-io/n8n	200; 0.219725s; TLS verify 0	Public, non-fork repository; created 2019-06-22T09:24:21Z; default master; size: 573529; has_wiki:false; pushed 2026-08-13T22:38:55Z. Excerpt began {"id":193215554,... "name":"n8n","full_name":"n8n-io/n8n"...}.
H-002	2026-08-13T23:07:33Z	GET https://api.github.com/repos/n8n-io/n8n/languages	200; 0.149276s; TLS verify 0	{"TypeScript":108660806,"Vue":7936791,"JavaScript":869028,"SCSS":368609,"Python":331418,...}
H-003	2026-08-13T23:07:33Z	GET https://api.github.com/repos/n8n-io/n8n/releases/latest	200; 0.140142s; TLS verify 0	Latest non-prerelease response: n8n@2.34.5, created 2026-08-12T07:27:18Z, published 08:04:40Z.
H-004	2026-08-13T23:07:33Z	GET https://api.github.com/repos/n8n-io/n8n/community/profile	200; 0.661431s; TLS verify 0	Health 100; links to CODE_OF_CONDUCT.md, CONTRIBUTING.md, pull request template, LICENSE.md, and README.md.
H-005	same collection window	Tool-mediated GET https://github.com/n8n-io/n8n	success; transport metadata not exposed	Page displayed public repo, master, 1,636 branches, 2,112 tags, and 23,011 commits; latest shown as 2578072.
G-HTTP-001	2026-08-13T23:07:32Z onward	Git smart-HTTP read-only clone/fetch https://github.com/n8n-io/n8n.git	Git operations succeeded	Initial blobless clone, branch-tip fetch, complete master commit graph, and tag-ref fetch. Smart Git transport may use POST to git-upload-pack, but only remote reads occurred.

3.4 Discovery searches

[OBSERVED] Two search-tool queries were used only to discover official/public links: `site:n8n.io` GitHub source code repository n8n, n8n official GitHub repository wiki architecture documentation, followed by `site:n8n.io "github.com/n8n-io/n8n"`. The tool reported success but did not expose its own HTTP endpoint, status, or timestamp; those transport details are [UNKNOWN]. Claims in this record rely on direct target/GitHub inspection, not third-party architectural summaries returned by search.

3.5 Initial live decision-gate answers

- [OBSERVED] `https://n8n.io/` responds publicly with HTTP 200 over successfully verified TLS.
 - [OBSERVED] The supplied URL renders a product/marketing page, not an authenticated workflow editor in the inspected response.
 - [OBSERVED] `robots.txt` disallows only `/cdn-cgi/` and points to a sitemap index.
 - [OBSERVED] At least 334,603 sitemap URL entries exist across 12 route families.
 - [UNKNOWN] A complete API map for the `n8n.io` website has not yet been established. Source history proves the n8n application has REST and webhook surfaces, but that does not establish which application APIs are deployed behind the marketing domain.
 - [UNKNOWN] Which public website pages require authentication has not been exhaustively tested. No credentials were supplied, and no login-wall content will be scraped.
 - [OBSERVED] A source repository is directly discoverable; target classification therefore escalated from TYPE 2 to TYPE 3.
-

4. Repository reconnaissance

4.1 Hosting metadata and scale

- [OBSERVED] Repository: `n8n-io/n8n`, public, non-fork, not archived.
- [OBSERVED] GitHub API creation timestamp: `2019-06-22T09:24:21Z`.
- [OBSERVED] Default branch: `master`.
- [OBSERVED] API description: “Fair-code workflow automation platform with native AI capabilities. Combine visual building with custom code, self-host or cloud, 400+ integrations.” This is [STATED] project wording, not independently validated here.
- [OBSERVED] API repository `size` field: 573,529 KiB-equivalent GitHub units (approximately 560 MiB); the local research clone was partial and not comparable to a full clone.
- [OBSERVED] GitHub API counts at collection: 200,531 stars, 60,119 forks/network count, 1,301 open issues, 1,156 subscribers.
- [OBSERVED] Issues were enabled; Projects, hosting-platform Wiki, and Discussions were reported disabled (`has_projects:false`, `has_wiki:false`, `has_discussions:false`).
- [OBSERVED] Latest API release: `n8n@2.34.5`, published 2026-08-12.
- [OBSERVED] Current `master` root `package.json` had version `2.35.0` and the head page identified a commit only about an hour old. This is source/release version drift, not a contradiction: the default branch can be ahead of the latest published stable release.

[INFERRED] For planning only, this is a **large, unusually branched, code-heavy monorepo with substantial documentation/test/schema/configuration surfaces**. These labels describe observable scale, not quality.

4.2 Commit graph and refs

- [OBSERVED] `master` head: `2578072643e84cb3ddd97e865656f9a233ddaa29`, committer timestamp `2026-08-13T22:20:30Z`, author `n8n-cat-bot[bot]`, message “chore: Bump nanoid 3.3.17 → 3.3.18 (patch) to clear 2 Aikido findings (#36266)”. Message content is [STATED]; the commit object and metadata are [OBSERVED].
- [OBSERVED] Root commit: `9cb9804eeec1576d935817ecda6bd345480b97fa`, authored `2019-06-23T12:35:23+02:00` by Jan Oberhauser, message “Initial commit to release”.
- [OBSERVED] One root commit; 23,011 commits reachable from `master`; 833 merge commits and 22,178 non-merge commits.
- [OBSERVED] Earliest authored timestamp reachable from `master`: `2019-06-23T12:35:23+02:00`. Latest inspected committer timestamp: `2026-08-13T22:20:30Z`.
- [OBSERVED] Current remote branch refs: 1,636. Current tags: 2,112.
- [OBSERVED] Deterministic ref-set digests for later drift checks:
 - sorted remote branch names SHA-256 `540f7c009b626684d29153ac711d057646ad8770e0cfb36b363c64093c4b66e3`;
 - sorted tag names SHA-256 `ed7658646de793c36a81e1664a9ab89749ad1851e3ca75668f215cefab5ce626`.

- [OBSERVED] 123 advertised branch-tip commits were ancestors of current `master` ; 1,513 tip commits were not. This is tip reachability only: it does **not** prove that every one of those branches was wholly unmerged or abandoned.

Commit counts grouped by author timestamp year

Year	Commits reachable from <code>master</code>
2019	1,078
2020	2,341
2021	1,798
2022	2,350
2023	2,385
2024	2,793
2025	4,332
2026 through collection	5,934

4.3 Branch sampling frame

Oldest extant tips by committer date

1. `origin/n8n-5173-microsoft-teams-node-roll-out-resource` — `ede4a354` , 2022-12-19T14:25:37+01:00.
2. `origin/check-data-processed` — `ca93689c` , 2023-07-27T22:04:17+02:00.
3. `origin/webhooks-caching` — `dc8eb778` , 2023-11-14T16:24:58+01:00.

Additional old tip: `origin/fix-workflow-history-throw-error-on-fail` (`b2e7cb44` , 2023-11-16). Release branches begin appearing among the next oldest tips (for example `release/1.50.1` in July 2024).

Newest non-default tips by normalized instant

1. `origin/ligo-965-git-connections-db` — `d94b3ef6` , 2026-08-13T23:28:43+01:00 (22:28:43Z).
2. `origin/ins-1186-forward-run-and-thread-ids-on-proxy-model-calls` — `c79f9a3a` , 2026-08-14T00:06:12+02:00 (22:06:12Z).
3. `origin/cat-3868-validate-unique-node-ids` — `70809f64` , 2026-08-13T23:02:26+02:00 (21:02:26Z).

[OBSERVED] `master` itself was the second-newest tip by normalized time (22:20:30Z), so it is handled separately as the default branch.

Keyword-bearing refs observed

The initial regex surfaced, among others:

- `DOC-1787`, `DOC-1962-context7-claim`, `DOC-2152`;
- `pay-2550-prototype-interactive-designs`;
- `prototype/improve-publish-state`;
- `cat-3685-create-n8nmodule-sdk-package-with-descriptor-v2-and...`;
- `hackmation-ai-agents-v2-rebased`;
- `node-4288-linear-v2-overhaul`;
- `core-item-grouping-utilities`;
- `make-cats-codeowners-of-core-execution-engine`;
- `cross-thread-memory-cli-persistence`, `cross-thread-memory-docs`, `cross-thread-memory-evals`, `cross-thread-memory-fe-integration`, `cross-thread-memory-profiles`, `cross-thread-memory-sdk-runtime`;
- `episodic-memory-cli-fe`, `episodic-memory-docs`, `episodic-memory-evals`, `episodic-memory-iteration`, `episodic-memory-sdk`;
- `observational-memory-cli-integration`, `n8n-agents-observational-memory-evals`, and several related memory branches.

[OBSERVED] Names and tip messages establish only terminology and a sampling lead. They do not establish that the named concepts were implemented, merged, active, or retained. Sprint 2 must inspect objects and diffs before classifying them.

4.4 Contributors visible in Git history

[OBSERVED] `git shortlog -sne master` produced 824 name/email identity lines after Git's mailmap behavior. That is not necessarily 824 distinct humans; aliases and bots can create multiple identities.

Top identity lines by reachable commit count:

	Count	Git identity
	4,426	Jan Oberhauser <jan.oberhauser@gmail.com>
	1,717	Iván Ovejero <ivov.src@gmail.com>
	1,227	🐞🐞🐞🐞🐞🐞🐞🐞🐞🐞🐞™ <netroy@users.noreply.github.com>
	762	Alex Grozav <alex@grozav.com>
	757	Ricardo Espinoza <ricardo@n8n.io>
	652	Michael Kret <88898367+michael-radency@users.noreply.github.com>
	559	Tomi Turtiainen <10324676+tomi@users.noreply.github.com>
	470	Csaba Tuncsik <csaba@n8n.io>

	Count	Git identity
	435	Mutasem Aldmour <4711238+mutdmour@users.noreply.github.com>
	393	Milorad Filipović <milorad@n8n.io>

[OBSERVED] A bot identity, `n8n-cat-bot[bot]`, appeared among the top 20 with 253 reachable commits. Counts are Git-history evidence, not employment or authorship attribution beyond recorded metadata.

4.5 Languages and present tree scale

[OBSERVED] GitHub language byte attribution totaled approximately 118.4 million bytes. Leading attributions were TypeScript 108,660,806; Vue 7,936,791; JavaScript 869,028; SCSS 368,609; Python 331,418; Handlebars 99,864; Shell 51,469; Dockerfile 23,919; MDX 22,505; HTML 17,590; HCL 6,776; CSS 6,185; Mermaid 2,535. This is byte attribution, not lines of code.

[UNKNOWN] A reliable current LOC count was deliberately not computed because hydrating every blob would conflict with the workspace budget.

[OBSERVED] Current `master` tree had 26,606 file entries. Top-level path counts were:

- `packages/` : 25,927
- `docs/` : 258
- `.github/` : 192
- `scripts/` : 67
- `.agents/` : 34
- `.claude/` : 32
- `patches/` : 18
- `.devcontainer/` : 14
- `docker/` : 13
- `security/` : 4
- `.vscode/` : 4
- `assets/` : 3

Path/extension heuristics found:

- 599 Markdown/MDX/RST/AsciiDoc files;
- 258 files under root `docs/` ;
- 723 image/diagram asset extensions;
- 626 paths under schema/migration-named locations;
- 9,434 test/spec/e2e-named paths (a path heuristic, not a test-case count);

- 93 GitHub Actions workflow files;
- 25 Docker-named paths;
- 6 Kubernetes/Helm-named paths;
- 87 `package.json` files, including packages and templates;
- no submodule entries;
- no `filter=lfs` declaration in current `.gitattributes`.

[OBSERVED] Documentation markup was concentrated in root `docs/` (258), `packages/@n8n` (181), `packages/frontend` (42), `.agents` (34), and `packages/testing` (22), with smaller counts elsewhere.

4.6 Current workspace/build signals (orientation only; detailed inspection deferred)

- [OBSERVED] Root package name `n8n-monorepo`, version `2.35.0`, `private:true`.
 - [OBSERVED] Package manager `pnpm@10.32.1`; engines Node `>=22.22`, `pnpm >=10.22.0`.
 - [OBSERVED] Workspace globs include `packages/*`, `packages/@n8n/*`, `packages/frontend/**`, `packages/modules/**`, `packages/extensions/**`, and `packages/testing/**`.
 - [OBSERVED] Root scripts invoke Turbo for build/test/typecheck and expose `start`, `webhook`, and `worker` commands. The script names are evidence of current commands; their runtime behavior is reserved for Sprint 3 code inspection.
 - [OBSERVED] Current root includes `pnpm-workspace.yaml`, `pnpm-lock.yaml`, `turbo.json`, Docker files, DB schema configs for PostgreSQL and SQLite, CI workflows, security material, `CHANGELOG.md`, and two license files (`LICENSE.md`, `LICENSE_EE.md`).
 - [OBSERVED] Current CI names include master/PR checks, DB tests, e2e tests, AI/instance-AI/MCP evals, Docker builds, release automation, SBOM generation, and security scans. Names alone do not establish execution success.
-

5. Root state: what existed at the first reachable commit

5.1 Root commit structure

[OBSERVED] Commit 9cb9804e was not an empty scaffold. It contained 257 files and these top-level entries:

```
.editorconfig .gitignore LICENSE README.md docker/
lerna.json package.json packages/
```

[OBSERVED] Extension counts included 144 TypeScript, 35 Vue, 22 JSON, 14 PNG, 12 JavaScript, 9 Markdown, and 2 SCSS files.

[OBSERVED] Root package scripts used Lerna 3.13.1:

```
{
  "name": "n8n",
  "private": true,
  "scripts": {
    "bootstrap": "lerna bootstrap --hoist --no-ci",
    "build": "lerna exec npm run build",
    "start": "cd packages/cli && node dist/index.js start",
    "watch": "lerna run --parallel watch"
  }
}
```

[OBSERVED] Seven package manifests existed:

Path	Name/version in root commit	Manifest description
root	n8n private	Lerna orchestration
packages/cli	n8n 0.1.2	"n8n Workflow Automation Tool"
packages/core	n8n-core 0.1.0	"Core functionality of n8n"
packages/editor-ui	n8n-editor-ui 0.1.1	"Workflow Editor UI for n8n"
packages/node-dev	n8n-node-dev 0.1.0	"CLI to simplify n8n credentials/node development"
packages/nodes-base	n8n-nodes-base 0.1.0	"Base nodes of n8n"
packages/workflow	n8n-workflow 0.1.0	"Workflow base code of n8n"

[STATED] The root README title was “n8n - Workflow Automation Tool” and said it allowed users to “easily and fast automate different taks,” was “still in beta,” and had little documentation. It directed installation/use to the CLI package and Docker instructions. The README described the license as “Apache 2.0 with Commons Clause.” These are historical project statements, not current status.

5.2 Root runtime/source form

[OBSERVED] `packages/cli/src` already contained `Server.ts`, `Db.ts`, `ActiveWorkflowRunner.ts`, `LoadNodesAndCredentials.ts`, `Push.ts`, `TestWebhooks.ts`, `WebhookHelpers.ts`, and `WorkflowExecuteAdditionalData.ts`.

[OBSERVED] `packages/core/src` already contained `ActiveExecutions.ts`, `ActiveWebhooks.ts`, `ActiveWorkflows.ts`, `Credentials.ts`, `NodeExecuteFunctions.ts`, and `WorkflowExecute.ts`.

[OBSERVED] `packages/workflow/src` contained `Workflow.ts`, `WorkflowDataProxy.ts`, `NodeHelpers.ts`, `ObservableObject.ts`, and interfaces.

[OBSERVED] The first `Server.ts` registered REST routes for workflows, credentials, executions, node metadata, active workflow IDs/errors, settings/timezones, current-execution stopping, and test webhooks. Examples:

```
POST /rest/workflows
GET /rest/workflows/from-url
GET /rest/workflows
GET /rest/workflows/:id
PATCH /rest/workflows/:id
DELETE /rest/workflows/:id
POST /rest/workflows/run
GET /rest/node-parameter-options
GET /rest/node-types
GET /rest/node-icon/:nodeType
GET /rest/active
GET /rest/active/error/:id
POST/PATCH/GET/DELETE /rest/credentials...
GET /rest/executions
GET /rest/executions/:id
POST /rest/executions/:id/retry
POST /rest/executions/delete
GET /rest/executions-current
POST /rest/executions-current/:id/stop
DELETE /rest/test-webhook/:id
GET /rest/options/timezones
GET /rest/settings
```

[OBSERVED] It also registered GET/POST wildcard handlers under configurable production and test webhook prefixes, then served the built `n8n-editor-ui` statically. These are historical source routes; no mutating route was called during research.

[OBSERVED] `ActiveWorkflowRunner.init()` queried the workflow collection for `{ active: true }`, created `ActiveWebhooks` and `ActiveWorkflows`, and registered active workflows. This source establishes that an `active` field had operational use in this first reachable state; a workflow name alone did not determine activity.

[OBSERVED] `WorkflowExecute.ts` and `Workflow.ts` exposed node/workflow execution functions, trigger and webhook execution, node hooks, and data proxy behavior. The target's own early terms included **workflow**, **node**, **connection**, **execution**, **active workflow**, **webhook**, **trigger**, **credential**, **staticData**, **runData**, and **execution mode**.

5.3 Root persistence model

[OBSERVED] The CLI imported both `./db/mongodb` and `./db/sqlite`. Its package dependencies included `mongodb`, `sqlite3`, and `typeorm`.

[OBSERVED] Both adapters defined three entities:

- **CredentialsEntity**: ID, name, encrypted/serialized `data` string, `credential type`, node-access list, created/updated timestamps.
- **ExecutionEntity**: ID, serialized `data`, `finished`, `execution mode`, retry linkage, start/stop timestamps, embedded workflow data, and `workflowId`.
- **WorkflowEntity**: ID, `name`, `active`, JSON/simple-JSON `nodes` and `connections`, created/updated timestamps, optional `settings`, optional `staticData`.

[OBSERVED] MongoDB used object IDs and JSON columns; SQLite used generated numeric IDs and TypeORM `simple-json`/text columns. This describes the root commit only.

5.4 Root node/integration surface

[OBSERVED] Forty `.node.ts`/`.trigger.ts` candidates existed. They included control/data nodes (`Start`, `If`, `Merge`, `Set`, `NoOp`, `SplitInBatches`), code/host nodes (`Function`, `FunctionItem`, `ExecuteCommand`), file/image/spreadsheet/PDF nodes, HTTP/Webhook/Cron/Interval/Error triggers, and integrations such as Asana, Chargebee, Dropbox, GitHub, Google Sheets, Mailgun, Nextcloud, OpenWeatherMap, Pipedrive-related support added immediately after root, Redis, Slack, Twilio, and others.

[OBSERVED] Twelve test/e2e-named paths existed, including core credential tests, workflow helper/model tests, and initial editor e2e/unit scaffolding. This is a path count, not a test count.

5.5 Root deployment evidence

[OBSERVED] The initial Dockerfile used `mhart/alpine-node:10`, installed GraphicsMagick, globally installed `n8n`, set `WORKDIR /data`, and ran `n8n`.

[OBSERVED] The initial Docker README exposed port 5678, described an optional `--tunnel`, stated SQLite was the default database in `/root/.n8n`, and described MongoDB as an alternative. It stated that the user folder also held settings, webhook-related data, and the credential encryption key.

[OBSERVED] `docker/compose/withMongo/docker-compose.yml` defined Mongo 4.0 and n8n services, port 5678, `~/.n8n:/root/.n8n`, and a `NODE_CONFIG` MongoDB URL.

5.6 Root-state architecture diagram

The boxes and edges below are a compact [INFERRED] reconstruction from observed imports, route registration, static serving, entity definitions, and execution classes. It does not claim a named architecture pattern.

MERMAID DIAGRAM SOURCE (RETAINED AS EVIDENCE)

```
flowchart LR
    UI["n8n-editor-ui\nVue 2 static build"] -->|HTTP /rest| S["packages/cli Server.ts\nExpress"]
    EXT["External callers"] -->|GET/POST webhook paths| S
    S --> A["ActiveWorkflowRunner\nActiveWorkflows / ActiveWebhooks"]
    S --> D["Db collections via TypeORM"]
    D --> SQ["SQLite entities\ndefault per Docker README"]
    D --> MO["MongoDB entities\noptional"]
    S --> E["n8n-core WorkflowExecute"]
    A --> E
    E --> W["n8n-workflow\nWorkflow / data proxy / interfaces"]
    E --> N["n8n-nodes-base\nnode and trigger implementations"]
    DEV["n8n-node-dev CLI"] --> N
```

6. Early-history orientation: changed, removed, attempted

This section records only transitions directly visible in commit metadata and bounded diff summaries inspected during Sprint 1. Detailed diff archaeology is deferred to Sprint 2.

6.1 First commits

- [OBSERVED] 9cb9804e (2019-06-23): root release state described above.
- [STATED] 40ed3f62 : “Fix link to README.”
- [STATED] 0f99d169 : “Fix indentation in docker-compose file.”
- [STATED] 08cc0155 : “Add repository entry to package.json files.”
- [STATED] 28262fdb : “Add information about how to use MongoDB as database.”
- [STATED] f26199f6 : “Add first basic support for Pipedrive.”
- [STATED] b776bd8b : added documentation for n8n run .
- [OBSERVED] 2b7382cd followed on 2019-06-24 with message “Rename cli command ‘run’ to ‘execute’.” The chronological relationship is observed; the exact diff will be checked in Sprint 2 before asserting every renamed symbol.
- [STATED] 646e1fc8 (2019-07-07): “Add first basic documentation how to create an own node.”

6.2 Configuration and CLI changes

- [OBSERVED] Commit d0275459 (2019-07-21) had message “Use convict for configuration to make n8n easier to configure,” and its diff summary deleted packages/cli/config/default.ts . The message is [STATED]; deletion is [OBSERVED]. Replacement files/behavior require bounded diff inspection.
- [OBSERVED] Commit 67afc883 (2019-08-28) had message “Replace voral with oclif in cli,” and its diff summary deleted packages/cli/index.ts . Root package dependencies had included Voral. The target therefore recorded a CLI-framework replacement attempt; exact retained command layout will be inspected next sprint.
- [OBSERVED] A related commit cd75c63f stated “Replace voral with oclif in node-dev” and deleted packages/node-dev/index.ts in the diff summary.

6.3 Node consolidation/removal and reverts

- [OBSERVED] c1e753e9 (2019-09-04) stated “Remove ‘Read File From Url’ node and add functionality to ‘HTTP Request’ node”; its diff summary deleted packages/nodes-base/nodes/ReadFileFromUrl.node.ts . The deletion is direct evidence; the HTTP Request addition still needs targeted diff verification.
- [OBSERVED] 70e972ff (2020-01-10) was explicitly a revert of an OAuth-support merge and deleted 0Auth2Api.credentials.ts and 0Auth.node.ts according to the diff summary. This is direct evidence of an attempted-and-reverted OAuth surface; later reintroduction is not yet reconstructed.
- [OBSERVED] 177c6f65 (2020-06-08) stated removal of no-longer-needed OAuth test files and deleted Test0Auth2Api.credentials.ts and 0Auth.node.ts in its summary. Relationship to the January revert remains [UNKNOWN] until targeted history inspection.

6.4 Database migration work

[OBSERVED] Several April 2020 commits changed initial migration files and their locations across SQLite, MySQL, and PostgreSQL paths:

- 032f4075 — message “postgresql initial migration”; deletion summary included a PostgreSQL initial migration under an uppercase Migrations path.
- 16bf916d — message “Migration configs, migration files, removed test queries”; deletion summary included MySQL and SQLite initial migration paths with capitalization/path changes.
- 3acd7792 — message “Fixed Queries”; deletion summary included a MySQL initial migration file.
- 48a1e0c4 — message “ormconfig to migrations folder”; deleted a checked-in packages/cli/database.sqlite file.

[INFERRED] These summaries establish migration-path churn and multi-SQL-database work in April 2020. They do not yet establish the final state of each database adapter at that date.

6.5 Documentation system movement

- [OBSERVED] In-repository docs/ existed historically.
- [OBSERVED] Commit 62612d0a (2020-07-05) stated “Removed docs-folders as they moved to n8n-io/n8n-docs.” Its diff summary deleted a documentation site including _sidebar.md, configuration.md, create-node.md, data-structure.md, database.md, development.md, docker.md, key-components.md, node-basics.md, security.md, server-setup.md, workflow.md, images, and other pages.
- [OBSERVED] The commit message exposes a second target-authored repository identifier, n8n-io/n8n-docs. It will be investigated as documentation evidence in Sprint 4 under the same disk budget; it must not be confused with the GitHub Wiki.
- [OBSERVED] GitHub API currently reports has_wiki:false for the main repository, while current master again contains 258 files under root docs/. These are different documentation surfaces and dates, not proof of a contradiction.

6.6 Annual boundary metadata

[OBSERVED] First/last commit metadata per author-timestamp year included:

- 2019: root 9cb9804e; last 8f253039, message about triggering every X minutes/hours.
- 2020: first 23985dbc, SSE Trigger node; last 29107f13, SplitInBatches description typo.
- 2021: first 403f1009, Phantombuster node; last 5c11eb52, digest-auth fix.
- 2022: first eeccc69b, merge; last 99e3eb63, CI linting fix.
- 2023: first 7e3f3c50, node lint rule; last ece48d6a, “Unify workflow controllers.”
- 2024: first e126ed74, user profile hook; last 4e9f2b3b, managed-credential UI behavior.
- 2025: first a484ea16, Cron workflow test; last 9329d55c, AI-builder recursion handling.
- 2026: first 07d4d285, year-related test; latest inspected 25780726, nanoid bump.

Commit-message descriptions are [STATED]; hashes/dates/order are [OBSERVED].

7. Terminology observed in target artifacts

7.1 Historical root terms

[OBSERVED] Source/manifests/routes used: workflow, node, node type, credentials, execution, active workflow, active execution, webhook, test webhook, trigger, connections, staticData, runData, workflow settings, execution mode, WorkflowExecute, WorkflowDataProxy, node parameter options, and credential types.

7.2 Current-tree orientation terms

[OBSERVED] Current package/path/script names additionally expose: editor-ui, task-runner, instance-ai, nodes-langchain, workflow-sdk, extension-sdk, module-sdk (on a branch), blob-storage, scheduler, permissions, telemetry, chat-hub, MCP, worker, webhook, queue/scaling mode in commit messages, benchmark, evals, and database schema. These names are evidence of vocabulary only until implementation inspection.

[STATED] The current repository description uses “fair-code,” “native AI capabilities,” “visual building,” “custom code,” “self-host or cloud,” and “400+ integrations.”

8. Contradictions, drift, and source-boundary cautions

1. Release versus branch version

- [OBSERVED] Latest GitHub stable release was `2.34.5`.
- [OBSERVED] `master` root package version was `2.35.0`.
- [INFERRED] This is normal branch/release drift; it must not be presented as the deployed website's version.

2. Repository creation versus root commit

- [OBSERVED] GitHub API says repository created 2019-06-22T09:24:21Z.
- [OBSERVED] earliest reachable root commit was authored 2019-06-23T12:35:23+02:00.
- [UNKNOWN] Whether earlier source history existed elsewhere or was squashed/imported is unresolved.

3. Documentation location over time

- [OBSERVED] Root `docs/` was removed in 2020 with a message pointing to `n8n-io/n8n-docs`.
- [OBSERVED] Current `master` again has a root `docs/` directory with 258 files.
- [UNKNOWN] Whether the present directory is a return of the same documentation, a different developer-doc subset, generated material, or another transition awaits Sprint 4.

4. Two wiki systems

- [OBSERVED] Hosting-platform Wiki is currently disabled (`has_wiki:false`).
- [OBSERVED] In-repository docs and a separate docs repository existed/exist.
- [INFERRED] Those must be investigated independently; the docs repo is not evidence that a GitHub Wiki existed.

5. Website versus application instance

- [OBSERVED] The supplied URL served a marketing/product homepage.
- [OBSERVED] Historical source contains an editor UI, REST API, and webhook server.
- [UNKNOWN] Which n8n application build/version, if any, runs at specific cloud/editor subdomains is not established by the homepage and will not be guessed.

6. Branch names versus implementation

- [OBSERVED] Numerous branch names contain `memory`, `prototype`, `v2`, `core`, and docs terms.
- [UNKNOWN] Merge, abandonment, unique commit count, and implementation content remain unresolved until Sprint 2.

9. Unified evidence matrix — Sprint 1

Evidence ID	Surface	State/date	Direct evidence	Supports	Limits
L-001	Live homepage	2026-08-13	HTTP 200, headers, title excerpt	Public product site, Cloudflare edge, repository discovery context	Not an authenticated app instance; no deployed n8n version
L-002	robots	current response	Only <code>/cdn-cgi/</code> disallowed; sitemap link	Crawl boundary and sitemap discovery	Does not enumerate private/login routes
L-003-021	sitemap family	current response	12 child maps, 334,603 entries	Live public-surface scale and sampling trigger	URL entries are not proof every page is healthy/unique
H-001	GitHub API	current response	Public repo metadata/default branch/wiki flag	TYPE 3 escalation, creation date, host features	API size is not local checkout size
H-002	GitHub languages	current response	Byte counts	Language mix	Not LOC; generated/vendored attribution not audited
H-003	Release API	2026-08-12 release	n8n@2.34.5	Latest stable release visible to API	Not deployed website/cloud version
G-001	Git refs	collection snapshot	1,636 heads, 2,112 tags, digests	Branch/tag scale and reproducibility	Refs can change after collection
G-002	master graph	through 25780726	23,011 commits, one root	Reachable history scale/range	Excludes unreachable/orphaned objects and deleted remote refs
G-003	current tree	25780726	26,606 paths and heuristic counts	Current code/document/test/config topology	Detailed implementation deferred
G-004	root tree	9cb9804e	257 files, seven manifests	Initial reachable architecture	Commit message says “release”; prehistory unknown
G-005	root <code>Server.ts</code>	9cb9804e	REST/webhook/static routes	Historical API and combined server/UI form	No live behavior established
G-006	root entities	9cb9804e	SQLite/Mongo workflow/execution/credential classes	Historical persistence model	Later migrations not yet reconstructed
G-007	root execution files	9cb9804e	Active workflow and workflow execution classes	Historical runtime terms/forms	Detailed control flow not exhaustively traced
G-008	62612d0a summary	2020-07-05	Docs deletions and message pointing to <code>n8n-docs</code>	Removed/moved documentation, second repo discovery	External docs history deferred
G-009	70e972ff summary	2020-01-10	Revert plus OAuth file deletions	Attempted/reverted material	Full parent diff and later reintroduction deferred
G-010	branch-tip refs	current snapshot	oldest/newest/keyword refs and reachability test	Sprint 2 sampling frame	Tip reachability is not branch lifecycle classification

10. Unknowns carried forward

- [UNKNOWN] Complete lifecycle, divergence, unique files, merge status, and abandonment classification for sampled branches.
 - [UNKNOWN] Architecturally consequential commits beyond the bounded early examples; revert chains and deleted architectural documents.
 - [UNKNOWN] Exact transition from Lerna/npm-era workspace to current pnpm/Turbo workspace.
 - [UNKNOWN] MongoDB adapter removal date/fate, SQL adapter history, and current schema/entity/migration model.
 - [UNKNOWN] Current REST/public API/controller map, authentication/authorization layers, process modes, queue/scaling implementation, webhooks, workers, task runners, and execution lifecycle.
 - [UNKNOWN] Current package dependency graph and whether any generated artifacts are committed.
 - [UNKNOWN] Hosting-platform wiki historical availability despite current `has_wiki:false`; GitHub API current flag cannot establish prior wiki history.
 - [UNKNOWN] Full history and relationship of `n8n-io/n8n-docs`, current root `docs/`, `docs.n8n.io`, website `llms.txt`, and changelog/release material.
 - [UNKNOWN] Live `n8n.io` origin/application framework behind Cloudflare, public API families, authentication boundaries, and deployed source/version drift.
 - [UNKNOWN] Reliable LOC count.
 - [UNKNOWN] Releases/tags not yet mapped into historical epochs despite all tag refs being counted.
 - [UNKNOWN] Issues/PRs materially tied to selected architectural changes; only current issue availability/count was recorded.
-

11. Sprint-end preservation and cleanup

[OBSERVED] The temporary bare/partial Git repository, hydrated promisor objects, and all scratch/cache material were deleted. Only this consolidated markdown evidence record is retained.

Sprint 1 of 6 completed. 5 research sprints remain.

Sprint 2 will reconstruct the sampled branches independently, fetch only bounded ancestry/trees, inspect important commit diffs and revert chains, and classify unique/deleted/abandoned material without checkout.

Sprint 2 — sampled branches, commit archaeology, and removed/abandoned material

12. Sprint 2 method and snapshot drift

[OBSERVED] Sprint 2 began from the Sprint 1 evidence file and used a new checkout-free, commit/tree-filtered Git acquisition. The remote snapshot had moved from 1,636 to 1,645 heads. `master` moved from 257807264 (23,011 reachable commits) to 2afc867f20d689891b516c0458d627bc52c979fd (23,074 reachable commits), dated 2026-08-15T00:00:29Z. The Sprint 1 head remained an ancestor. The branch-name digest changed from 540f7c... to f6915c...

[OBSERVED] Two Sprint 1 “newest” refs (`ins-1186-forward-run-and-thread-ids-on-proxy-model-calls` and `cat-3868-validate-unique-node-ids`) disappeared before Sprint 2. Their recorded commits remained retrievable from the public GitHub commit API and are analyzed separately below. The other four fixed-frame oldest/newest refs were unchanged.

[OBSERVED] Token-delimited matching of `arch`, `architecture`, `doc`, `docs`, `spec`, `prototype`, `experiment`, `v2`, `rewrite`, `core`, or `memory` selected 51 current branch refs. Adding the fixed oldest/newest refs and representative release lines produced 60 current refs for independent metadata/tree comparison. `master` is the comparison base; branch creation time remains unknowable, so the merge base is reported as the reconstructable divergence point.

12.1 Current sampled branch table — batch A

[OBSERVED] `ahead/behind` are commit counts against Sprint 2 `master`; `tip-merged` means the exact tip is an ancestor of `master`; `unmerged-tip` means it is not. `A/M/D` and file counts are no-rename tree diffs from merge base to branch tip. Documentation/test counts are path heuristics. These facts do not alone establish abandonment or patch-equivalent partial merges.

Flags	Branch	Head / tip date	Merge base	Ahead / behind	Exact-tip status	A/M/D; files	Docs/tests	Leading path groups	Tip subject
R	1.x	e267e4576 2026-08-14T23:50:30 +01:00	6ceef1391	299 / 6462	unmerged-tip	337/613/50; 1000	16/317	packages/nodes-base:238,packages/cli:206,packages/@n8n:199	chore: Bump @xmldom/xmldom 0.8.13 → 0.8.14 (patch) on 1.x to clear 2 Aikido findings (#36357)
K	DOC-1787	0b1491868 2026-03-20T10:37:26 +01:00	9b6888ff6	1 / 4572	unmerged-tip	0/488/0; 488	1/0	packages/nodes-base:487,packages/@n8n:1	fix(docs): broken doc link
K	DOC-1962-context7-claim	97cd70296 2026-05-19T14:42:08 +02:00	b9d8452d8	1 / 3194	unmerged-tip	1/0/0; 1	0/0	context7.json:1	chore: add context7.json to claim Context7 library ownership (DOC-1962)
K	DOC-2152	90a5fb986 2026-08-11T16:11:09 +02:00	6bd31e5e1	4 / 164	unmerged-tip	0/3/0; 3	0/0	packages/@n8n:3	Merge branch 'master' into DOC-2152
K	agent-100-feature-memory-compression-with-sub-agents	41ceaf60a 2026-05-05T18:15:45 +02:00	74c256c1c	131 / 3629	unmerged-tip	335/172/25; 532	1/175	packages/frontend:216,packages/@n8n:160,packages/cli:139	add complex example of om

Flags	Branch	Head / tip date	Merge base	Ahead / behind	Exact-tip status	A/M/D; files	Docs/tests	Leading path groups	Tip subject
K	agent-429-agents-episodic-memory-is-not-tuned-well-enough-to-remember	20978e707 2026-07-22T11:38:16+01:00	87115d6cb	5 / 924	unmerged-tip	1/30/0; 31	2/15	packages/ @n8n:27,packages/ frontend:2,packages/ cli:2	Merge branch 'master' of github.com:n8n-io/n8n into agent-429-agents-episodic-memory-is-not-tuned-we
K	aws-ai-agent-node-ui-prototype	13a756a26 2026-07-21T11:09:16+02:00	2b846aa2b	11 / 1037	unmerged-tip	4/1/0; 5	0/1	packages/nodes-base:5	fix(AWS AI Agent Node): use the AWS credential alone for identity
K	branch-private-credentials-v2	b4a16cc46 2026-07-09T00:19:46+02:00	1614f9c7a	5 / 1392	unmerged-tip	2/28/0; 30	0/11	packages/ cli:15,packages/ core:7,packages/ workflow:4	test: Update assertions for end-user credentials wording
K	burivuhster/messenger-platforms-settings-v2	ccf082fd1 2026-05-13T17:14:07+02:00	7ab1acae0	3 / 3342	unmerged-tip	22/15/0; 37	0/5	packages/ cli:17,packages/ frontend:11,packages/ @n8n:9	feat(core): Gate chat integrations behind account linking
K	cat-3685-create-n8nmodule-sdk-package-with-descriptor-v2-and	21752ca8d 2026-07-10T19:20:39+03:00	3b24f493d	1 / 1292	unmerged-tip	32/18/4; 54	1/7	packages/ frontend:53,pnpm-lock.yaml:1	feat(editor): Create @n8n/module-sdk package with descriptor v2 and registries (no-changelog)
O	check-data-processed	ca93689c1 2023-07-27T22:04:17+02:00	d050b99fb	32 / 14079	unmerged-tip	9/21/0; 30	0/1	packages/ cli:22,packages/ core:4,packages/ nodes-base:3	Merge branch 'check-data-processed' of github.com:n8n-io/n8n into check-data-processed
K	claude/node-type-memory-analysis-i9tldz	85ba4e159 2026-07-02T17:57:06Z	b9e344404	1 / 1589	unmerged-tip	3/0/0; 3	1/0	packages/cli:3	chore(cli): add node-type description memory baseline harness (no-changelog)
K	core-item-grouping-utilities	eba9a0b66 2026-08-12T12:33:59+01:00	5bfcde593	1 / 136	unmerged-tip	1/1/0; 2	0/0	packages/nodes-base:1,packages/cli:1	feat(core): Add execution digest endpoint and item grouping utility
K	cross-thread-memory-cli-persistence	a05037561 2026-05-10T21:24:20+02:00	ffc63691	30 / 3496	unmerged-tip	47/61/20; 128	1/59	packages/ @n8n:79,packages/ cli:26,packages/ frontend:22	feat(agents): persist memory profiles
K	cross-thread-memory-docs	ea5b531a2 2026-05-11T14:12:44+02:00	ffc63691	33 / 3496	unmerged-tip	49/65/20; 134	2/62	packages/ @n8n:80,packages/ cli:28,packages/ frontend:25	docs(agents): update memory profile docs
K	cross-thread-memory-evals	c2042f165 2026-05-10T20:55:15+02:00	ffc63691	50 / 3496	unmerged-tip	50/60/20; 130	1/64	packages/ @n8n:84,packages/ cli:23,packages/ frontend:22	test(agents): Score injected memory recall separately
K	cross-thread-memory-fe-integration	d157a33c4 2026-05-10T21:31:50+02:00	ffc63691	31 / 3496	unmerged-tip	48/62/20; 130	1/61	packages/ @n8n:79,packages/ cli:26,packages/ frontend:24	feat(agents): add memory profile toggle
K	cross-thread-memory-profiles	ba453f91a 2026-05-12T08:05:52+02:00	e96872380	27 / 3405	unmerged-tip	38/66/20; 124	1/62	packages/ @n8n:78,packages/ frontend:25,packages/ cli:20	fix(agents): dedupe profile loads by key

Flags	Branch	Head / tip date	Merge base	Ahead / behind	Exact-tip status	A/M/D; files	Docs/tests	Leading path groups	Tip subject
K	cross-thread-memory-sdk-runtime	7ee2867d4 2026-05-10T21:18:30+02:00	ffc63691	29 / 3496	unmerged-tip	44/60/20; 124	1/59	packages/ @n8n:78,packages/ cli:23,packages/ frontend:22	feat(agents): add memory profiles
K	docs-package-typecheck-unbuilt-deps	f912b35ef 2026-08-06T09:05:11+02:00	28d929dc2	2 / 357	unmerged-tip	0/1/0; 1	1/0	AGENTS.md:1	Merge branch 'master' into docs-package-typecheck-unbuilt-deps
K	docs/n8n-plugin-install-guide	a3878951d 2026-06-17T10:01:40+02:00	285e4a983	2 / 2270	unmerged-tip	1/1/0; 2	2/0	.claude:2	docs: Improve recommendation for checking out copy
K	ent-348-extract-shared-ms-graph-transport-kernel-from-teams-v2	752f161dd 2026-08-14T16:37:45+02:00	a0f350ed2	7 / 14	unmerged-tip	2/2/1; 5	0/3	packages/nodes-base:5	chore: Fix biome formatting in MS Graph kernel tests (no-changelog)

12.2 Current sampled branch table — batch B

Flags	Branch	Head / tip date	Merge base	Ahead / behind	Exact-tip status	A/M/D; files	Docs/tests	Leading path groups	Tip subject
K	ent-354-bug-kafka-trigger-v2-activates-on-a-missing-topic-then	d4aa2ffd7 2026-08-14T11:51:45+03:00	5c51a0c15	1 / 53	unmerged-tip	2/5/0; 7	0/4	packages/nodes-base:7	fix(Kafka Trigger Node): Refuse activation when the topic does not exist
K	ent-364-bug-sharepoint-v2-renders-unclassified-list-columns-incl	a7c2b82d8 2026-08-14T11:52:19+01:00	a59d84271	1 / 41	unmerged-tip	0/2/0; 2	0/1	packages/nodes-base:2	fix(Microsoft SharePoint Node): Render unclassified list columns as text fields
K	ent-365-bug-sharepoint-v2-site-by-id-accepts-malformed-ids-and	91b3ccf10 2026-08-14T11:56:25+01:00	a59d84271	1 / 41	unmerged-tip	0/16/0; 16	0/15	packages/nodes-base:16	fix(Microsoft SharePoint Node): Validate the site ID before sending it to Graph
K	episodic-memory-cli-fe	63a706b69 2026-05-12T08:29:36+02:00	e96872380	51 / 3405	unmerged-tip	46/69/20; 135	1/65	packages/ @n8n:82,packages/ frontend:30,packages/ cli:22	fix(agents): clean advanced profile loading
K	episodic-memory-docs	b053801b2 2026-05-12T08:30:01+02:00	e96872380	63 / 3405	unmerged-tip	48/69/20; 137	3/65	packages/ @n8n:84,packages/ frontend:30,packages/ cli:22	Merge branch 'episodic-memory-cli-fe' into episodic-memory-docs
K	episodic-memory-evals	de10dd7c6 2026-05-12T08:30:10+02:00	e96872380	73 / 3405	unmerged-tip	54/69/20; 143	4/65	packages/ @n8n:90,packages/ frontend:30,packages/ cli:22	Merge branch 'episodic-memory-docs' into episodic-memory-evals
K	episodic-memory-iteration	e0b1146ac 2026-05-12T10:36:26+02:00	e96872380	74 / 3405	unmerged-tip	56/69/20; 145	4/66	packages/ @n8n:92,packages/ frontend:30,packages/ cli:22	Improve Episodic Memory
K	episodic-memory-sdk	8fff55d7b 2026-05-12T08:06:20+02:00	e96872380	38 / 3405	unmerged-tip	41/66/20; 127	1/64	packages/ @n8n:81,packages/ frontend:25,packages/ cli:20	Merge branch 'cross-thread-memory-profiles' into episodic-memory-sdk
K	eval-ci-lane-memory-headroom	f5837da29 2026-07-14T13:55:02+01:00	6cb9a2f13	14 / 1262	unmerged-tip	1/13/0; 14	1/4	packages/ @n8n:12,,github:2	ci: Restore 14 eval lanes and widen the high-N job ceiling (no-changelog)

Flags	Branch	Head / tip date	Merge base	Ahead / behind	Exact-tip status	A/M/D; files	Docs/tests	Leading path groups	Tip subject
K	feat/observation-memory-optimize	a94a8aa93 2026-07-03T13:29:43+02:00	af8acbfab	5 / 1559	unmerged-tip	7/2/0; 9	1/1	scripts:4,packages/@n8n:2,.agents:2	chore: fix test
K	feature/instance-ai-workflow-setup-rewrite	a62d17756 2026-05-07T14:45:32+02:00	5c7921f71	10 / 3516	unmerged-tip	77/26/27; 130	1/26	packages/testing:64,packages/frontend:56,packages/@n8n:8	Limit shown credentials to current project or workflow (#29968)
K	fix-package-doc-dependency-policy-guidance	e2f5b50fb 2026-07-22T07:48:19Z	43c6f329f	17 / 939	unmerged-tip	4/12/0; 16	1/8	packages/cli:13,packages/@n8n:3	docs(cli): Clarify sub-workflow dependency handling with policy flag
K	hackmation-ai-agents-v2-rebased	3cad43bae 2026-06-05T12:09:12+01:00	06cf1d9a7	1 / 2601	unmerged-tip	78/34/0; 112	5/26	packages/cli:44,packages/@n8n:28,packages/frontend:17	feat: Rebase A2A agentic-users feature onto master
K	hackmation-memory-guard	716f4d9e3 2026-08-14T15:51:45+02:00	a3fefb257	1 / 188	unmerged-tip	6/9/0; 15	0/1	packages/cli:9,packages/@n8n:4,packages/workflow:2	feat(core): Shed load under memory pressure
K	ins-198-hitl-pending-confirmations-are-in-memory-only	450413037 2026-05-06T19:12:12+03:00	64079ad98	11 / 3555	unmerged-tip	6/15/0; 21	0/7	packages/cli:10,packages/@n8n:10,packages/frontend:1	Merge branch 'master' into ins-198-hitl-pending-confirmations-are-in-memory-only
K	instance-ai-agents-pr6-docs-polish	3d5760f9c 2026-05-20T14:08:07+02:00	ba35fc81c	4 / 3166	unmerged-tip	0/38/0; 38	9/14	packages/@n8n:29,packages/frontend:9	fix(instance-ai): align native tool contract references
K	kafka-v2-credential-connection-test	8ed02bbae 2026-08-14T03:06:29-07:00	b2df4c885	5 / 45	unmerged-tip	1/3/0; 4	0/3	packages/nodes-base:4	Merge branch 'master' into kafka-v2-credential-connection-test
N	ligo-965-git-connections-db	d94b3ef61 2026-08-13T23:28:43+01:00	bab26492b	4 / 65	unmerged-tip	5/2/0; 7	4/1	docs:4,packages/cli:2,packages/@n8n:1	test(core): Cover git_connection CHECK constraints and down migration (no-changelog)
K	ligo-workflow-testing-v2	5e51b07b6 2026-08-12T15:25:32+02:00	a319c7626	13 / 140	unmerged-tip	19/1/0; 20	5/2	packages/n8n-test:15,tasks:2,.agents:2	chore(n8n-test): Apply v2 verification findings

12.3 Current sampled branch table — batch C

Flags	Branch	Head / tip date	Merge base	Ahead / behind	Exact-tip status	A/M/D; files	Docs/tests	Leading path groups	Tip subject
K	make-cats-codeowners-of-core-execution-engine	41ddea0ed 2026-06-24T17:06:37+03:00	c8ef7b808	1 / 1929	unmerged-tip	0/1/0; 1	0/0	.github:1	Make cats codeowners of packages/core/src/execution-engine
O	n8n-5173-microsoft-teams-node-roll-out-resource	ede4a3546 2022-12-19T14:25:37+01:00	5059c57f4	27 / 15892	unmerged-tip	0/7/0; 7	0/0	packages/nodes-base:6,packages/core:1	[-] - remove unused loadOptions method getBuckets
K	n8n-agents-episodic-memory	5301c6f9c 2026-05-19T19:31:04+02:00	cdfc9053f	98 / 3184	unmerged-tip	24/41/0; 65	0/16	packages/cli:25,packages/@n8n:25,packages/frontend:15	refactor(Agents): clean up episodic memory CLI integration

Flags	Branch	Head / tip date	Merge base	Ahead / behind	Exact-tip status	A/M/D; files	Docs/tests	Leading path groups	Tip subject
K	n8n-agents-episodic-memory-evals	4b390803a 2026-05-20T14:53:45+02:00	cd9c9053f	24 / 3184	unmerged-tip	35/44/0; 79	1/19	packages/@n8n:38,packages/cli:27,packages/frontend:14	test(Agents): update episodic eval harness logging
K	n8n-agents-episodic-memory-ui-card	bd1ee8300 2026-05-27T18:37:44+01:00	e9649d21f	13 / 2935	unmerged-tip	0/0/0; 0	0/0		Merge branch 'master' into n8n-agents-episodic-memory-ui-card
K	n8n-agents-observational-memory-evals	22b52d58b 2026-05-18T18:03:12+02:00	89ffd90e8	71 / 3258	unmerged-tip	26/62/18; 106	1/45	packages/@n8n:70,packages/frontend:19,packages/cli:16	test(Agents): add observational memory golden evals
K	named-version-prototype	5b83dc24d 2026-01-13T13:53:43Z	7b74533a1	119 / 5935	unmerged-tip	12/118/2; 132	0/49	packages/frontend:80,packages/testing:32,packages/cli:11	improvements
K	node-4288-linear-v2-overhaul	a65cb19c3 2026-07-19T17:57:45+01:00	0a4dddeb2	8 / 1147	unmerged-tip	128/6/2; 136	0/11	packages/nodes-base:136	feat(Linear Node): Migrate Issue and Project selectors to resource-locator
K	node-5068-community-issue-mongodb-chat-memory-connection-string	bd519aa48 2026-08-14T11:53:14+02:00	01b01ab77	1 / 46	unmerged-tip	1/1/0; 2	0/1	packages/@n8n:2	fix(MongoDB Chat Memory Node): Sanitize messages
K	observational-memory-cli-integration	b4f0b0e4a 2026-05-12T12:43:16+02:00	d5d51731d	25 / 3372	unmerged-tip	4/53/4; 61	1/23	packages/@n8n:25,packages/frontend:22,packages/cli:13	Merge branch 'master' into observational-memory-cli-integration
K	pass-agent-id-to-memory	1e538fb71 2026-05-20T16:33:11+02:00	df5a1c445	1 / 3148	unmerged-tip	0/7/0; 7	0/3	packages/cli:7	fix(core): Pass agent id to memory implementation
K	pay-2550-prototype-interactive-designs	a1c3d93b2 2025-02-03T09:09:59+01:00	0063bbb30	2 / 10042	unmerged-tip	5/6/0; 11	0/0	packages/editor-ui:11	Merge branch 'master' into pay-2550-prototype-interactive-designs
K	prototype/improve-publish-state	d1de75096 2026-06-03T17:10:10+02:00	7bdf9e900	1 / 3263	unmerged-tip	2/3/0; 5	0/0	packages/frontend:5	feat(editor): Prototype clearer workflow publish states (no-changelog)
R	release-candidate/2.34.x	d743b0ad2 2026-08-14T07:33:28Z	b533ecd68	22 / 462	unmerged-tip	20/213/0; 233	3/73	packages/@n8n:71,packages/nodes-base:48,packages/cli:44	:rocket: Release 2.34.6 (#36277)
R	release-candidate/2.35.x	0dccd14a4 2026-08-14T09:39:19Z	5102c608e	14 / 213	unmerged-tip	10/134/0; 144	5/39	packages/@n8n:60,packages/cli:36,packages/frontend:26	:rocket: Release 2.35.3 (#36288)
R	release/1.50.1	485181647 2024-07-16T13:56:50+02:00	de9590e10	3 / 11915	unmerged-tip	8/7/8; 23	1/0	packages/editor-ui:19,packages/cli:2,package.json:1	:rocket: Release 1.50.1 (#10072)
R	release/2.34.5	f745babb4 2026-08-12T07:27:18Z	b533ecd68	18 / 462	unmerged-tip	20/204/0; 224	3/69	packages/@n8n:71,packages/cli:44,packages/nodes-base:42	:rocket: Release 2.34.5 (#36103)

Flags	Branch	Head / tip date	Merge base	Ahead / behind	Exact-tip status	A/M/D; files	Docs/tests	Leading path groups	Tip subject
K	rly-37-introduce-docs-for-using-n8n-skills-inside-cursor	2cb5a3a6e 2026-05-05T15:24:05+02:00	c742a85b3	1 / 3616	unmerged-tip	0/2/0; 2	2/0	AGENTS.md:1,.claude:1	docs: add Cursor setup instructions for n8n skills plugin
O	webhooks-caching	dc8eb7782 2023-11-14T16:24:58+01:00	a9c7188c4	3 / 13466	unmerged-tip	0/10/0; 10	0/2	packages/cli:8,packages/workflow:2	cleanup

12.4 Significant unmerged-head census

[OBSERVED] A commit-graph-only pass examined all 1,645 advertised heads. **504** non-default heads had more than 10 commits not reachable from current `master`; 29 of those are in the 60-ref tree sample above. A mutually exclusive branch-name grouping counted 237 `release/*` refs, 12 `release-candidate/*` refs, 17 backport/1.x-build refs, 39 agent/TRUST-named refs, and 199 other refs. These are name groupings, not implementation classifications.

[OBSERVED] Every >10-ahead head was therefore noticed at commit-graph metadata level. Exhaustively hydrating/diffing 504 tips would conflict with the mandatory large-surface sampling rule and disk ceiling, so tree inspection remains stratified. Release/backport families are represented by `1.x`, `release/1.50.1`, `release-candidate/2.34.x`, `release-candidate/2.35.x`, and `release/2.34.5`.

Largest ahead counts among non-release/non-backport heads

Ahead	Branch	Behind	Tip	Date	Subject
300	cat-bot/run-31855574975	6462	d59af9af4	2026-08-15T01:14:17Z	chore: Bump @zone-eu/maillsplit 5.4.8 → 5.4.15 (patch) on 1.x to clear 1 Aikido finding
257	fix/aikido-security-update-packages-72152345-4zot	6462	8ba62b27f	2026-07-23T13:42:24+01:00	fix: Scope brace-expansion override to preserve v2 consumers
257	fix/aikido-security-update-packages-78929934-pl6e	6462	9c889a8d0	2026-08-01T23:12:28Z	fix(security): update dependencies
250	fix/aikido-security-update-packages-63715811-cthm	6462	9e80e0a1c	2026-07-09T23:10:42Z	fix(security): update numpy from 2.3.4 to 2.4.6
233	cat-bot/run-27673672366	6462	0b129429f	2026-06-17T18:37:40+01:00	chore(deps): bump tmp, form-data, protobufjs, ws, element-plus
194	fix/aikido-security-update-packages-39112761-ptwb	6462	5bf370bbd	2026-05-25T15:02:32+01:00	fix(SAML): adapt express request to ESamlHttpRequest type
182	fix/aikido-security-update-packages-33090525-x2w8	6462	33c1f234a	2026-05-11T23:12:56Z	fix(security): update urllib3 from 2.6.3 to 2.7.0
173	fix/aikido-security-update-packages-28231892-du5t	6462	faf5b4e70	2026-05-05T11:59:59+01:00	Merge branch '1.x' into fix/aikido-security-update-packages-28231892-du5t
157	feat/crdt-abstraction-layer	5463	ae347cbb5	2026-02-02T11:00:50+01:00	fix(crdt): clean up awareness states on editor disconnect
138	fix-resolve-llm-tool	3594	01f827e31	2026-05-06T18:11:17+02:00	fix(core): Validate agent builder LLM model ids against provider's live list
137	n8n-agents	3594	f330a2c11	2026-05-06T15:39:47+01:00	Tidy up guards and redundant checks.
131	agent-100-feature-memory-compression-with-sub-agents	3629	41ceaf60a	2026-05-05T18:15:45+02:00	add complex example of om

Ahead	Branch	Behind	Tip	Date	Subject
122	fix-export-entities-v1	6462	517b496e1	2026-03-09T17:09:31+01:00	fix(cli): prevent self-inclusion of output ZIP during export
122	nodes/nocodb-v4	3378	d8fa18142	2026-05-12T10:31:50+01:00	Merge branch 'master' of github.com:/n8n-io/n8n into nodes/nocodb-v4
119	named-version-prototype	5935	5b83dc24d	2026-01-13T13:53:43Z	improvements
103	sidebar-virtual	7889	f3f098054	2025-09-25T16:26:11+01:00	icon
101	ds-289-component-combobox	99	8fcdbad5f	2026-08-13T13:46:03+01:00	Merge branch 'master' into ds-289-component-combobox
98	n8n-agents-episodic-memory	3184	5301c6f9c	2026-05-19T19:31:04+02:00	refactor(Agents): clean up episodic memory CLI integration
97	cat-3871-error-loop-and-rate-limit-backoff	227	3cc6d2fa0	2026-08-11T11:14:27+01:00	fix(core): Skip backing off a poll node that was deactivated mid-poll
74	episodic-memory-iteration	3405	e0b1146ac	2026-05-12T10:36:26+02:00	Improve Episodic Memory
73	episodic-memory-evals	3405	de10dd7c6	2026-05-12T08:30:10+02:00	Merge branch 'episodic-memory-docs' into episodic-memory-evals
73	fix-iai-mcp-tool-loading	1587	929637193	2026-07-03T04:55:49Z	feat: Always show save settings button as enabled on init
71	add-agent-to-slack-stack	20	ab4ae3eca	2026-08-14T17:08:42+02:00	fix formatting
71	n8n-agents-observational-memory-evals	3258	22b52d58b	2026-05-18T18:03:12+02:00	test(Agents): add observational memory golden evals
64	fix-iai-mcp-tool-loading-3	1673	0b9fb35dd	2026-07-01T06:52:04Z	feat: Always show save settings button as enabled on init

13. Fixed-frame branch reconstruction

13.1 Three oldest extant branch tips

n8n-5173-microsoft-teams-node-roll-out-resource

- [OBSERVED] Head `ede4a3546`, tip date 2022-12-19; merge base `5059c57f4` dated 2022-11-23; 27 commits not in Sprint 2 `master`, 15,892 `master` commits absent from the branch.
- [OBSERVED] Its tree delta is confined to seven modified files: `packages/core/src/NodeExecuteFunctions.ts` and six Microsoft Teams node description/implementation files.
- [STATED] The branch commits repeatedly use “RLC” and describe rolling out Chat, Team, Channel, Group, Plan, Member, and Bucket parameters “by URL” or “by ID”; commits remove corresponding `loadOptions` methods.
- [OBSERVED] Current `master` contains `resourceLocator` declarations under the Microsoft Teams trigger and `Microsoft/Teams/v2/descriptions/rlc.description.ts`.
- [OBSERVED] The exact branch tip is not an ancestor of `master`; its old paths differ from current v2 paths.
- [UNKNOWN] Whether current Teams v2 resource-locator code was ported from this branch, independently rewritten, or partially cherry-picked cannot be established from exact-tip ancestry and path names alone. Status: **historical, extant, exact tip unmerged; later same-domain implementation exists.**

check-data-processed

- [OBSERVED] Head `ca93689c1`, tip date 2023-07-27; merge base `d050b99fb`; 32 commits ahead and 14,079 behind current `master`; 9 added and 21 modified files.
- [OBSERVED] The branch added `ProcessedDataManager`, `ProcessedDataManagers/NativeDatabase`, a `ProcessedData` entity/repository, PostgreSQL and SQLite migrations, an integration test, and a `CheckProcessed` node; it modified command initialization, DB wiring, core execute functions, workflow interfaces, and `RssFeedRead`.
- [STATED] Commit messages describe checking whether data “got already processed,” recording it, keeping contexts independent, cleaning DB data on workflow update/delete, `maxEntries`, and mode `latest`.
- [OBSERVED] The node description implemented operations to add, filter, or remove a processed value and returned `new` and `processed` item groups. Tests exercised context separation and pruning by a maximum saved-entry count.
- [OBSERVED] Current `master` does **not** contain the branch’s `CheckProcessed.node.ts`, `ProcessedDataManager.ts`, or old CLI entity/repository paths. It does contain `packages/@n8n/db/src/migrations/common/1726606152711-CreateProcessedDataTable.ts`; the current path’s visible history includes `249b7f855` (2025-05-14), “Move migrations to `@n8n/db`.”
- [UNKNOWN] The branch-specific node/manager was not exactly merged. The later current processed-data table may represent a related, partial, or independent implementation; evidence inspected does not prove derivation. Status: **historical, extant, exact tip unmerged; some same-term persistence exists later under different paths.**

webhooks-caching

- [OBSERVED] Head `dc8eb7782`, tip date 2023-11-14; merge base `a9c7188c4`; 3 commits ahead and 13,466 behind; ten modified files.

- [STATED] Its three messages were “bench,” “cleanup (webhook.service test still failing),” and “cleanup.” The test-failure wording is a target-authored statement, not an independently run test result.
- [OBSERVED] The diff removed `CacheService/DB` lookup code from `WebhookService`, added a nested in-memory `Map<string, Map<method, RegisteredWebhook>>`, and introduced `registerWebhook`, `unregisterWebhook`, synchronous `findWebhook`, and synchronous `getWebhookMethods`.
- [OBSERVED] Registered values included the `Workflow`, node, node type, and pre-resolved webhook response fields. A source TODO said to update Redis and publish a message for other instances to pull the cache.
- [OBSERVED] `ActiveWorkflowRunner` changed from awaiting repository/cache lookup to reading the registration map and using preconstructed workflow/webhook/node objects. The branch also moved response-mode/data/code/binary resolution into `NodeHelpers.getNodeWebhooks`.
- [OBSERVED] The exact tip is unmerged. Current `master` still has `packages/cli/src/webhooks/webhook.service.ts`, but a current path alone does not establish that this Map implementation survived.
- [INFERRED] The branch is evidence of a bounded caching/registration experiment. Its exact retention status is **unmerged**; abandonment versus later rewrite is [UNKNOWN].

13.2 Three newest Sprint 1 non-default tips

`ligo-965-git-connections-db` (ref still present)

- [OBSERVED] Head `d94b3ef61`; merge base `bab26492b`; 4 commits ahead and 65 behind; seven changed files (five additions, two modifications).
- [OBSERVED] It added a reversible `CreateGitConnectionTable` migration, an enterprise-module `GitConnection` entity, migration tests, and generated PostgreSQL/SQLite schema pages.
- [OBSERVED] Table/entity fields included string ID, name, repository URL, nullable branch name, connection type (`ssh/https`), public/private-key material fields, encrypted username/password fields, key-generator type (`ed25519/rsa`), base commit, and timestamps. DB CHECK constraints limited connection and key-generator types.
- [OBSERVED] No `git_connection/git-connection` path existed in current `master` at the inspected head.
- [OBSERVED] This was the first layer of a visible stack: `d94b3ef61` is ancestor of `ligo-965-git-connections-module` (`aed4ab70a`), which adds 21 commits for DTOs, validation, Git utilities/service, module/service, tests, working-copy reset, known-host handling, URL guards, and timeouts.
- [OBSERVED] `ligo-965-git-connections-cli` (`5b4aa5c6a`) diverged from the module line at `c84b6b4c9`, incorporating public-API and CLI work. `ligo-965-create-and-update-a-git-connection-v1` (`44fdc83a2`) is an ancestor of the CLI tip by two commits.
- [UNKNOWN] The stack was active and unmerged at collection; eventual merge/removal cannot be determined.

Removed ref `ins-1186-forward-run-and-thread-ids-on-proxy-model-calls`

- [OBSERVED] The ref disappeared between sprints. GitHub still returned commit `c79f9a3a5387c28f11793f488de98c219176e898`.
- [OBSERVED] Its divergence comparison exposed 12 branch commits and 30 changed files. Messages described Kimi/MoonshotAI proxy routing, provider quirks/defaults, model configuration, Agent Builder/Instance AI settings, and finally forwarding run/thread IDs.
- [STATED] The final commit says `x-n8n-run-id` and `x-n8n-thread-id` headers let the proxy derive sticky-session routing without n8n knowing provider infrastructure details.

- [OBSERVED] Against Sprint 2 `master`, GitHub reported `diverged`: `master` 65 commits ahead of the merge base and 12 commits from this branch absent from `master`.
- Status: **remote branch ref removed; exact commits not merged at collection; reason for ref deletion [UNKNOWN]**.

Removed ref `cat-3868-validate-unique-node-ids`

- [OBSERVED] This ref also disappeared between sprints; commit `70809f6415e416951f3c1adbfb3c7f89c6d8e0b` remained public.
 - [OBSERVED] Four branch commits changed eight workflow service/controller/validation test files. They added duplicate node-ID validation, skipped structure validation on metadata-only updates, treated empty-string IDs as absent, and expanded create-path integration tests.
 - [OBSERVED] GitHub comparison to Sprint 2 `master` reported `diverged`: 83 `master` commits beyond the merge base and 4 branch commits absent from `master`.
 - Status: **remote branch ref removed; exact commits not merged at collection; reason [UNKNOWN]**.
-

14. Keyword branch families and experiments

14.1 Memory branch family

[OBSERVED] The memory-named refs are not one linear branch. Exact ancestry established this partial graph:

MERMAID DIAGRAM SOURCE (RETAINED AS EVIDENCE)

```

flowchart LR
    SDK["SDK[\"cross-thread-memory-sdk-runtime\"]"] --> CLI["CLI[\"cross-thread-memory-cli-persistence\"]"]
    CLI --> FE["FE[\"cross-thread-memory-fe-integration\"]"]
    FE --> DIVERGED["FE .. diverged"]
    DIVERGED --> DOCS["DOCS[\"cross-thread-memory-docs\"]"]
    DIVERGED --> PROFILES["PROFILES[\"cross-thread-memory-profiles\"]"]
    PROFILES --> ESDK["ESDK[\"episodic-memory-sdk\"]"]
    ESDK --> ECLI["ECLI[\"episodic-memory-cli-fe\"]"]
    ECLI --> EDOCS["EDOCS[\"episodic-memory-docs\"]"]
    EDOCS --> EEVALS["EEVALS[\"episodic-memory-evals\"]"]
    EEVALS --> ITER["ITER[\"episodic-memory-iteration\"]"]
  
```

- [OBSERVED] `cross-thread-memory-sdk-runtime` → `cli-persistence` → `fe-integration` is exact ancestry. `fe-integration` and `cross-thread-memory-docs` share merge base `2ed961cb8` but each has commits absent from the other.
- [OBSERVED] `cross-thread-memory-profiles` → `episodic-memory-sdk` → `episodic-memory-cli-fe` → `episodic-memory-docs` → `episodic-memory-evals` → `episodic-memory-iteration` is exact ancestry. Each later tip adds commits; `iteration` is one commit beyond `evals`.
- [OBSERVED] Cross-thread branches added memory-profile runtime code/tests, observation cursor/lock/store/cycle code, DB migrations, profile/observation entities and repositories, agent-service integration, and frontend memory panels/toggles. `cross-thread-memory-docs` added `memory-stack.html`.
- [OBSERVED] Episodic branches added `episodic-memory-implementation.md`, `episodic-memory.html`, runtime and integration tests, runtime episodic-memory code, SDK memory-builder tests, frontend panels, and evaluations.
- [OBSERVED] Separate `n8n-agents-episodic-memory*` and `n8n-agents-observational-memory-evals` tips were **not** descendants of the earlier chain. For example, `n8n-agents-episodic-memory` and `...-evals` each diverged from common merge base `cd9c9053f`; neither contains the other's tip.
- [OBSERVED] The observational-eval branch added memory scenarios and scoring for active-fact recall/precision, stale-fact suppression, exact identifier recall, lifecycle accuracy, contamination, final-answer correctness, and answer faithfulness. It also contained observer/reflector code operating on observation logs and token thresholds.
- [OBSERVED] Current `master` contains episodic-memory and observation-log files under `packages/@n8n/agents/src/runtime/memory/`, plus Instance AI observation-log storage and generated schema pages. The current episodic path has later visible commits including runtime refactoring, secret sanitization, cancellation, and OpenTelemetry tracing.
- [OBSERVED] None of the five representative memory branch tips tested was an ancestor of `master`. Therefore, existence of same-term current files does not make the old branch tips “merged.”
- [OBSERVED] `agent-429...episodic-memory...` is a later five-commit tuning branch; it modified episodic-memory runtime/defaults/tests, cassette scenarios, prompt-caching docs, tool schema behavior, and UI message display. Its exact tip was also unmerged.
- [UNKNOWN] Patch-level lineage from each May branch to current memory code is unresolved. The evidence supports multiple parallel attempts and later same-domain current implementation, not a single proven branch-to-main path.

14.2 Prototype/rewrite branches

Publish-state prototype

[OBSERVED] `prototype/improve-publish-state` is one unmerged commit (`d1de75096`) changing five frontend files. It added `PublishStatePrototypeSwitcher.vue` and `usePublishStatePrototype.ts` and extensively modified `draft/publish` header actions. Both added paths return 404 at current `master`. Status: **branch-only prototype at inspected current state**.

Insights/dashboard prototype

[OBSERVED] `pay-2550-prototype-interactive-designs` contains two branch commits and eleven frontend changes, adding an Insights store/router/summary/view and Chart.js utilities. The historical added path under `packages/editor-ui/src/features/insights` is absent in current `master`; current source has an insights feature under `packages/frontend/editor-ui/src/features/execution/insights`. [UNKNOWN] Whether current insights code descends from this prototype.

Named-version prototype

[OBSERVED] `named-version-prototype` has 119 branch-only commits and a 132-file tree delta. Commit messages span autosave, save-button removal, collaboration/read-only behavior, workflow-history compaction, activity recording, tags, import dirty-state behavior, and clearer published-state indicators. The generic tip message “improvements” does not identify one final concept. Status: **large composite unmerged prototype line**; partial cherry-picks or later independent implementations are [UNKNOWN].

Instance AI setup rewrite

[OBSERVED] `feature/instance-ai-workflow-setup-rewrite` has ten commits and 130 changed files (77 added, 26 modified, 27 removed). Messages explicitly delete “legacy FE,” improve credential handling, add parameter support and node grouping, fix displayed-version selection, and scope displayed credentials to project/workflow. The delta includes service tests, frontend setup components, Playwright pages/specs, and many added/removed recorded expectation artifacts. Exact tip: unmerged.

A2A “agentic users” branch

[OBSERVED] `hackmotion-ai-agents-v2-rebased` is one large unmerged commit with 112 changed files (78 added). It adds an `agentic-users` API/controller/service area, A2A adapter/router, external-agent registration entity/migration/repository, DTOs, public-API schemas, permissions, frontend views/stores/components, tests, scripts, and `docs/agent-a2a-guide.md`.

[OBSERVED] The commit also contains branch-only artifacts such as `.playwright-mcp` logs/screenshot, `err.log`, critique Markdown files, and research/demo files. Current `master` returns 404 for `packages/cli/src/agentic-users` but 200 for `packages/cli/src/modules/agents`. This establishes non-retention of that path, not equivalence between the two concepts.

14.3 Frontend module SDK v2 branch and later mainline form

- [OBSERVED] `cat-3685-create-n8nmodule-sdk-package-with-descriptor-v2-and` is one unmerged commit changing 51 files. It adds `packages/frontend/@n8n/module-sdk`, descriptor tests/types, command/modal/push/resource registries, a module manifest, and integration changes; it removes the prior `module.types.ts` path.
- [OBSERVED] Current `master` returns 404 for `@n8n/module-sdk/package.json` and 200 for `@n8n/frontend-module-sdk/package.json`.
- [OBSERVED] Later mainline commit `c84ada02` (2026-07-20) is titled “Add descriptor v2 fields and module SDK registries,” changing 28 files under `packages/frontend`; current path history includes that commit.
- [INFERRED] The branch records an earlier package/name/form for the same target vocabulary. The evidence does not establish whether `c84ada02` was a rewrite, selective port, or separately authored implementation.

14.4 Other keyword refs

[OBSERVED] The 60-row tables independently record every strict keyword ref plus fixed oldest/newest and release representatives. Examples of branch-local forms include:

- docs/context ownership and install guidance (`D0C-*`, `docs/*`);
- private credentials v2, messenger settings v2, Kafka/SharePoint/Linear v2 work;
- a node-type description “memory baseline harness” (not agent memory implementation);
- execution digest/item-grouping utilities;
- a memory-pressure load-shedding branch (`hackmation-memory-guard`);
- codeowner changes for `packages/core/src/execution-engine`;
- tests/evals and CI memory-headroom changes.

[OBSERVED] All 60 exact tips were unmerged relative to the Sprint 2 `master`. One (`n8n-agents-episodic-memory-ui-card`) had 13 commits not in `master` but a zero-file merge-base-to-tip tree delta, showing that unique commit ancestry need not imply unique final tree content.

14.5 Release branches

- [OBSERVED] `1.x` was 299 commits ahead and 6,462 behind `master`, with a 1,000-file merge-base delta; it is a separately maintained line, not merely a stale pointer.
- [OBSERVED] `release/1.50.1` was 3 ahead/11,915 behind; `release/2.34.5` 18/462; `release-candidate/2.34.x` 22/462; `release-candidate/2.35.x` 14/213 at the new snapshot.
- [OBSERVED] Exact release tips are not ancestors of current `master`; that does not mean the releases were abandoned. Their divergent patch histories and backports make “merged to master” an unsuitable synonym for “released.”

15. Consequential commit archaeology

Commit messages below are [STATED]; hashes, dates, and inspected tree deltas are [OBSERVED]. “Files” is no-rename changed-file count.

Commit	Date	Message	Files	Direct structural evidence
2b7382cd	2019-06-24	Rename CLI <code>run</code> to <code>execute</code>	4	Deleted <code>commands/run.ts</code> , added <code>commands/execute.ts</code> , updated <code>README/index</code> ; the command body remained workflow-file/ID execution.
d0275459	2019-07-21	Use Convict for configuration	15	Deleted <code>config/default.ts</code> , added <code>config/index.ts</code> , added <code>convict</code> dependency/types, updated <code>server/DB/workflow</code> helpers and Docker docs/config.
67afc883	2019-08-28	Replace Vorpall with <code>oclif</code> in CLI	8	Added CLI launchers, removed CLI index, converted <code>execute/start</code> command classes/imports/dependencies from Vorpall to <code>oclif</code> .
c1e753e9	2019-09-04	Remove Read File From URL node; add functionality to HTTP Request	4	Deleted <code>ReadFileFromUrl.node.ts</code> ; modified <code>HttpRequest.node.ts</code> , package manifest, breaking-change file.
70e972ff	2020-01-10	Revert OAuth-support merge	12	Deleted OAuth node/credential files and removed OAuth callback/auth code and dependency/UI routes.
48a1e0c4	2020-04-29	Move <code>ormconfig</code> to migrations folder	3	Deleted a committed <code>database.sqlite</code> and old <code>ormconfig.ts</code> ; added <code>migrations/ormconfig.ts</code> .
62612d0a	2020-07-05	Move docs to <code>n8n-io/n8n-docs</code>	29	Deleted the complete then-current in-repo docs site, including setup, architecture-adjacent “key-components,” DB, workflow, security, node, and image files.
b33a5fcd	2021-01-23	Remove MongoDB support	18	Deleted nine MongoDB entities/migrations/index files; modified CLI and Docker files.
7a3aaf8a	2021-02-09	Unify execution ID + Queue system	26	Added worker command and queue-related files; modified execution, webhook, core workflow <code>execute</code> , UI settings/list paths.
e53efdd3	2021-02-09	Separate webhooks from core	18	Added webhook command, <code>WebhookServer.ts</code> , and <code>Queue.ts</code> ; modified <code>worker/start/execution/webhook</code> and UI paths.
7264239b	2022-03-14	Add User Management	265	Added JWT/auth/routes/email, user/role/shared-credential entities and migrations, UI flows, design-system components, and tests.
521cf51e	2022-03-17	Change to Sustainable Use License	19	Replaced Apache 2.0 + Commons Clause text with Sustainable Use License text across root/packages/docs/UI; README wording changed accordingly.
a18081d7	2022-06-08	Add n8n Public API	113	60 additions concentrated in CLI plus editor/design-system changes; establishes an explicit Public API code transition.
a9fa830b	2022-08-06	Migrate Lerna to Turborepo	15	Added <code>turbo.json</code> , deleted Lerna config, and modified package/CI/Docker workspace build files.
97cd564f	2022-09-21	User Management and Credentials sharing	168	Added credential controllers/services, role/user service and migrations; reorganized auth/credential endpoints and sharing UI/tests.

Commit	Date	Message	Files	Direct structural evidence
73677738	2022-11-09	Switch npm to pnpm	34	Added pnpm workspace/lock/config material, removed npm lock/config material, modified CLI/packages/patch handling.
fc7aa8bd	2023-07-26	Environments release using source control	51	Five additions and 44 modifications, concentrated in CLI/editor source-control surfaces.
dd6a4c95	2023-07-28	Migrate Design System and Editor UI to Vue 3	461	436 modifications plus additions/deletions across editor, design system, and Cypress support.
c857e426	2023-11-07	Coordinate workflow activation in multiple-main scenario	15	Internal-API/CLI/core/workflow files changed; this is early direct multi-main coordination evidence.
27d83e0d	2024-10-02	Initial Code Task Runners support	39	25 additions across CLI and @n8n packages, with nodes/core/build changes.
5a055ed5	2024-12-12	Move execution-engine code out of n8n-workflow	46	Deleted RoutingNode from workflow and added it plus triggers/pollers to core; modified WorkflowExecute, webhooks, active workflows, node loading, editor, and tests.
d80b49d6	2025-03-20	Set up backend modules	10	Added module decorator/config and tests; integrated module setup into base/start/webhook/worker commands and lifecycle hooks.
11593457	2025-07-18	Set up backend modules	11	Added backend-module guide/templates/setup script plus root script/config changes; a tooling/form expansion distinct from the March runtime setup.
c84ada02	2026-07-20	Add descriptor v2 fields and module SDK registries	28	Added/modified frontend module SDK registry/descriptor paths and removed five prior paths.

15.1 Historical forms established by the sequence

[OBSERVED] The inspected history contains these distinct structural forms over time:

- 1. Lerna/Vorpal-era TypeScript packages with combined CLI REST/webhook/static UI server (2019 root).
- 2. Convict configuration and oclif commands (2019).
- 3. SQLite/MongoDB dual persistence, followed by explicit MongoDB removal (2021).
- 4. Queue worker and separate webhook command/server (2021).
- 5. User/auth/role/shared-credential entities/routes/UI (2022).
- 6. Public API, Turborepo, pnpm, and source-control “Environments” (2022–2023).
- 7. Vue 3 editor/design system and multi-main activation coordination (2023).
- 8. Code task runners and movement of execution-engine code from workflow to core (2024).
- 9. Backend module runtime/decorator/config and module-generation tooling (2025).
- 10. Frontend descriptor-v2/module-registry forms, including an unmerged package-name variant and later mainline frontend-module-sdk (2026).

These are observed forms, not an assertion that one named architectural pattern governed the project.

16. Deleted, reverted, branch-only, and unknown-status material

16.1 Removed from reachable `master` history

- [OBSERVED] `ReadFileFromUrl.node.ts` was deleted in `c1e753e9`, with HTTP Request modified in the same commit.
- [OBSERVED] OAuth node/credential and server callback/auth code introduced by an external branch merge were removed by explicit revert `70e972ff`.
- [OBSERVED] A committed SQLite database file was deleted in `48a1e0c4`.
- [OBSERVED] The in-repository documentation site was deleted in `62612d0a` with a message pointing to `n8n-io/n8n-docs`.
- [OBSERVED] MongoDB persistence support (not the MongoDB integration node) was removed in `b33a5fcd` by deleting its CLI entities/migrations.
- [OBSERVED] `RoutingNode` implementation/tests were removed from `packages/workflow` and corresponding code added under `packages/core` in `5a055ed5`.
- [OBSERVED] License text changed from Apache 2.0 + Commons Clause to Sustainable Use License in `521cf51e`; this is replacement of license material, not code deletion.

16.2 Branch refs removed during the two-sprint observation

[OBSERVED] Two newest refs vanished while their commits remained public and diverged from `master`: the proxy run/thread-ID branch and unique-node-ID validation branch. The disappearance proves ref removal, not author intent or permanent abandonment.

16.3 Branch-only at current snapshot

[OBSERVED] Exact branch-only material includes the in-memory webhook-registration experiment, `CheckProcessed` node/manager, Git connection table/module/CLI stack, publish-state switcher/composable, `@n8n/module-sdk` package-name variant, A2A `agentic-users` area, and multiple memory branch versions. “Branch-only” means the exact tip/path is absent or unmerged at the inspected `master`; it does not mean no related current feature exists.

16.4 Unknown status

[UNKNOWN] For most extant unmerged refs, Git alone cannot distinguish active work, paused work, abandoned work, intentionally retained release history, or a branch whose patches were cherry-picked under different hashes. The report therefore avoids converting age/name into abandonment.

17. Sprint 2 HTTP/Git interaction ledger

All direct HTTP requests were public read-only GETs. Response bodies were not persisted; only parsed structure and excerpts up to 500 characters were retained.

17.1 Commit/compare API

- [OBSERVED] GET commit endpoints for `c79f9a3a` and `70809f64` : 200 at 2026-08-15T01:29:02Z; 0.246s and 0.205s; TLS verify 0.
- [OBSERVED] GET comparisons from each removed tip to Sprint 2 `master` : 200 at 01:37:48-49Z; approximately 1.18s/1.13s; both `diverged`.
- [OBSERVED] GET merge-base-to-tip comparisons for those removed refs: 200 at 01:37:59-01:38:00Z; 0.517s/0.404s; 12 and 4 branch commits respectively.
- [OBSERVED] GET comparisons for publish-state, named-version, insights, module-SDK, Instance-AI rewrite, and A2A branches: 200 at 02:09:33-34Z. Three bounded comparisons (insights/module-SDK/rewrite) were repeated at 02:09:48Z to recover compact parsed evidence after output truncation; all repeated requests returned 200.
- [OBSERVED] GET comparisons across the stacked Git-connection branches: 200 at 02:11:55-56Z; 0.645s, 0.506s, and 0.348s.

Representative comparison excerpt: `{"url":"https://api.github.com/repos/n8n-io/n8n/compare/...", "html_url":...}`. Exact URLs and parsed results are recorded in the sections above.

17.2 Contents API at pinned `master`

[OBSERVED] Ten GETs at 02:11:35-38Z tested exact paths at `2afc867f...` :

Path	Status
<code>packages/frontend/@n8n/module-sdk/package.json</code>	404
<code>packages/frontend/@n8n/frontend-module-sdk/package.json</code>	200
<code>.../PublishStatePrototypeSwitcher.vue</code>	404
<code>.../usePublishStatePrototype.ts</code>	404
current frontend execution/insights directory	200
old editor insights-summary path	404
<code>packages/cli/src/agentlic-users</code>	404
<code>packages/cli/src/modules/agents</code>	200
<code>CheckProcessed.node.ts</code>	404
current common CreateProcessedDataTable migration	200

200 excerpts began with returned JSON name/path/directory entries; 404 excerpts were `{"message": "Not Found", ... "status": "404"}`. TLS verification succeeded for all.

17.3 Path-scoped commit API

[OBSERVED] Four GETs at 02:12:08–09Z queried up to five commits for exact current paths. All returned 200 with TLS verify 0:

- current processed-data migration: visible path history included `249b7f855`, “Move migrations to `@n8n/db`”;
- current episodic-memory runtime path: visible commits included runtime refactor, cancellation, secret sanitization, and tracing;
- current frontend-module-SDK package: visible history included `c84ada02`, descriptor v2/registries;
- current webhook service: visible recent commits concerned publish conflicts, native field resolvers, OAuth resources, and utility movement.

17.4 Git smart-HTTP sessions

[OBSERVED] Multiple read-only Git smart-HTTP fetch sessions against `https://github.com/n8n-io/n8n.git` were used between 01:26Z and 02:12Z. Each successful bounded session fetched refs/commit graphs with `tree:0`; no remote mutation or checkout occurred. Git transport HTTP status codes are not exposed by Git CLI; operation success/failure was logged instead.

18. Resource audit and methodological incidents

[OBSERVED] The sprint repeatedly deleted and recreated the partial bare repository to keep tree hydration bounded. Normal commit-only clones were ~17–24 MB; sampled tree batches peaked at 67 MB and were cleaned before the next batch.

[OBSERVED] Two unplanned promisor-object hydration incidents occurred and are disclosed:

1. Fetching commits whose refs had disappeared caused a partial clone to grow to 136–140 MB; it was immediately deleted, and those commits were subsequently inspected through bounded GitHub API GETs.
2. A path-history `git log` on current episodic-memory code caused a lingering promisor fetch to grow to approximately **3.3 GB and 14,902 files**, far beyond the requested 128 MB/10,000-file ceiling. The process was killed and the repository deleted immediately; the workspace returned to 108 KB/two evidence files. No hydrated object was retained. The needed path history was then obtained through bounded GitHub API GETs.

[OBSERVED] These incidents violated the requested operating budget even though the final retained workspace does not. They are evidence-handling failures, not properties of the target repository, and are recorded to preserve research-chain transparency.

19. Sprint 2 contradictions and unresolved questions

1. [OBSERVED] All 60 sampled exact tips were unmerged, yet several same-term capabilities exist in current `master` (Teams resource locators, processed-data table, agent memory, frontend module SDK, insights). [UNKNOWN] Exact cherry-pick/rewrite lineage.
 2. [OBSERVED] The module-SDK branch package path is absent, while a later mainline package with a longer name and descriptor-v2/registry commit is present. [UNKNOWN] Whether this is a direct rename/supersession.
 3. [OBSERVED] Memory branches form parallel and chained lines; current memory code exists under reorganized paths. [UNKNOWN] Which branch commits, if any, were selectively merged.
 4. [OBSERVED] Release refs are not ancestors of `master`. This contradicts any simplistic use of “merged” as a release-status test, but not the existence of the releases.
 5. [OBSERVED] 504 current heads exceed `master` by more than ten commits; 237 are release refs and many are backports/long-lived lines. [UNKNOWN] Lifecycle status of the remaining heads without issue/PR context.
 6. [UNKNOWN] Hosting PR merge/squash metadata and linked issues for every sampled branch were not fetched; commit/tree evidence is primary in this sprint.
 7. [UNKNOWN] Orphaned/unadvertised branch refs beyond commits already known cannot be enumerated from public Git refs.
-

20. Sprint 2 evidence-matrix additions

Evidence ID	Surface	Direct evidence	Reconstruction	Limit
G2-001	1,645 branch refs	Commit graph and ahead/behind census	504 heads >10 ahead; 60 tree-inspected	Ref snapshot changed during research
G2-002	Oldest branch trio	Merge bases, commits, tree diffs	Teams RLC, processed-data manager/node, webhook Map experiment	Patch-equivalent merges unknown
G2-003	Removed newest refs	Commit and compare APIs	Two refs removed; 12/4 commits unmerged	Reason for removal unknown
G2-004	Git connection stack	Branch diffs and pairwise compares	DB → module/service → public API/ CLI layered work	Stack still unmerged
G2-005	Memory refs	Ancestry graph, paths, code symbols	Parallel cross-thread, episodic, observational implementations/ evals	Mainline lineage unresolved
G2-006	Prototype refs	Compare API file/commit lists	Publish, insights, named-version, setup rewrite, A2A forms	Prototype names do not prove production use
G2-007	c84ada02 and contents API	Later mainline descriptor-v2 SDK paths	Current longer-name package contrasts with branch variant	Direct derivation unknown
G2-008	Consequential commits	Bounded no-rename tree diffs	Historical structural timeline	Selected, not exhaustive 23k-commit audit
G2-009	Deleted/reverted commits	Direct deletion/revert diffs	OAuth revert, node/docs/Mongo removal, license replacement	Later reintroduction may use new forms
G2-010	Resource ledger	Disk/file/process observations	Research-chain integrity disclosure	Incidents exceeded requested transient ceiling

21. Sprint 2 completion state

[OBSERVED] Branches examined: all 1,645 advertised heads at commit-graph/ahead-count level; 60 current refs at independent merge-base/tree-diff level; two disappeared Sprint 1 refs via commit/compare API; representative high-ahead, release, prototype, memory, docs, v2, rewrite, oldest, and newest branches at deeper content level.

[OBSERVED] Important commits inspected with direct bounded tree diffs: 24 listed in Section 15, plus branch commits and removed/ref comparisons above.

Sprint 2 of 6 completed. 4 research sprints remain.

Sprint 3 will inspect the current source architecture at a pinned default-branch state: package roles/dependencies, runtime commands/processes, REST/public API organization, execution lifecycle, queue/scaling, persistence/schema/migrations, task runners, tests, and deployment/configuration.

22. Sprint 3 method, pinned state, and coverage boundary

22.1 Current-source snapshot

- [OBSERVED] Sprint 3 pinned `refs/remotes/origin/master` to commit `2afc867f20d689891b516c0458d627bc52c979fd` , timestamped `2026-08-15T00:00:29Z` . All current-source statements in Sections 22–32 refer to that object unless another ref is named.
- [OBSERVED] The pinned tree has 26,727 tracked paths, 26,040 under `packages/` , and 87 `package.json` paths. It was inspected through a filtered bare promisor clone; no checkout was created.
- [OBSERVED] Path names were enumerated across the whole pinned tree. Content inspection covered all 87 manifests; all 56 files in `packages/@n8n/config/src/configs` ; all 76 hand-written Public API path-spec YAML files; all 32 CLI backend-module declarations; all tracked Dockerfiles/Compose paths; all conventional test-file path names; and selected implementation/test artifacts in the runtime, API, execution, persistence, scaling, task-runner, Engine 2.0, module, deployment, and CI areas.
- [OBSERVED] This was a stratified current-tree inspection, not a claim that every one of the 26,727 blobs was read. Broad content scans were avoided after prior partial-clone hydration incidents.
- [OBSERVED] No live HTTP request was made in Sprint 3. No tests, build, container, migration, server, workflow, or package installation were run. Test source is evidence of tracked assertions, not proof that the pinned suite passes.

22.2 Current status terms used in this sprint

Status	Meaning in Sections 22-32
CURRENT — primary runtime	Tracked and directly wired into the <code>n8n</code> CLI/runtime path at the pinned commit.
CURRENT — conditional	Tracked and wired, but activated only by process role, execution mode, configuration, instance role, or license/module eligibility.
CURRENT — experimental/not ready	Tracked on <code>master</code> , but target-authored source explicitly says it is in development or shows unimplemented boundaries.
CURRENT — tracked, activation unknown	Present in the tree, but the bounded evidence did not establish a runtime composition path.
RETAINED — reachability unknown	Older-form implementation remains in current source, but its active call path was not established.

[OBSERVED] These current-source labels do not overwrite Sprint 2’s HISTORICAL, BRANCH-ONLY, REMOVED, or UNKNOWN statuses.

23. Workspace and package architecture at the pinned commit

23.1 Workspace census and declared dependency graph

[OBSERVED] All 87 manifests parsed as JSON. They comprise 61 under `packages/@n8n/`, 24 under other `packages/` paths, the root manifest, and one `.github` template manifest. Eighty-one distinct package names were declared; 15 manifests set `private: true`.

[OBSERVED] The root manifest declares version `2.35.0`, Node `>=24.0.0`, pnpm `>=10.22.0`, and Turbo-driven workspace tasks. The production Dockerfiles pin Node `24.18.1`.

[OBSERVED] Matching dependency names to the 81 named manifests yielded 423 unique directed internal-package edges across `dependencies`, `devDependencies`, `peerDependencies`, and `optionalDependencies` (439 declarations before duplicate-section collapse). In the graph restricted to `dependencies` and `optionalDependencies`, no strongly connected component with more than one package was found. This is a static manifest result, not a runtime import graph.

[OBSERVED] The most widely referenced internal names across all dependency sections were development foundations `@n8n/typescript-config` (67 declarations) and `@n8n/vitest-config` (60), followed by `n8n-workflow` (35), `@n8n/utils` (23), `@n8n/eslint-config` (17), and `@n8n/config`, `@n8n/di`, and `@n8n/constants` (15 each). Name frequency establishes manifest centrality only.

23.2 Directly observed package roles

Package/path	Current form directly observed
<code>packages/cli</code> (<code>n8n</code>)	CLI bootstrap, commands, HTTP servers/controllers, Public API, module registry consumers, execution orchestration, Bull/Redis scaling, lifecycle hooks, persistence services, scheduling, and task broker.
<code>packages/workflow</code> (<code>n8n-workflow</code>)	Workflow/node data types, expression/workflow abstractions, and the stack-driven <code>WorkflowExecute</code> engine consumed by <code>n8n-core</code> /CLI.
<code>packages/core</code> (<code>n8n-core</code>)	Node execution contexts, <code>WorkflowExecute</code> , lifecycle hook class, binary/storage services, runners-related execution plumbing, and supporting runtime services.
<code>packages/@n8n/db</code>	TypeORM entities, repositories, DB options, database-specific migration indexes, and generated schema documentation inputs.
<code>packages/@n8n/config</code>	Decorator-driven typed configuration classes and environment bindings.
<code>packages/@n8n/api-types</code>	Shared API contract/types; its name alone is not treated as proof of every endpoint implementation.
<code>packages/frontend/editor-ui</code>	Vue editor shell and current feature UI. The tree contains 7,096 editor-ui paths.
<code>packages/nodes-base</code> and <code>packages/@n8n/nodes-langchain</code>	Built-in node/credential implementations and AI/LangChain node implementations respectively.
<code>packages/@n8n/backend-common</code> , <code>@n8n/decorators</code> , <code>@n8n/di</code>	Backend shared services, metadata/decorator registration, and dependency injection used by commands, controllers, modules, lifecycle events, and pub/sub handlers.

Package/path	Current form directly observed
packages/@n8n/task-runner and packages/@n8n/task-runner-python	Separate JavaScript and Python Code-node runner implementations. The Python runner is a Python project, not a JavaScript workspace package.
packages/@n8n/engine	Separate Engine 2.0 package/server with its own Postgres persistence model and Docker image. Its explicit not-ready status is reconstructed in Section 28.
packages/@n8n/node-engine-compatibility	Adapter from v1 workflow/node forms to the Engine 2.0 graph/executor interfaces.

[OBSERVED] The root workspace also includes test, benchmark, tooling, SDK, extension, and frontend packages. A package name is recorded as target-native terminology, not converted into an implementation claim without inspected content.

24. Primary runtime: bootstrap, process roles, and HTTP/API composition

24.1 Bootstrap and command surface

[OBSERVED] The CLI command tree contains 26 non-test TypeScript implementation/support files, including `start` , `worker` , `webhook` , `execute` , `execute-batch` , `audit` , `DB revert` , `import/export` , `list/update/publish/unpublish workflow` , `license` , `LDAP` , `MFA` , `user-reset` , and `test-workflow` commands. One of the 26 is `base-command.ts` , so this is a file count rather than a count of invocable commands.

[OBSERVED] The executable bootstrap validates Node compatibility, loads configuration before DB entity construction, applies network settings, and hands control to a DI-managed command registry. `BaseCommand.init()` then establishes logging/reporting, node and credential loading, optional Redis locking, DB connection/migrations, instance identity, binary/execution storage, task runners, telemetry/event relays, and expression-engine initialization. Sources: `packages/cli/bin/n8n` , `packages/cli/src/command-registry.ts` , and `packages/cli/src/commands/base-command.ts` .

24.2 Long-running process roles

Role	CURRENT wiring observed
<code>start</code> / <code>main</code>	Serves the editor and REST surfaces, initializes modules and execution services, manages active workflows, scheduling and wait tracking, and conditionally initializes queue scaling and multi-main behavior.
<code>worker</code>	Forces <code>EXECUTIONS_MODE=queue</code> , initializes DB, nodes, task runners and scaling dependencies, then consumes Bull jobs only after readiness. Worker health endpoints are optional.
<code>webhook</code>	Refuses to run outside queue mode and serves webhook traffic as a dedicated process.

[OBSERVED] Role names describe executable composition. They do not imply that every module/controller is active on every role; module metadata, license checks, `instanceTypes` , and command-specific initialization filter the actual runtime.

24.3 Internal REST and Public API forms

- [OBSERVED] A bounded static scan examined 110 non-test files under `packages/cli/src` selected as controller files. Of these, 109 contained a recognized controller/route decorator, and the files contained 537 HTTP method-decorator calls. This is a source-call census, **not** a deduplicated or runtime endpoint count.
- [OBSERVED] Internal controllers are registered through decorator metadata and DI. Modules may dynamically import controllers during `init()` , so tree presence does not by itself prove route activation.
- [OBSERVED] Public API v1 is hybrid. Decorated Public API controllers are mounted first. Express OpenAPI Validator then validates and dispatches the older operation-handler surface generated from YAML operation IDs.
- [OBSERVED] The 76 hand-written path-spec YAML files contain 66 distinct path templates and 103 operations. Two operations are marked deprecated in the inspected YAML.

- [OBSERVED] The merged OpenAPI artifact generated from hand-written specs plus decorated routes contains 70 path templates and 113 operations. Generation tests include a committed-spec drift check.
- [OBSERVED] Public API mounting, resolver logic, and OpenAPI generation live under `packages/cli/src/public-api/`; decorated controller infrastructure is shared with the broader controller system.

[INFERRED] The ten-operation/four-path increase from hand-written to merged output is attributable to decorator-generated additions because the generation code performs that merge. It does not establish which routes are enabled under every license or process role.

25. Primary execution architecture and lifecycle persistence

25.1 Local and queue-mode path

MERMAID DIAGRAM SOURCE (RETAINED AS EVIDENCE)

```
graph LR
    T["Trigger / REST / manual request"] --> WR["WorkflowRunner.run"]
    WR --> C["Construct execution context + persist execution"]
    C --> M["EXECUTIONS_MODE and manual-offload rule"}
    M -->|regular/local| WE["WorkflowExecute"]
    M -->|queue| B["Bull job in Redis"]
    B --> W["worker JobProcessor"]
    W --> R["Reload persisted execution + construct same workflow/additional-data/hooks"]
    R --> WE
    WE --> H["Lifecycle hooks"]
    H --> P["DB/blob persistence, events, push, statistics, external/module hooks"]
```

- [OBSERVED] `WorkflowRunner.run()` establishes execution metadata and persistence, performs credential-permission checks, and then either starts the execution locally or enqueues it.
- [OBSERVED] Queue mode normally offloads non-manual runs. `OFFLOAD_MANUAL_EXECUTIONS_TO_WORKERS=true` additionally offloads manual runs.
- [OBSERVED] The primary `WorkflowExecute` implementation is cancellable and stack-driven. It checks input readiness and timeouts, records run data, applies retry/continue-on-fail behavior, executes lifecycle hooks, and schedules downstream or wait-node work.
- [OBSERVED] Queue mode uses Bull over Redis. `JobProcessor` reloads the persisted execution, reconstructs workflow/additional-data/lifecycle state, and invokes `WorkflowExecute` or the manual-execution path.
- [OBSERVED] Bull progress messages relay execution chunks, webhook responses, summaries/failures, cancellation signals, and MCP responses between worker and main-side services.
- [OBSERVED] A separate Redis pub/sub layer handles cross-process commands and relays; it is not the Bull job queue itself.

25.2 Lifecycle hook contract and role split

[OBSERVED] `packages/core/src/execution-engine/execution-lifecycle-hooks.ts` defines eight hook families: `nodeExecuteBefore`, `nodeExecuteAfter`, `nodeFetchedData`, `workflowExecuteBefore`, `workflowExecuteResume`, `workflowExecuteAfter`, `sendResponse`, and `sendChunk`. Callbacks run sequentially in registration order.

[OBSERVED] CLI composition under `packages/cli/src/execution-lifecycle/` adds target-specific behavior:

Composition	Hook responsibilities observed
Regular main	Workflow/node events, final-status normalization, save/delete decisions, UI push, optional per-node progress save, statistics, external hooks, and module-contributed handlers.

Composition	Hook responsibilities observed
Scaling worker	Node events, final-status normalization, worker-side data save, optional progress save, statistics, external hooks, manual push when appropriate, and module handlers.
Scaling main	Workflow-level events, progress/external/finalization hooks, retention deletion or metadata save after worker completion, and module handlers; node-before/after handlers are explicitly cleared.
Subexecution	Workflow/node events, status finalization, save/progress/statistics/external/module hooks, plus binary-data duplication to the parent execution when required.

[OBSERVED] The role split is deliberate in source comments: a scaling worker saves execution data without metadata, while the main applies deletion decisions and saves metadata to avoid foreign-key ordering problems.

25.3 Save policy and persistence adapter

- [OBSERVED] Default save policy is all production errors, all production successes, manual executions enabled, and per-node progress saving disabled. Workflow settings may override those values. Source: `packages/@n8n/config/src/configs/executions.config.ts` and `packages/cli/src/execution-lifecycle/to-save-settings.ts`.
- [OBSERVED] Unsaved manual executions are soft-deleted so binary data remains briefly accessible; comments state hard deletion occurs on a later pruning cycle after the configured buffer.
- [OBSERVED] For non-manual executions that policy says not to retain, lifecycle hooks delete in-flight execution data once no wait state requires it. Error-workflow dispatch is handled separately from retention.
- [OBSERVED] Optional progress save writes `lastNodeExecuted`, run data, and `running` status with guards requiring the execution not already be finished or canceled. Save errors are reported/logged rather than allowed to leave the execution in an unknown state.

[OBSERVED] `ExecutionPersistence` is the current adapter for an execution entity plus its serialized data bundle:

- `database` mode (`storedAt=db`) writes entity and data together in a DB transaction;
- `filesystem`, `s3`, and `azure` modes (`fs`, `s3`, `az`) write the entity to the relational DB and the serialized bundle to the configured blob store;
- reads dispatch according to each row's recorded `storedAt`, not merely the current global mode;
- create/update operations track serialized and binary-data byte sizes;
- missing DB data is reported as a corrupt record, whereas missing out-of-band blob data is surfaced as a missing-data error on a single read.

[OBSERVED] `StorageConfig` defaults to `database`, exposes `N8N_EXECUTION_DATA_STORAGE_MODE`, and maps the four target-native mode names to the four compact tags. Filesystem storage defaults to `~/.n8n/storage`. The old `~/.n8n/binaryData` rename is opt-in through `N8N_MIGRATE_FS_STORAGE_PATH=true` and is stated to become default in v3.

25.4 Relational model

- [OBSERVED] The primary runtime uses `@n8n/typeorm` with SQLite or PostgreSQL. `DB_TYPE` defaults to SQLite; PostgreSQL defaults include database/user `n8n/postgres`, host `localhost`, port `5432`, and schema `public` as recorded in the config classes.
 - [OBSERVED] Runtime DB options combine 59 core decorated entity classes with entities supplied by the module registry, then select database-specific migrations. `synchronize` and TypeORM automatic `migrationsRun` are false; startup invokes the project's migration path. PostgreSQL migration startup uses an advisory lock, while SQLite executes migration units individually.
 - [OBSERVED] Core DB source contains 61 non-test repository files and 329 migration TypeScript files. Execution metadata and serialized execution/workflow data are split between `execution_entity` and one-to-one `execution_data` in the primary schema.
 - [OBSERVED] Generated schema documentation has 128 PostgreSQL tables/996 columns and 127 SQLite tables/992 columns. `workflow_statistics_delta` is the only documented PostgreSQL-only table in the path comparison.
-

26. Configuration, Redis orchestration, and backend modules

26.1 Typed configuration system

[OBSERVED] The bounded package census covered all 56 files in `packages/@n8n/config/src/configs/`: 89 declared classes, 87 decorated with `@Config`, 467 `@Env` bindings, and 465 distinct environment names. Adding the root `GlobalConfig` produces 481 bindings and 479 distinct names. The duplicate names are the SAML and OIDC login-enable variables.

[OBSERVED] Resolution checks a direct environment value and then `${NAME}_FILE`. It parses numbers, booleans, dates, strings, custom constructors, and Zod schemas. Invalid values warn and retain the declared default. Nested config objects are resolved through DI.

[OBSERVED] This is a **config-package census**, not a repository-wide environment-variable census. `StorageConfig` in `n8n-core` and the Python runner's own env parser are current examples outside that package.

[OBSERVED] Selected defaults that materially shape architecture:

Setting	Pinned default
<code>EXECUTIONS_MODE</code>	<code>regular</code>
Execution timeout	<code>-1</code> (unlimited)
Execution pruning	enabled; 336-hour age; 10,000-count limit
Bull retained completed/failed jobs	remove immediately (<code>0</code>)
Task-runner mode	<code>internal</code>
JS runner allowlisted built-ins/external modules	empty strings
JS runner insecure mode	<code>false</code>
Primary runner max concurrency/task timeout	10 / 300 seconds
Python runner native defaults when not overridden	5 / 60 seconds
Engine 2.0 host/port	<code>0.0.0.0:3000</code>

26.2 Multi-main leadership and Redis services

- [OBSERVED] `MultiMainSetup` designates one main as leader and others as followers through a Redis lease keyed to the instance `hostId`. Initialization attempts `SET if absent`; periodic checks renew only when the stored value matches the host.
- [OBSERVED] Losing the key to another host triggers step-down. A missing key causes a re-acquisition attempt. Gain/loss emits `leader-takeover` or `leader-stepdown` so services start or stop triggers, pollers, pruning, wait tracking, license renewal, queue recovery, insights, and other leader work.

- [OBSERVED] Source comments explicitly acknowledge possible inconsistency during Redis/network-partition failure and a shutdown TODO to verify ownership before deleting the leader key. These are target-authored limitations, not inferred failures.
- [OBSERVED] `RedisLockService` implements namespaced SHA-256-keyed leases with random ownership values, `NX` acquisition, periodic compare-and-renew, compare-and-delete release, jittered retry, optional wait timeout, and an `AbortSignal` when renewal loses the lock or exceeds the lease window.

[OBSERVED] Scaling pub/sub uses deployment-prefixed command, worker-response, and MCP-relay channels. The typed event map contains 40 event names: 39 command names plus the worker-response name `response-to-get-worker-status`. They cover configuration reloads, community-package actions, worker status, stop execution, trigger/poller changes, workflow publication, execution/chat/AI relays, test cancellation, agent changes, and redaction cache invalidation.

[OBSERVED] Publisher messages include sender ID, target/self-send semantics, and immediate/debounced handling. Subscriber filtering suppresses same-sender or non-targeted commands unless self-send is set, and debounces eligible event names for 300 ms. Decorated handlers are additionally filtered by instance type and, for mains, current leader/follower role.

[OBSERVED] Top-level Bull jobs use Bull's abort path; inline subworkflow executions have no Bull job, so `ExecutionStopService` broadcasts `stop-execution`, and the worker holding that ID cancels it through its local `ActiveExecutions` registry.

26.3 Module registry and module-contributed persistence

- [OBSERVED] The CLI tree contains 32 backend module declaration files. `MODULE_NAMES` lists those same 32 names.
- [OBSERVED] `ModuleRegistry.defaultModules` contains 30; `agents` and `agent-evals` are not defaults and become eligible only through `N8N_ENABLED_MODULES` (unless later configuration changes the inspected code). `N8N_DISABLED_MODULES` removes names from the deduplicated default-plus-enabled set; listing the same name in both sets throws.
- [OBSERVED] `loadModules()` imports each eligible `<name>.module.js` or `<name>.ee/<name>.module.js` before datasource creation, then collects entity classes and node loaders and registers commands. `initModules()` later applies license and instance-type filters before calling module `init()`, settings, and context hooks.
- [OBSERVED] A current source census found 62 `@Entity/@ViewEntity` declarations in 14 CLI module directories: `agents` 22; `instance-ai` 13; `oauth-server` 5; `chat-hub` 4; `dynamic-credentials` 3; `insights` 3; `token-exchange` 3; `community-packages` 2; `data-table` 2; and one each in `favorites`, `instance-version-history`, `log-streaming`, `MCP registry`, and `workflow builder`.
- [OBSERVED] Fourteen module classes implement `entities()`. This aligns with the 14 module directories above, but decorator count is not a guarantee that every entity is loaded in every deployment: eligibility and license/role logic still applies.
- [OBSERVED] The opt-in `agent-evals` module states that it is main-only, cannot cross the queue boundary because of its tool-mock substrate, and remains dark unless feature flag `101_agent_evals` is on for the user.

[OBSERVED] `packages/modules/README.md` describes a future/organizational form: frontend feature modules are intended under `packages/modules/<name>/frontend`, while backend discovery still uses `packages/cli/src/modules/<name>`. At the pinned tree, `packages/modules` contains only that README; the existing backend modules remain in CLI.

26.4 Extension SDK artifact

- [OBSERVED] `packages/@n8n/extension-sdk` is a current versioned package (`0.17.0`) with backend and frontend exports plus a generated schema.

- [OBSERVED] `packages/extensions/insights` contains an `n8n.manifest.json` declaring backend/frontend setup files, an event, and a workflow-view extension. Its backend setup only logs the supplied object; its frontend registers an `InsightsDashboard` component.
 - [UNKNOWN] No inspected `n8n` manifest dependency or primary bootstrap path established that this example extension is loaded by the current runtime. It is therefore **CURRENT — tracked, activation unknown**, separate from the established CLI `insights` backend module.
-

27. Task runners, Code-node integration, and sandbox forms

27.1 Broker topology and protocol

MERMAID DIAGRAM SOURCE (RETAINED AS EVIDENCE)

```
sequenceDiagram
    participant Code as Code node / IExecuteFunctions
    participant Req as Local task requester
    participant Broker as n8n task broker
    participant Runner as JS or Python runner
    Code->>Req: startJob(taskType, settings)
    Req->>Broker: requester:taskrequest
    Runner->>Broker: runner:taskoffer
    Broker->>Runner: broker:taskofferaccept
    Runner->>Broker: runner:taskaccepted
    Broker->>Req: broker:taskready
    Req->>Broker: requester:tasksettings
    Broker->>Runner: broker:tasksettings
    Runner-->>Broker: data/node-type/RPC request (as needed)
    Broker-->>Req: broker request
    Req-->>Broker: requester response
    Broker-->>Runner: broker response
    Runner->>Broker: runner:taskdone or runner:taskerror
    Broker->>Req: broker:taskdone or broker:taskerror
```

- [OBSERVED] The TypeScript protocol declares nine broker-to-runner message variants, seven broker-to-requester variants, six requester-to-broker variants, and ten runner-to-broker variants.
- [OBSERVED] Runners advertise available capacity with expiring offers. The broker pairs a requester and runner, handles independent accept timeouts, forwards settings/data/node-type/RPC traffic, arms execution timeout, and routes completion/error/cancellation.
- [OBSERVED] Internal mode starts n8n-managed JavaScript and Python child processes. External mode requires an auth token and exposes the broker to separately launched runners. Both authenticate through short-lived grant tokens before the WebSocket task protocol.
- [OBSERVED] Timeout handling differs: internal mode emits a lifecycle event that can restart the managed runner; external mode sends task cancellation and relies on the external launcher/orchestrator.

27.2 Python package and runner

- [OBSERVED] `packages/@n8n/task-runner-python/pyproject.toml` names `task-runner-python` version `0.1.0`, requires Python `>=3.13`, and declares `urllib3==2.7.0` and `websockets>=15.0.1`; Sentry is optional. Development groups contain `Ruff`, `ty`, `pytest`, `coverage`, `asyncio testing`, `aiohttp`, and `setuptools`.

- [OBSERVED] The runner defaults to five concurrent tasks, 60-second task timeout, one-GiB payload limit, no auto-shutdown, empty standard-library/external import allowlists, denied runner-environment access, and disabled transitive-import trust unless launcher/n8n env overrides them.
- [OBSERVED] Each accepted Python task executes in a `forkserver` multiprocessing child with a one-way result pipe. Parent timeout terminates and then kills if necessary. Cancellation reaches the task process. Results/errors and bounded print output return over the pipe/protocol.
- [OBSERVED] Before `exec`, source is parsed and transformed as AST. Validation blocks configured imports, dangerous names/attributes/introspection paths, dynamic/relative import forms, name-mangled access, unsafe formatting patterns, and other listed constructs. Built-ins are filtered and exposed through a read-only wrapper; imports route through guarded callables; `sys.modules` is pruned; `importlib` entry points are wrapped; and the child clears `os.environ` by default.
- [OBSERVED] Allowlisting is still configurable. These mechanisms are observed controls, not a claim that the sandbox is invulnerable.

27.3 JavaScript runner

- [OBSERVED] JavaScript tasks default to `secure` mode. Allowed built-in and external modules are parsed from `NODE_FUNCTION_ALLOW_BUILTIN` and `NODE_FUNCTION_ALLOW_EXTERNAL`; defaults are empty.
- [OBSERVED] Secure startup replaces unsafe Buffer allocation with zero-filled allocation, preloads explicitly allowed external packages before freezing, freezes global constructors/prototypes and internal workflow/expression classes, and wraps module exports in a write-blocking proxy membrane.
- [OBSERVED] Secure task execution uses Node `vm.createContext / runInContext` with a synchronous VM timeout and protocol cancellation. Generated wrapper code blocks prototype get/set operations, removes/locks `Error.prepareStackTrace` and `captureStackTrace`, disables property-definition helpers, and freezes constructors.
- [OBSERVED] `N8N_RUNNERS_INSECURE_MODE=true` selects direct execution via `new Function` and `with(context)` for normal Code-node task modes. This is a configured alternate path, not the default.
- [OBSERVED] The runner statically estimates which n8n built-ins a script uses, requests only needed execution data when possible, and requests node-type descriptions when paired-item behavior requires them. Per-item Code execution is chunked by the Code-node adapter in groups of 1,000.

27.4 Code-node integration

- [OBSERVED] The current Code node offers JavaScript and native Python and two modes: once for all items or once for each item. Python can be disabled with `N8N_PYTHON_ENABLED=false` and checks runner availability before dispatch.
- [OBSERVED] The current `Code.node.ts` execution path always constructs `JsTaskRunnerSandbox` or `PythonTaskRunnerSandbox`; both dispatch through `IExecuteFunctions.startJob` rather than executing user code directly in the workflow process.
- [OBSERVED] JavaScript sends code/settings first and then requests only required workflow/input/env/node-type data over the broker. Python sends its input items and workflow/node identifiers with task settings. Both validate returned item shape in the Code-node adapter.
- [OBSERVED] The JS context exposes seven Code-node RPC helpers: binary assertion/read/write/prepare/string conversion, `helpers.httpRequest`, and deprecated `helpers.request`; `logNodeOutput` is the eighth runner-supported RPC method. Stream, arbitrary filesystem-write, authenticated-request, SSH-client, and other listed helpers are explicitly replaced by unsupported-function errors.
- [OBSERVED] The runner image's launcher config overrides the external image defaults to allow JS built-in `crypto` and external `moment`, while leaving Python stdlib/external allowlists empty. JavaScript and Python health ports are 5681 and 5682 inside the image.

27.5 Retained vm2 form and bounded uncertainty

[OBSERVED] `packages/nodes-base/nodes/Code/JavaScriptSandbox.ts` remains in the current tree and implements a `vm2` `NodeVM` path with legacy module allowlists. The current Code node does not import it.

[UNKNOWN] Whether another current runtime path still reaches that retained class. Two bounded reference-search attempts across the large nodes tree timed out in the partial clone and were stopped; no result was retained. The accurate status is **RETAINED — reachability unknown**, not “removed” and not “active Code-node runtime.”

28. Engine 2.0: current but explicitly not ready

28.1 Separate composition root

- [OBSERVED] `@n8n/engine` version `0.15.0` describes itself as “n8n workflow execution engine (v2),” has a standalone `start` script and `build:docker`, and depends on Express, PostgreSQL, TypeORM, config, DI, UUID, and Zod.
- [OBSERVED] `EngineConfig.databaseUrl` is documented: “This is in development and not ready for use.” When unset, `serve.ts` starts healthcheck-only mode. This is the strongest current status statement and controls the classification here.
- [OBSERVED] With a DB URL, the standalone server initializes a **PostgreSQL-only** datasource, runs two engine-owned migrations, starts in-memory orchestration and step queues/workers, and mounts `POST /api/workflow-executions`; `/healthz` is always mounted.
- [OBSERVED] The start schema accepts graph step types `trigger`, `v1-node`, `wait`, `subworkflow`, and `batch`, validates the graph, applies an admittance service, persists status `queued`, then publishes `execution:enqueued`.
- [OBSERVED] The engine’s own tables are `workflow_execution` and `workflow_step_execution`, with JSONB graph/trigger/output/error fields and per-execution/node uniqueness for step rows. Those table names are absent from the primary generated PostgreSQL/SQLite schema docs, consistent with a separate data plane.
- [OBSERVED] The standalone process supplies no v1 executor because the compatibility package depends on the engine. Source comments state that `v1-node` steps therefore fail as unimplemented unless an integrated host supplies the executor.

28.2 Explicit incomplete boundaries

- [STATED] Persisting the execution and publishing the enqueue message are not atomic; a crash between them leaves a queued record. The source says a reconciliation sweep is “not yet built.”
- [STATED] Shutdown stops workers but drops work behind the currently handled item because queues are in memory; draining is a TODO.
- [OBSERVED] The M1 acceptance integration suite has three explicit `it.todo` cases: If/Merge routing, fixed-count loop/back-edge execution, and mid-flight cancellation. Comments state that back-edge graphs are rejected and no cancellation mechanism exists yet.
- [OBSERVED] The v1 workflow converter documents deliberate semantic differences: sibling branches may run in parallel rather than v1 connection order, and disabled-node output handling differs from v1.

28.3 Compatibility layer and primary-runtime separation

- [OBSERVED] `@n8n/node-engine-compatibility` version `0.3.0` depends on `@n8n/engine`, `n8n-core`, and `n8n-workflow`. It converts a deliberately small supported v1 workflow topology and adapts executable v1 nodes to Engine 2.0 step interfaces.
- [OBSERVED] Its `V1StepExecutor` currently requires the VM expression implementation, resolves registered node types with `execute`, adapts v1 execution/workflow/context objects, settles close functions, and maps step input/output slots.
- [OBSERVED] The primary `n8n` CLI manifest has no dependency on `@n8n/engine` or `@n8n/node-engine-compatibility`. The current primary execution path in Section 25 remains `WorkflowRunner` + `WorkflowExecute`.

- [INFERRED] The separate package, config, database, Dockerfile, tests, and compatibility layer establish an active tracked architectural attempt, while explicit status and missing CLI dependency prevent classifying it as the current primary engine.

Status: **CURRENT — experimental/not ready**. It is neither removed nor proven deployed by the main product runtime.

29. Deployment forms and deployment tests

29.1 Tracked image and stack forms

Artifact	Current observed composition
<code>docker/images/n8n-base/Dockerfile</code>	Pinned DHI Node 24.18.1 Alpine base; fonts, OpenSSH, GraphicsMagick, tini, timezone, CA, libc compatibility and librdkafka; exposes 5678.
<code>docker/images/n8n/Dockerfile</code>	Builds native <code>isolated-vm</code> and <code>sqlite3</code> , copies compiled deployment, runs as <code>node</code> , and enters through <code>tini + docker-entrypoint.sh</code> on 5678.
<code>docker/images/runners/Dockerfile</code>	Multi-stage Node 24.18.1 + Python 3.13 sidecar; builds JS/Python runners, downloads launcher 1.4.7 with checksum verification, runs as UID 1000, exposes 5680, launches JavaScript and Python.
<code>docker/images/runners/Dockerfile.distroless</code>	Debian/glibc build and <code>gcr.io/distroless/cc-debian13</code> runtime with no shell/package manager/system utilities; UID 65532; same two runners.
<code>docker/images/engine/Dockerfile</code>	Separate Engine 2.0 image; compiled production package, non-root Node, port 3000, <code>dist/serve.js</code> .
Other tracked forms	Node-PC image, development Compose stacks, benchmark Compose stacks, build scripts/actions, and testing-container infrastructure.

[OBSERVED] The n8n entrypoint optionally trusts certificates mounted at `/opt/custom-certificates`, then executes the `n8n` CLI with supplied arguments or no arguments.

29.2 `get-n8n.sh` stack

[OBSERVED] The current one-script installer creates a version-pinned `.env`, a locally owned Compose file, and SearXNG settings. It resolves the latest stable GitHub release with fallback `2.32.0`; existing installations are never silently rewritten, and `--upgrade` changes only `N8N_VERSION` before pull/recreate.

[OBSERVED] The tracked Compose stack has six services:

1. one-shot `sandbox-certs` mTLS bootstrap;
2. `sandbox-api`;
3. privileged Docker-in-Docker `sandbox-runner-1`;
4. `n8n` with persistent `/home/node/.n8n`;
5. external JavaScript/Python `runners` connected to broker port 5679;
6. internal SearXNG for Instance AI web search.

[OBSERVED] Only host port 5678 is published. Source comments explicitly say not to expose sandbox, runner, or SearXNG ports. The installer generates separate sandbox API, runner, registration, runner-auth, runner-broker, and SearXNG secrets and writes `.env` mode 0600.

[OBSERVED] The installer sets external task-runner mode and Instance AI/sandbox variables. It sets `N8N_ENABLED_MODULES=instance-ai`; the pinned module registry already lists `instance-ai` among defaults and deduplicates enabled names.

[OBSERVED] Main n8n and runners use `${N8N_VERSION}` tags, while sandbox and SearXNG image references in the Compose file use `latest`. This is a directly observed versioning asymmetry, not an evaluation.

29.3 Helm boundary

- [OBSERVED] No production Helm chart was found in the pinned n8n repository path inventory.
- [OBSERVED] Helm deployment tests are nevertheless present. `packages/testing/containers/HELM-TESTING.md` and `helm-stack.ts` use privileged K3s in Testcontainers, preload an n8n image, and install a chart cloned by default from `https://github.com/n8n-io/n8n-hosting.git`.
- [OBSERVED] The Helm CI workflow accepts `standalone` or `queue`; queue mode includes PostgreSQL, Redis, and workers through the external chart. It then runs the Playwright building-block suite against the NodePort service.
- [OBSERVED] The Docker no-cache smoke workflow builds the full `pnpm/application/base/n8n/runners` chain and invokes a smoke command described as validation against the live n8n-cloud Helm chart.

[INFERRED] Chart **testing integration** is current in this repository, while chart **source ownership/location** is external to this tree. “No chart found here” does not mean “no Helm deployment exists.”

30. Current test topology

30.1 Path census

[OBSERVED] A whole-tree path census found 5,790 conventionally named test/spec files: 5,738 `.ts`, 37 `.mjs`, and 15 `.py`. This counts files, not test cases. Naming/path categories overlap.

Bounded category	Files observed
Under <code>__tests__</code>	2,315
Path/name includes <code>integration</code>	454
Path/name includes <code>e2e</code>	219
Path/name includes <code>benchmark</code>	22
Under <code>packages/testing/playwright</code>	262
<code>packages/cli</code>	1,325
<code>packages/frontend</code>	1,277
<code>packages/nodes-base</code>	1,106
<code>packages/testing</code>	311
<code>packages/core</code>	99
<code>packages/@n8n/engine</code>	16
<code>packages/@n8n/node-engine-compatibility</code>	6 conventionally named tests
<code>packages/@n8n/task-runner</code>	11
<code>packages/@n8n/task-runner-python</code>	13

30.2 Architecture-confirming test areas

- [OBSERVED] Public API tests cover decorated-controller registration/resolution, global middleware, scope parity, generated-spec drift, OpenAPI column names, workflow publication errors, and selected handlers/services.
- [OBSERVED] Primary execution tests cover `WorkflowExecute`, node contexts, partial execution graph operations, lifecycle hooks, persistence, recovery, redaction, permissions, external request behavior, and SSRF integration.
- [OBSERVED] Scaling tests cover job processing, multi-main setup, Redis leases, worker server/status, pub/sub publisher/subscriber/registry, execution stop, webhook response relay, and scaling service.

- [OBSERVED] JavaScript runner tests cover task behavior, data reconstruction, built-in parsing, module resolution, secure module views, prototype hardening, and error mapping. Python runner tests cover execution, RPC logging, health, env parsing, import validation, sandbox hardening, transitive-import trust, analyzer/executor/config, cancellation, and Sentry behavior.
- [OBSERVED] Engine 2.0 tests cover graph validation/queries, in-memory queue, admittance, execution start/settlement/step workers, Postgres entity stores, HTTP server, and end-to-end engine startup. Compatibility tests include v1 adapters/conversion/execution and the M1 Postgres acceptance suite.

30.3 Deployment validation

- [OBSERVED] `docker/test-get-n8n.sh` has fast configuration checks and an optional full-stack/upgrade E2E mode. It checks generated files, Compose validation, permission mode, unique/distinct secrets, idempotency, stack-version notice, unavailable source failure, version override, and full-stack health/upgrade behavior.
- [OBSERVED] The nightly/on-change Docker smoke workflow disables application/Docker cache for the build chain specifically to expose native `isolated-vm/sqlite3` compilation failures that caching might hide.
- [OBSERVED] Helm E2E builds and publishes an ephemeral image, starts K3s and the external chart, runs building-block tests, records Kubernetes diagnostics on failure, and removes the image.

[UNKNOWN] Pass/fail state of the pinned commit. No suite was executed, and hosted CI run results are outside this sprint's source-only scope.

31. Current architecture status matrix, contradictions, and unresolved points

31.1 Status matrix

Form	Status at pinned commit	Evidence boundary
WorkflowRunner + WorkflowExecute, regular mode	CURRENT — primary runtime	Direct CLI composition and defaults.
Bull/Redis workers and dedicated webhook process	CURRENT — conditional	Activated by queue mode and command role.
Multi-main leader/follower orchestration	CURRENT — conditional	Redis lease and role handlers; configuration/license path applies.
Internal/external JS and Python task runners	CURRENT — primary/conditional form	Runner mode defaults internal; external sidecar and installer wiring present.
CLI backend modules	CURRENT — conditional	Thirty defaults plus two opt-ins; license/instance-type filters.
Extension SDK Insights example	CURRENT — tracked, activation unknown	Manifest and code present; no primary bootstrap dependency established.
Engine 2.0 + compatibility package	CURRENT — experimental/not ready	Explicit config statement, separate package/DB/image/tests, incomplete boundaries, no CLI dependency.
vm2 JavaScriptSandbox	RETAINED — reachability unknown	File present; current Code node uses task runners; broad reference scan inconclusive.
Historical/branch experiments from Sprint 2	HISTORICAL / BRANCH-ONLY / REMOVED as previously labeled	Current same-term files do not alter exact ancestry results.

31.2 Directly evidenced tensions or contradictions

- Code-node network wording.** [STATED] The Code node's builder hint says the sandbox has no network access and `fetch`, `axios`, `XHR`, and `HTTP-module` require fail. [OBSERVED] The JS runner simultaneously exposes `helpers.httpRequest` and deprecated `helpers.request` over RPC. [UNKNOWN] Whether the wording intends "no direct network primitives" rather than "no network-capable helper"; the inspected source does not reconcile it.
- Runner default comments versus current version.** [STATED] TypeScript runner config comments say n8n v2 "will reduce" max concurrency from 10 to 5 and timeout from 300 to 60 seconds. [OBSERVED] The pinned root is version 2.35.0 and those TypeScript defaults remain 10/300. The Python runner independently defaults to 5/60. The comments are therefore stale or scoped differently; exact intended interpretation is [UNKNOWN].
- Schema breadth versus default module activation.** [OBSERVED] Generated primary DB schema docs include many `agent_*` tables. [OBSERVED] `agents` is not in `defaultModules` and must be enabled, whereas its entity method supplies 22 entities. [UNKNOWN] Which module profile the schema generator used. Schema-page existence is not proof of default runtime activation.

4. **Engine 2.0 maturity signals.** [OBSERVED] A package, image, API, migrations, and integration suite all exist on `master`. [STATED] Its config says not ready, its standalone host lacks v1 executors, and acceptance TODOs cover loop/routing/cancellation. Both sets of facts are retained; “current” is not equated with “primary” or “ready.”
5. **Module location vocabulary.** [STATED] `packages/modules/README.md` reserves a future co-located frontend/backend module shape. [OBSERVED] Current backend discovery remains under `packages/cli/src/modules`, and `packages/modules` contains no implemented module package at this commit.
6. **Installer module duplication.** [OBSERVED] The installer explicitly enables `instance-ai`, while the registry already includes it in `defaultModules`; the registry’s `Set` removes duplication. Reason is [UNKNOWN].
7. **Pinned core versus floating auxiliaries.** [OBSERVED] `get-n8n.sh` pins n8n/runners through `N8N_VERSION` but composes sandbox and SearXNG images at `latest`. This records the current artifact form only.

31.3 Remaining unknowns deferred or bounded

- [UNKNOWN] Runtime endpoint count after all license, feature, module, instance-role, and configuration filters.
- [UNKNOWN] Which current controllers and modules are exposed in each hosted deployment tier; source capability is not live deployment evidence.
- [UNKNOWN] Whether the retained vm2 sandbox is reachable outside the current Code node.
- [UNKNOWN] Runtime activation/loading of the extension SDK Insights example.
- [UNKNOWN] Hosted CI result for the pinned commit and actual production image/chart combinations.
- [UNKNOWN] External `n8n-hosting` chart internals; only this repository’s integration boundary was inspected in Sprint 3.
- [UNKNOWN] Deployment prevalence of regular, queue, multi-main, external-runner, and Engine 2.0 forms.
- [UNKNOWN] Patch lineage from Sprint 2 prototypes to similarly named current packages except where exact ancestry/commit evidence was already established.
- [UNKNOWN] Documentation/product claims and live public-surface drift; these are the planned subjects of Sprints 4 and 5.

32. Sprint 3 completion state and resource audit

32.1 Coverage accounting

[OBSERVED] Current tree analyzed: all 26,727 path names; all 87 manifests; 423 unique internal declared edges; all 56 config-package config files; 76 Public API path-spec files; 110 bounded internal controller files; 59 core DB entities; 61 repositories; 329 migration files; 62 module entity declarations across 14 modules; 32 module declarations; all conventional test path names; and all tracked deployment artifact path names. Selected implementation blobs were inspected for every architecture described above.

[OBSERVED] Sprint 3 adds current architecture without changing Sprint 2's branch sampling totals: all 1,645 heads graph-scanned, 60 refs tree-sampled, 504 heads more than ten commits ahead, and 24 consequential commits directly inspected.

32.2 Method incidents and cleanup

- [OBSERVED] Two attempted broad reference searches for `JavaScriptSandbox` across the nodes tree timed out in the promisor clone. One left a `git grep/promisor-fetch` process after the command wrapper timed out; exact PIDs were terminated. Workspace size peaked at approximately 5.2 MB during this incident, remaining below the 80 MB research target.
- [OBSERVED] No full source body, HTTP body, bundle, source map, raw log, binary asset, or standalone diagram was retained.
- [OBSERVED] No PDF was created; the final PDF remains reserved for Sprint 6.
- [OBSERVED] End-of-sprint cleanup deleted the filtered clone and the Sprint 2 copy. Final workspace audit: one file, `/home/user/working_sprint_3.md`, and `du -sh /home/user` = 144 KB.

Sprint 3 of 6 completed. 3 research sprints remain.

Sprint 4 will inspect target-authored documentation independently of source: the GitHub wiki state, repository documentation systems, docs.n8n.io information architecture and selected current/historical pages, release/changelog evidence, terminology, architecture/deployment claims, and contradictions with the pinned current source.

Sprint 4 — documentation, wiki, releases, terminology, and source/ document cross-check (replacement investigation)

33. Sprint 4 method, pins, and bounded coverage

33.1 Pinned evidence states

Surface	Pinned or observed state	Status use in this sprint
Main implementation repository	n8n-io/n8n@2afc867f20d689891b516c0458d627bc52c979fd , authored 2026-08-15T00:00:29Z	Current discoverable source pin; its packages/cli/package.json says 2.35.0 , which is a package field, not a release-channel classification.
Latest stable semantic release inspected	n8n@2.34.6 , commit d743b0ad2e0488663f858503d8dc7de08e667c1f , published 2026-08-14T08:12:03Z	Current stable release artifact observed through GitHub Releases.
Latest 2.x prerelease inspected	n8n@2.35.3 , commit 0dccd14a4a4e590b8641a12a95c9c76f8d525dc , published 2026-08-14T10:21:09Z	Current prerelease artifact observed through GitHub Releases.
Documentation repository	n8n-io/n8n-docs@ec418cd6a498f2a7f78755e386bd98a73cf9da24 , 2026-08-14T12:06:17Z	Current main documentation source.
Documentation wiki	n8n-io/n8n-docs.wiki@d8b53a7cfe7ce2e7b4c3695dd4bc14b02f886be8 , 2025-08-28T09:59:44Z	Current advertised wiki tip; complete reachable wiki history was inspected.
Rendered documentation	https://docs.n8n.io/ , final canonical batch 2026-08-15T16:38:29-36Z	Current public response surface, not assumed to be byte-identical to repository source.

[OBSERVED] GitHub repository metadata reports n8n default branch master , has_wiki=false , and n8n-docs default branch main , has_wiki=true . Both return license SPDX NOASSERTION . These hosted fields are recorded independently of repository license text.

33.2 Acquisition and access boundary

[OBSERVED] Work remained passive and read-only. Live requests used GET only. There was no login, API-key use, workflow execution, mutation, credential collection, private-page access, exploitation, or load test. Git research used filtered bare repositories, explicit refs, git show , ls-tree , cat-file , and bounded tree diffs; no checkout or worktree was created.

[OBSERVED] HTTP response bodies were parsed in memory. No full response, JavaScript bundle, source map, binary asset, workflow export, or raw HTTP log was retained. The compact ledger in Section 43 limits every retained body excerpt to 160 characters; all are below the requested 500-character ceiling.

33.3 Scale gates, census, and sampling

Three surfaces exceeded the mandatory exhaustive-inspection threshold:

- [OBSERVED] Documentation `main` has 10,913 reachable commits. The commits API page-1 `Link` identified page 10,913 as last at one commit per page; page 10,913 returned root `7e853e2462599907f080b7921329416b6ffe0b7f` and page 10,914 returned `[]`.
- [OBSERVED] Documentation exposed 249 branch names and 247 unique tip objects. All 249 names/tips were enumerated, but branch bodies were sampled.
- [OBSERVED] The current sitemap exposes 1,334 unique public URLs. Endpoint bodies were therefore stratified rather than exhaustively fetched.

[OBSERVED] Documentation branch selection was deterministic and mutually exclusive: four lexically even-spaced names from each of seven strata, plus `main`, for 29 tips. The strata were release/changelog, navigation, generated-version update, platform/style, ticket/general, product/feature, and lexical residual. All 29 tip trees and metadata were inspected; ten representative branches received GitHub compare calls, and eight of those received a second, detailed compare parse.

[OBSERVED] Other coverage:

- all 1,863 current documentation-repository path names; all 1,473 Markdown blobs in the repository-wide Markdown syntax scan; all 1,437 Markdown files under `docs/` in the collection reconciliation;
- all 48 reachable wiki commits and all four wiki paths;
- 13 consequential documentation commits plus required parent trees, without checking them out;
- all 26,727 path names at the implementation pin; the stable/prerelease/pin OpenAPI roots and their referenced operation fragments; all 257 generated database Markdown artifacts; all 16 v3 rule implementation files;
- current sitemap index and every one of its nine partitions; rendered routes spanning every public collection; selected old redirects, `.md` variants, and a synthetic nonexistent route.

33.4 Status vocabulary

- **CURRENT** means present at a pinned current source/ref or directly returned by the live surface.
- **HISTORICAL** means directly observed in a dated reachable commit or archived page.
- **REMOVED** means an exact later tree/diff deletes the form.
- **REVERTED** means a later commit restores the earlier tree; it does not mean the attempt never existed.
- **FUTURE — STATED** means target-authored material explicitly schedules or proposes a later state.
- **FUTURE — CANDIDATE SOURCE** means code exists for a future transition but is not enabled/composed as current behavior.
- **UNKNOWN** is used where publication, deployment, optional-module, or maintenance status cannot be established.

Names, branch labels, directories, and test names are not treated as proof that a behavior is active.

34. Current documentation hosting and information architecture

34.1 Current GitBook form

- [STATED] Current `n8n-docs/README.md` names GitBook as the documentation framework and Vale, Lychee, and GitHub Actions as quality tooling.
- [STATED] The same README says local build/preview is not supported; contributors must push and use GitBook's pull-request preview. This is a current operational statement, not a recommendation from this report.
- [OBSERVED] The live root returns HTML titled `n8n Docs`, canonical `https://docs.n8n.io`, with GitBook deployment identifiers and preconnects to `static-2v.gitbook.com` and `api.gitbook.com/cache/`.
- [OBSERVED] Current repository tree: 1,863 paths, including 1,473 Markdown, 268 PNG, 19 JSON, nine GIF, four JPG, four JPEG, two PDF, two SVG, and 70 YAML/YML files. Under `docs/` are 1,746 paths and 1,437 Markdown files.
- [OBSERVED] Current repository root has no `mkdocs.yml`, `nav.yml`, `requirements.txt`, `runtime.txt`, or `netlify.toml`. Navigation is represented by collection-local `SUMMARY.md` files and GitBook constructs.

34.2 Exact repository-to-sitemap reconciliation

[OBSERVED] The nine public sitemap partitions reconcile exactly to current collection Markdown plus GitBook-generated OpenAPI pages:

Public partition	Repository Markdown in collection	Non-page SUMMARY.md	Repository-backed public pages	Generated OpenAPI pages	Sitemap URLs
Root / get-started	6	1	5	0	5
deploy	120	1	119	0	119
build	135	1	134	0	134
integrations	946	1	945	0	945
connect	48	1	47	24	71
administer	43	1	42	0	42
contribute	6	1	5	0	5
privacy-and-security	5	1	4	0	4
changelog	10	1	9	0	9
Total public	1,319	9	1,310	24	1,334

[OBSERVED] `docs/reusable-content/` contains another 118 Markdown files, including its `SUMMARY.md`; these are reusable include sources and have no public sitemap partition. The arithmetic is therefore not evidence that content is missing.

[OBSERVED] docs/connect/SUMMARY.md contains a builtin:openapi block whose dependency is an OpenAPI spec named n8n , with models, a download link, and by-tag grouping. The 24 generated routes are 23 resource groups—Audit through Workflow, including Role—plus Models. They are live but have no corresponding 24 Markdown source files in the docs repository.

34.3 Sitemap, llms.txt , and machine-readable page forms

- [OBSERVED] The sitemap index names nine child sitemaps. Their counts are 5, 119, 134, 945, 71, 42, 5, 4, and 9; total and unique total are both 1,334.
- [OBSERVED] llms.txt is Markdown, 279,594 bytes in the principal parse, and contains 1,334 documentation links, all unique.
- [OBSERVED] Raw URL sets do not intersect because llms.txt uses .md forms. After stripping .md and mapping welcome.md to the root, 1,326 routes match. The eight sitemap-only entries are the collection roots; the eight normalized llms.txt counterparts use readme for six collections and contribute-to-n8n / privacy for two. Totals remain equal; the representations are not identical.
- [OBSERVED] Appending .md to an existing page returns text/markdown with a preamble pointing to llms.txt . A nonexistent HTML route returns HTTP 404, while a nonexistent .md route returns HTTP 200 containing a generated # Page Not Found document. HTTP status alone is therefore insufficient to establish existence for the Markdown form.
- [OBSERVED] Old /hosting/ and /workflows/ forms each produced a 302 slash-normalization hop then a 307 redirect to /deploy/host-n8n and /build/understand-workflows , respectively; final pages were 200 with the new canonical URL.

34.4 Current repository content mechanics

[OBSERVED] A repository-wide scan of 1,473 Markdown blobs found 1,435 frontmatter openings, 1,542 GitBook hint blocks, 986 include blocks, 52 tab blocks, 15 embed blocks, 25 HTML details elements, 119 ordinary Markdown image links, and three Mermaid fences in two docs pages. These are source-syntax counts, not rendered-component counts.

[OBSERVED] Migration-management frontmatter is pervasive: 1,250 files have originalFilePath , 1,250 have originalUrl , 1,256 have url , 1,257 have nodeTitle , and 1,238 have contentType . The current style guide explicitly says these fields support migration management on pre-existing pages and must not be added to new pages. They are not treated as independent live routing truth.

34.5 Publication and quality automation

[OBSERVED] Current .github/workflows/ has nine workflows:

Workflow	Trigger classes	Job
Auto-comment and handle PRs/issues on label	issues , pull_request_target	handle-labels
GitBook preview links	status , workflow_dispatch	preview-links
Internal link check	pull_request	internal-links
Report readability	pull_request	report-readability

Workflow	Trigger classes	Job
Run Vale	pull_request	vale
Lychee PR external linkcheck	pull_request	link-check
Scheduled/manual Lychee	schedule , workflow_dispatch	link-check
Script tests	pull_request , workflow_dispatch	gitbook-preview-links
Support Requests	issues	action

[OBSERVED] At the docs pin, the checks API returned 72 preview-links check runs: 21 success and 51 skipped. Combined commit status was success , with 16 success statuses. The latest 30 main Actions rows were all named GitBook preview links : six success, nine skipped, and 15 cancelled. These are hosted observations; no local test was run.

35. Historical documentation forms, migration, repair, and removals

35.1 Three publishing/build epochs

Epoch	Direct evidence	Exact transition evidence
VuePress	Root commit <code>7e853e2...</code> , 2020-05-23, has 14 paths and eight Markdown files. <code>package.json</code> uses VuePress 1.5 with <code>vuepress dev docs / vuepress build docs</code> .	Initial reachable form.
MkDocs Material	Commit <code>9821b9c061f...</code> , 2022-04-13, subject <code>new ia and new site</code> , introduces <code>mkdocs.yml</code> , <code>requirements.txt</code> , <code>runtime.txt</code> ; removes VuePress paths.	Raw parent-to-commit tree diff: 3,618 entries—1,819 added, 1,780 deleted, 19 modified. Parent had 1,807 paths/651 Markdown; result had 1,846 paths/675 Markdown.
GitBook	Commit <code>4f6df3123af...</code> , 2026-06-24, subject <code>GitBook Docs Release (#4876)</code> , reorganizes content into collection spaces and GitBook syntax.	Pre-tree 1,918 paths/1,422 Markdown; post-tree 1,947 paths/1,553 Markdown. Raw path diff: 2,422 entries (784 A, 755 D, 883 M). Rename-aware diff: 2,040 entries (402 A, 373 D, 382 R, 883 M).

[STATED — HISTORICAL] The pre-GitBook README supported local `mkdocs serve --strict`, but said organization members could use a full Material Insiders theme unavailable to external contributors. Current GitBook README instead says local preview is unsupported. These statements belong to different epochs.

35.2 Consequential migration sequence

UTC/date order	Commit	Exact observed change
2026-06-19	<code>b9246290d26f...</code>	Course/video-course removal: 140 entries, 134 deleted and six modified. Deleted 27 Markdown, 100 PNG, and seven GIF files; <code>nav.yml</code> was modified.
2026-06-24	<code>4f6df3123af...</code>	GitBook migration described above.
2026-06-26	<code>5db109aa3f33...</code>	Post-migration cleanup: 191 entries, 190 deleted and <code>.gitignore</code> modified. Deleted 115 <code>_snippets</code> , 58 <code>archive</code> paths, 10 <code>_overrides</code> , two <code>_doctools</code> , <code>mkdocs.yml</code> , <code>nav.yml</code> , <code>main.py</code> , <code>_yaml</code> , and a <code>feature-test</code> path.
2026-06-26	<code>06bec291b500...</code>	<code>fix migration-broken internal links</code> : 14 files, 19 lines added and 19 deleted. Changes replaced moved relative links with GitBook space links or new <code>README.md</code> paths.
2026-06-29	<code>44bb160d5151...</code>	<code>restore migration-dropped assets</code> : 22 entries—18 added, four modified. Added 16 workflow JSON artifacts and two Okta PDF artifacts.

UTC/date order	Commit	Exact observed change
2026-07-15	56233905ee82...	Renamed feature-feed.md to release-notes.md and feeds/feature-feed.xml to feeds/release-notes.xml ; five entries total, including three modifications and two high-similarity renames.
2026-07-23	4998c2287cc9...	Deleted residual netlify.toml , requirements.txt , and runtime.txt .
2026-07-31	a90822f7dde7...	Added v30-breaking-changes.md and modified changelog navigation.

35.3 Dropped and restored artifacts

[OBSERVED] The GitBook migration tree deletes 28 `.json` workflow artifacts and two PDFs without same-path additions. The restoration commit later adds 12 workflows at their old `_workflows` paths, four AI/code workflow files under `docs/build/.gitbook/assets` , and the two PDFs under `docs/administer/.gitbook/assets` .

[OBSERVED] Twelve deleted JSON files were not restored by that commit: three advanced-AI tutorial exports, eight level-one course exports, and the very-quick-quickstart export. Eight of those course exports had survived the earlier prose/image course-removal commit as residue. Their later absence is exact tree evidence; no reason beyond the target-authored commit subjects is inferred.

[OBSERVED] Current docs source contains 16 workflow-shaped JSON artifacts. In memory, they parse to 168 nodes, 88 top-level connection-map entries, and 45 distinct node-type strings. Nine nodes have a `credentials` field; values were not retained. Eight exports say `active:false` , one says `active:true` , and seven omit `active` . A stored flag is not evidence that documentation artifacts are running anywhere.

35.4 A fully reverted redirect-cleanup attempt

[OBSERVED] On 2026-05-06, `3d3c38501691...` added a submodule pointer and changed `docs/_redirects` by 13 additions and 776 deletions. The next commit, `21299f1d0af...` , is titled `Revert "Cleaning up redirects"` . The tree before the attempt and the revert tree are exactly the same SHA, `8424d7245827c534ed9f9cd3ee88574996ae5697` ; the attempt tree is `c4e96f6d170349413e47ee6eb538d8fcad734de8` .

[OBSERVED] The revert message names original SHA `8cecc29b...` , which the remote no longer served by direct SHA (`upload-pack: not our ref`). The reachable attempt commit has the changed tree and exact parent; the absent named object is not reconstructed beyond that.

36. Documentation branch sample and commit-scale reconstruction

36.1 Complete ref census and deterministic sample

[OBSERVED] The API returned 249 unique names, 247 unique tips, and two protected branches (`api-spec-updates` , `main`). Duplicate-tip groups were:

- `update-breaking-changes-21044125` and `update-breaking-changes-21049145` → `3b9d79c226ce9b386ecc235c7f9498b0e89458ef` ;
- `update-latest-next-71221422` and `update-latest-next-71221434` → `e2129c300693bbb60ec9b5512d749d627e2d5435` .

[OBSERVED] The 29 selected names were:

- baseline: `main` ;
- release/changelog: `changelog-2x-gitbook` , `release-note-2.20.10` , `release-notes/REL-34-2026-05-27` , `release-notes/REL-76-2026-07-14` ;
- navigation: `june-nav-changes` , `nav-work-in-progress` , `new_nav_exploration` , `test-new-nav` ;
- generated version: `update-breaking-changes-21044125` , `update-latest-next-53130768` , `update-latest-next-65819572` , `update-latest-scopes` ;
- platform/style: `DOC-22200-preview` , `fix-vale-endtab` , `int-test-preview` , `ligo-690-import-export-docs-preview` ;
- ticket/general: `DOC-1093` , `DOC-795` , `doc-2048-update-version-snippet` , `ligo-933-write-customer-facing-docs-for-workflow-reviews` ;
- product/feature: `NODE-4253-crypto-credentials` , `chore/openapi-sync-16167794424-1` , `docs/remove-otel-self-hosted-only-hint` , `update-webhooks` ;
- lexical residual: `DesireeLindsey-patch-1` , `docs-discord-moderation-actions` , `nik8n-patch-1` , `update-variables-plans` .

36.2 All sampled tip snapshots

The `snapshot Δ` column is a direct current-`main` versus tip-tree path comparison, not a claim about branch-only commits.

Branch	Tip date	Tip subject (bounded)	Snapshot Δ paths
<code>main</code>	2026-08-14	bring feature-availability refs in line with style guide	0
<code>changelog-2x-gitbook</code>	2026-06-23	embed changelog release videos / README landing	2,835
<code>release-note-2.20.10</code>	2026-05-18	merge <code>main</code>	2,831
<code>release-notes/REL-34-2026-05-27</code>	2026-07-01	OpenTelemetry custom telemetry tags	511
<code>release-notes/REL-76-2026-07-14</code>	2026-08-11	update changelog README	161
<code>june-nav-changes</code>	2025-09-16	use index pages instead of overviews	2,710
<code>nav-work-in-progress</code>	2025-06-11	rename pages and sections	2,686
<code>new_nav_exploration</code>	2025-02-11	More messing around	2,673
<code>test-new-nav</code>	2026-05-28	new navigation script	2,838
<code>update-breaking-changes-21044125</code>	2025-04-30	transforming-data update	2,700
<code>update-latest-next-53130768</code>	2026-04-15	update latest/next snippet	2,813

Branch	Tip date	Tip subject (bounded)	Snapshot Δ paths
update-latest-next-65819572	2026-05-19	update latest/next snippet	2,832
update-latest-scopes	2026-02-10	Microsoft Agent 365 Trigger docs	2,795
D0C-22200-preview	2026-08-14	normalize preview-status information	27
fix-vale-enttab	2026-07-07	omit enttab from spellcheck	430
int-test-preview	2026-07-23	test(test): test	616
ligo-690-import-export-docs-preview	2026-06-16	rename workflowPublishingPolicy values	2,839
D0C-1093	2024-09-24	merge main	2,645
D0C-795	2024-05-28	fix title case	2,702
doc-2048-update-version-snippet	2026-06-29	update latest/next snippet	535
ligo-933-write-customer-facing-docs-for-workflow-reviews	2026-08-05	add workflow reviews documentation	167
N0DE-4253-crypto-credentials	2026-01-27	Crypto credential v2 security docs	2,789
chore/openapi-sync-16167794424-1	2025-07-09	update Public API schema	2,727
docs/remove-otel-self-hosted-only-hint	2026-07-01	merge main	519
update-webhooks	2026-05-29	update webhook for production flow	2,831
DesireeLindsey-patch-1	2025-09-26	update release notes	2,744
docs-discord-moderation-actions	2026-07-02	document Discord member moderation	508
nik8n-patch-1	2024-11-27	clarify AI-usage privacy docs	2,650
update-variables-plans	2023-04-19	merge unrelated execution-timeout PR	3,450

[OBSERVED] The last row is direct evidence that a branch name is not an implementation fact: `update-variables-plans` points to an old merge whose subject is about execution timeout.

36.3 Ten comparisons to current `main`

Branch	Compare status	Ahead	Behind	Net files returned
changelog-2x-gitbook	diverged	2	299	2
new_nav_exploration	diverged	3	1,660	5
update-breaking-changes-21044125	behind	0	1,469	0
D0C-22200-preview	ahead	1	0	27
D0C-1093	diverged	17	1,935	0
ligo-690-import-export-docs-preview	diverged	9	363	8
chore/openapi-sync-16167794424-1	diverged	1	1,334	1

Branch	Compare status	Ahead	Behind	Net files returned
docs/remove-otel-self-hosted-only-hint	diverged	3	227	0
docs-discord-moderation-actions	diverged	1	210	2
update-variables-plans	behind	0	5,518	0

[OBSERVED] Detailed compare evidence separates attempts from current forms:

- changelog-2x-gitbook added a 364-line docs/release-notes/README.md and one nav.yml entry in June 2026. It is a branch precursor, not the current path layout.
- new_nav_exploration moved navigation from mkdocs.yml to a new 1,289-line nav.yml and renamed three pages in February 2025. The current GitBook form later removed both MkDocs navigation files.
- DOC-22200-preview is the only sampled branch strictly ahead of main; it adds docs/administer/.gitbook/tags.yaml and changes 26 files to distinguish preview status from general feature availability.
- ligo-690-import-export-docs-preview adds seven docs/import-export/ pages and a navigation entry. Current main instead documents the later n8n-package form under docs/build/manage-workflows/n8n-packages/.
- chore/openapi-sync-16167794424-1 modifies the historical docs-repository docs/api/v1/openapi.yml. Current docs main has no such path; GitBook uses its named OpenAPI dependency, while the implementation repository contains current OpenAPI source.
- DOC-1093 has 17 ahead commits but zero net compare files; docs/remove-otel-self-hosted-only-hint has three ahead commits but zero net files. Commit existence is not treated as surviving content difference.
- docs-discord-moderation-actions has two net modified files adding Ban/Get Many/Kick and troubleshooting content, while being 210 commits behind.

37. Independent hosting-platform wiki reconstruction

37.1 Complete wiki history

[OBSERVED] The wiki has 48 reachable commits from 61a75c3960df... (2022-04-20, Initial Home page) through d8b53a7cfe7c... (2025-08-28, Updated Styles). Exactly four paths have ever existed:

Path	First event	Change-event count	Current bytes / state
Home.md	added 2022-04-20	1	30 bytes; welcome sentence
Styles.md	added 2022-04-20	37	7,903 bytes
Screenshots.md	added 2022-08-15	8	3,591 bytes
Templates.md	added 2022-08-17	1	296 bytes

[OBSERVED] No fifth path, rename, or deletion appears in complete reachable wiki history.

37.2 Current artifact with historical platform guidance

- [OBSERVED] Styles.md still describes Material for MkDocs admonitions, SuperFences, preserved tabs, and MkDocs attribute syntax.
- [OBSERVED] Screenshots.md gives screenshot-maintenance, accessibility, framing, spacing, and annotation-color guidance.
- [OBSERVED] Templates.md points to repository document templates.
- [OBSERVED] Current repository and live pages use GitBook, and MkDocs build/config files are removed. The wiki is therefore a current hosted artifact containing platform-specific guidance from a historical build system.
- [UNKNOWN] Whether the wiki remains an intended contributor authority. has_wiki=true and a cloneable wiki establish availability, not maintenance intent.

37.3 Main-repository wiki boundary

[OBSERVED] n8n-io/n8n metadata says has_wiki=false ; n8n-io/n8n-docs says has_wiki=true . A generic 200 HTML response from a hosting-platform wiki route is not used to override the API/repository distinction.

38. Current public documentation claims and architectural forms

38.1 Rendered-route sample

[OBSERVED] The final canonical HTTP batch sampled root and every public collection. Canonical rendered pages for choose/how-to-use, Docker Compose, save/publish, Webhook, API authentication, RBAC, terminology, license, release notes, v3 changes, and learning paths all returned 200 with expected titles/canonicals. The synthetic nonexistent HTML route returned 404. Old hosting/workflow routes redirected as recorded in Section 34.3.

[OBSERVED] The large public surface was not exhaustively rendered. The sitemap/source arithmetic is exhaustive for route names; body assertions below are limited to requested pages and pinned source.

38.2 Architecture, deployment, and execution claims

- [STATED] How n8n works says architecture knowledge is useful for embedding n8n or customizing default databases and labels the section a work in progress.
- [STATED] Queue mode uses a main instance, Redis message broker, workers, and a database. Production executions are processed by workers; main/webhook processes receive webhook HTTP requests and pass actual execution to workers. The page says distributed queue operation over SQLite is unsupported.
- [STATED] Task runners comprise requester, broker, and runner over a WebSocket. The n8n main/worker instance is broker and Code node is requester. Internal and external modes are distinct; external mode commonly uses an independent n8nio/runners sidecar, one per queue worker. The live page says to use external mode for production/sensitive instances.
- [STATED] Save/publish documentation says edits autosave to draft versions, publishing fixes the production version, production executions use that published version, and unpublish removes it from production.
- [OBSERVED] These lifecycle terms align with publication entities reconstructed in Sprint 3, while compatibility active fields/routes remain in current source. Both are retained rather than silently equated.

38.3 Code, API, CLI, and machine interfaces

- [STATED] The Code page supports JavaScript and Python; self-hosted JavaScript modules require installation/allowlisting, Cloud module imports are restricted, Pyodide is legacy and unsupported in n8n 2, and native Python uses task runners and is stated stable as of n8n 2.
- [STATED] Public API pages describe API-key authentication in X-N8N-API-KEY, scoped Enterprise keys, full-access non-Enterprise keys, pagination, and a self-hosted API playground. They state the API is unavailable during a free trial.
- [STATED] n8n CLI is a lightweight remote client over the Public API and API key; the page distinguishes it from the server CLI that runs on the instance machine with database-adjacent administration commands. Source frontmatter says status: beta.
- [STATED] Current connect navigation distinguishes Public API, n8n CLI, instance MCP server, documentation MCP server, and node-creation documentation. Similar names are not collapsed into one interface.

38.4 Diagrams and downloadable artifacts

- [OBSERVED] Current docs tree has 268 PNG, two SVG, two PDF, and 19 JSON files. Of the JSON files, 16 are workflow-shaped artifacts; the other three are `context7.json` and two Vale style metadata files.
 - [OBSERVED] Docs source has three Mermaid fences in two pages. It also references binary diagrams for queue mode, task runners, the task-runner sidecar, and the hosted database ERD; binary bodies were not retained.
 - [OBSERVED] Main source contains 298 Mermaid fences across 271 Markdown files. Of these, 257 occur one-per-file in generated PostgreSQL/SQLite schema documentation. Remaining diagrams occur in source-local technical specifications and architecture documents, including expression runtime and Instance AI.
 - [OBSERVED] Mermaid/source artifacts establish documented structure, not deployment prevalence or successful execution.
-

39. Release, changelog, and future-transition reconstruction

39.1 Current published release signals

[OBSERVED] The first 100 GitHub Releases rows at 2026-08-15T16:14:36Z contained:

Release line	Highest semantic release observed	Published UTC	API classification
Stable 2.x	n8n@2.34.6	2026-08-14T08:12:03Z	non-prerelease
2.x prerelease	n8n@2.35.3	2026-08-14T10:21:09Z	prerelease
Maintained 1.x	n8n@1.123.71	2026-08-13T08:06:46Z	non-prerelease

[OBSERVED] Filtered Git fetches of the two semantic tags resolve to release commits `d743b0ad...` and `0dccd14a...`. The later source pin `2afc867f...` has package version `2.35.0`; chronology and package text are recorded separately from release-channel labels.

39.2 Current and archived documentation ledgers

Documentation artifact	Version-heading count	Observed range	Page status
Current feature-oriented <code>release-notes.md</code>	39	n8n 2.0 through n8n 2.34	Current curated release feed; not every patch.
<code>release-notes-2.x.md</code>	169	2.0.0-rc.0 through 2.29.1	Explicitly says no longer updated.
<code>release-notes-1.x.md</code>	440	1.0.1 through 1.123.62	Explicitly says no longer updated.
<code>release-notes-0.x.md</code>	299	0.70.0 through 0.237.0	Explicitly says no longer updated.

[OBSERVED] The archived ledgers lag the release API (`1.123.62` versus `1.123.71`, and `2.29.1` versus `2.34.6`) while explicitly directing readers to GitHub Releases. This is scoped archival behavior, not an unstated missing-release inference.

39.3 Historical change epochs recorded by target documentation

- [STATED — HISTORICAL] The retained 0.x notes include the transition to webhook paths of the form `https://{hostname}/webhook/{path}`.
- [STATED — HISTORICAL] The v1 migration page records Python/Pyodide support, execution-order changes, database/driver deprecations, container-user and endpoint changes, owner setup, WebSocket defaults, retention changes, and Node 16 removal.
- [STATED — HISTORICAL/CURRENT] The v2 page records the publish/unpublish lifecycle, task runners enabled by default, Pyodide removal, database/driver removals, tunnel/config/CLI changes, risky-node defaults, OAuth callback changes, and release-channel changes.

- [OBSERVED] Compatibility source and current configuration remain the authority for accepted current values; a migration page is not treated as a complete current reference.

39.4 n8n 3.0 remains future at the research date

- [STATED] The public page schedules n8n 3.0 for October 2026 and says it will be updated with full details and migration guides.
 - [STATED] It announces Docker-only supported self-hosting; removal of Function, Function Item, and Item Lists; changed Execute Workflow behavior; removal of `$getPairedItem`; stronger but not fully detailed security defaults; retirement of Chat hub; removal of editor URL import; and removal of non-functional nodes.
 - [OBSERVED] Current stable, prerelease, and pinned trees each contain 16 imported v3 rule files. Current pin's replacement-rule source names 28 node types with replacements and a separate rule names seven without replacements—35 total—plus rules for deployment, runner timeout, queue manual execution, binary mode, compression, unverified packages, Gmail Trigger, embedded chat JSON, Execute Sub-workflow modes, AI Transform, and URL import.
 - [OBSERVED] Current source still contains Function/Function Item implementations, 37 Item Lists paths, 127 chat-hub-named paths, `ImportWorkflowUrlModal.vue`, its test, and an import-from-URL DTO. These are CURRENT 2.x source artifacts, not evidence that future removals already occurred.
 - [OBSERVED] `MIGRATION_REPORT_TARGET_VERSION` is `null` in stable 2.34.6, prerelease 2.35.3, and the later pin. `BreakingChangesModule.init()` returns before rule import/registration when that constant is null.
 - [INFERRED] The v3 rule set is **FUTURE — CANDIDATE SOURCE** and more detailed than the intentionally incomplete public summary. It is not classified as current n8n 2.x behavior or guaranteed final v3 scope.
-

40. Target-native terminology and observed drift

40.1 Current target-authored terminology

[STATED] The live/repository terminology page calls itself the source of truth. Its principal pairs are:

Use	Avoid in ordinary current prose
workflow	flow, automation as the whole-system noun, scenario, pipeline, zap
node	step, block, action, module
execution (noun)	run, job, invocation
publish / unpublish; published workflow	activate/deactivate/enable/disable/turn on/go live/deploy
app node; core node; cluster/root/sub-node; community node	integration/service node; system node; parent/child; custom/third-party/plugin
canvas; sticky note; sub-workflow	board/diagram/workspace/graph; comment; subworkflow/subflow/child workflow
self-hosted; n8n Cloud; instance	on-premises; hosted version/SaaS; server/deployment/box
project; folder; environment variable; data table	workspace; directory in the app; env as noun; database table/datatable
n8n , GitHub, npm, JavaScript, OAuth2	N8N/N8n; Github; NPM; Javascript; oAuth

[OBSERVED] Vale substitution files enforce only a subset: self-hosted/on-premises, sub-workflow forms, GitHub, and JavaScript. Absence from an automated rule is not absence from the prose standard.

40.2 Terms are surface- and time-dependent

- [OBSERVED] Current prose standard says publish/unpublish; current OpenAPI still exposes `/workflows/{id}/activate`, `/deactivate`, `/publish`, and `/unpublish`. Live API scope documentation says `workflow:activate` means “Activate or deactivate” and also calls this publish/unpublish in the Public API. API identifiers are retained verbatim rather than rewritten.
- [OBSERVED] Historical pages and source fields use `active`; current UI prose uses published. These coexist as historical/compatibility and current product terms.
- [STATED] Current style guide says Preview, not beta, is the feature-maturity term.
- [OBSERVED] Current frontmatter has eight `status: preview` and six `status: beta` pages. Beta appears on AI Workflow Builder, Chat hub, two MCP/connect surfaces, the Connect landing page, and n8n CLI. This is current terminology drift; labels alone do not establish runtime maturity.
- [OBSERVED] The sampled strictly-ahead `DOC-22200-preview` branch changes 27 paths to regularize Preview presentation, but an ahead branch is not current `main`.

41. Main-repository documentation, OpenAPI, schemas, and architecture artifacts

41.1 Current source/documentation inventory

[OBSERVED] The implementation pin has 26,727 paths: 19,283 TypeScript, 1,244 Vue, 592 Markdown, 606 SVG, and 3,567 JSON. Path-name census finds 87 `package.json` files, 94 GitHub workflow YAML files, 9,619 conventional test/spec paths, 348 migration-related TypeScript paths across the repository, 64 TypeScript paths directly under `@n8n/db/src/entities`, 556 `*.node.ts` implementations under `nodes-base`, and 405 `*.credentials.ts` implementations.

[OBSERVED] These are path counts, not proof that every package, test, migration, node, or credential type is enabled in every build or deployment.

41.2 Public OpenAPI form and three near-current source states

The source build generates decorator fragments, bundles both roots with Redocly, merges at method granularity, rejects duplicate method/path operations, and writes one distribution `openapi.yml`.

Source state	Handwritten root	Decorator root	Merged literal path keys	Operations	Root tags	Consumed/declarable API-key scopes
Stable 2.34.6	68 keys / 109 ops	absent	68	109	22	86
Prerelease 2.35.3	67 keys / 106 ops	4 keys / 4 ops	70	110	23	88
Pin 2afc867f...	66 keys / 103 ops	8 keys / 10 ops	72	113	24	91

[OBSERVED] At the pin, parameter-name normalization collapses 72 literal keys to 71 URI shapes because `/workflows/{id}` and `/workflows/{workflowId}` represent the same shape. All 113 method/path pairs remain unique. Methods total 41 GET, 35 POST, 14 PUT, eight PATCH, and 15 DELETE. Every operation has `x-required-scope`; two say `none` and none is missing the field.

[OBSERVED] Change across the three states is monotonic at normalized method/path level in this bounded comparison:

- prerelease adds `POST /roles` with `role:manage,role:manageProject`;
- the later pin adds `GET /roles`, `GET /roles/{slug}`, and `POST /role-mapping-rules`, with `role:list`, `role:read`, and `roleMappingRule:create`.

[OBSERVED] Live GitBook exposes a Role group with only “Create a custom role,” no Role Mapping Rule group, and 23 resource-group pages plus Models. This aligns with the prerelease tag list for these newly changing groups; it does not prove that every live operation byte-for-byte matches that tag.

[OBSERVED] This Sprint corrects Sprint 3's earlier provisional literal path count. Current pin accounting is **72 merged literal keys / 71 normalized shapes / 113 operations**, not 70 paths. The operation total was already correct.

41.3 API scope-document drift

- [OBSERVED] Live/current `authentication.md` lists 67 unique scope rows. It explicitly marks `user:enforceMfa` as reserved and says no `/api/v1/` endpoint consumes it.
- [OBSERVED] Stable source declares and consumes 86 scopes; prerelease declares/consumes 88; the later pin declares/consumes 91.
- [OBSERVED] Relative to the prerelease—the observed state matching the live Role group—the page includes the one explicitly reserved scope and omits 22 consumed scopes. Omitted groups include event-bus destinations, LDAP/OIDC/OTel/SAML/security settings, project export, test runs, workflow deactivate/export/import, and the two Role-create scopes.
- [OBSERVED] At the later pin the omission set grows to 25 through role list/read and role-mapping creation.
- [INFERRED] The page's statement that its tables list available scoped-key scopes is materially incomplete relative to inspected release/source OpenAPI. The reserved row is not itself a contradiction because the page discloses its non-consumption.

41.4 Generated relational-schema documentation

- [OBSERVED] Source has 129 PostgreSQL schema Markdown files including README, representing 128 table pages; SQLite has 128 files including README, representing 127 table pages.
- [OBSERVED] After removing PostgreSQL's `public.` filename prefix, `workflow_statistics_delta` is the only PostgreSQL-only table page.
- [OBSERVED] Every one of the 257 generated schema files contains a Mermaid ER diagram fence. The READMEs describe generated source state; they do not prove a particular live database has every optional/module table.
- [OBSERVED] Current hosted database explanation has 24 table headings and one static ERD image. It says `role` is “Not currently used. For use in future work on custom roles.” Current source has role, role-scope and role-mapping tables, role APIs, and current Role documentation. The hosted sentence is directly stale relative to current source wiring.

41.5 Other source-local architecture/status artifacts

- [STATED] `packages/@n8n/engine/README.md` calls Engine 2.0 a scaffold whose API/interfaces are still being defined and says it is not yet wired into n8n.
- [STATED] Expression Runtime documentation describes evaluator/bridge layers across isolated-vm, Web Workers, and task runners, but explicitly labels task-runner integration TBD, Phase 2+, and an architecture decision pending between two options.
- [OBSERVED] Instance AI architecture Markdown contains two Mermaid diagrams and names frontend, backend, storage, planned-task, workflow-loop, workspace, memory, and MCP boundaries. Labels in that document are retained; directory names are not converted to completion claims.
- [STATED] `docker/images/runners/README.md` labels the `n8nio/runners` image (PREVIEW), while live task-runner docs say to use external task runners in production. The scopes differ—image packaging versus operational guidance—so this is a status tension, not automatically a contradiction.

42. Contradiction and drift register

42.1 Binary-data lifecycle

1. [STATED] The v2 change page says n8n 2 removes binary mode `default` and `N8N_AVAILABLE_BINARY_DATA_MODES`, with filesystem/database automatic defaults.
2. [STATED] Current environment reference still lists `N8N_AVAILABLE_BINARY_DATA_MODES`, says `N8N_DEFAULT_BINARY_DATA_MODE` defaults to `default`, and says `default` keeps data in memory.
3. [OBSERVED] Current `BinaryDataConfig` accepts `default` but, when unset, chooses filesystem in regular mode and database in queue mode. `availableModes` defaults in code but has no inspected `@Env('N8N_AVAILABLE_BINARY_DATA_MODES')` binding.
4. [OBSERVED] A disabled v3 rule targets in-memory/default mode for future removal.

[INFERRED] **CURRENT DOCUMENTATION CONTRADICTION; PINNED SOURCE RESOLVES CURRENT DEFAULT; FUTURE REMOVAL CANDIDATE.** The v2 page overstates completed removal, while the current environment page is stale about automatic default/binding.

42.2 Task-runner launcher path

- [STATED] One live paragraph says `/etc/task-runners.json`.
- [STATED] Its immediately following mount and custom-image examples use `/etc/n8n-task-runners.json`.
- [OBSERVED] Current runners Dockerfile copies to `/etc/n8n-task-runners.json`.

Status: **CURRENT DOCUMENTATION TYPO/CONTRADICTION, RESOLVED BY CURRENT IMAGE SOURCE.**

42.3 Queue mode and filesystem storage

- [STATED] Queue page says filesystem binary storage is unsupported in queue mode.
- [STATED] Its n8n 2.34 large-webhook-response section allows filesystem offload if every instance mounts the same disk, while saying this is not recommended.

Status: **CURRENT SCOPE AMBIGUITY.** [UNKNOWN] Whether relay offload is a narrow exception to the general support statement. No operational test was performed.

42.4 Hosted database page

Hosted 24-table page and unused-role sentence conflict with 127-128 generated table pages and wired role APIs. Status: **CURRENT PUBLIC PAGE STALE/INCOMPLETE.** Optional-table deployment breadth remains [UNKNOWN].

42.5 Documentation repository license signals

- [STATED] Current README calls `n8n-docs` fair-code under the Sustainable Use License and links the main-repository license.
- [OBSERVED] Current `n8n-docs/LICENSE.md` begins with Commons Clause v1.0, names Apache 2.0 as underlying license, and names software `n8n`.
- [OBSERVED] Hosted SPDX is `NOASSERTION`.

Status: **CURRENT REPOSITORY-INTERNAL TEXT CONTRADICTION**. No legal conclusion or interpretation is made.

42.6 GitBook versus wiki syntax

Current repository/live form is GitBook; current enabled wiki still teaches MkDocs Material/SuperFences syntax. Status: **CURRENT HOSTED WIKI WITH HISTORICAL PLATFORM GUIDANCE**. Intended authority is [UNKNOWN].

42.7 Frontmatter versus live routing

Migration fields appear in at least 1,250 files. Current style guide identifies them as migration-management fields. Live canonical routes can differ—for example collection `README.md` source paths map to collection roots. Status: **CURRENT MIGRATION RESIDUE, NOT BY ITSELF A ROUTING FAILURE**.

42.8 Hidden/outdated Docker page

`install-with-docker.md` has `hidden:true` and a warning that it is outdated, yet its direct route remains public 200. Status: **CURRENT ADDRESSABLE BUT HIDDEN/OUTDATED**; it is not treated as current recommended installation guidance.

42.9 v3 summary versus candidate source

Public page intentionally names only three removed legacy nodes and promises later details; disabled source rules cover 35 node types and more behavior/config changes. Status: **FUTURE PUBLIC SUMMARY VERSUS DISABLED, MORE DETAILED CANDIDATE SOURCE**, not current-product contradiction.

42.10 Preview/beta labels

Style guide says use Preview, while six current page frontmatters say beta. Status: **CURRENT TERMINOLOGY DRIFT**. The labels do not establish implementation maturity.

42.11 Public API temporal drift

Live generated Role content matches the prerelease's one new Role operation, while later source has three further role/mapping operations. Live scope prose lists 67 rows versus 88 consumed prerelease scopes. Status: **CURRENT PUBLIC GENERATED/PROSE SURFACES AT DIFFERENT OBSERVED SYNCHRONIZATION DEPTHS**. Exact GitBook spec-sync mechanism/version is [UNKNOWN].

42.12 Markdown not-found status

A nonexistent HTML route returns 404, but a nonexistent `.md` route returns 200 with `# Page Not Found`. Status: **CURRENT MACHINE-SURFACE STATUS/CONTENT MISMATCH**. Existence checks must inspect body content.

43. Compact Sprint 4 request and Git ledger

43.1 Request policy and common profiles

All requests below were GET. No credential-bearing header was sent. Documentation responses commonly returned

Cache-Control: public, max-age=0, must-revalidate and Server: cloudflare; exact final-batch headers and body excerpts are in Section 43.4. GitHub API responses were application/json; charset=utf-8, with ETag and pagination Link where applicable; quota headers are retained below. Excerpts/parsed fields are all under 500 characters.

43.2 GitHub API reconnaissance and metadata

GitHub paths in Sections 43.2–43.3 are relative to `https://api.github.com/repos/n8n-io/`, except `/rate_limit`, which is relative to `https://api.github.com`. Compare calls use the exact pattern `n8n-docs/compare/main...<URL-encoded-branch>?per_page=<1-or-100>&page=1`.

UTC timestamp(s)	Method/URL	Status and relevant headers	Body excerpt / parsed result
16:13:31.107307Z	GET /rate_limit	200; JSON	core limit=60, remaining=60
16:14:04.714536, .983233, 16:14:05.260838Z	GET n8n-docs/branches?page=100&page=1..3	200; remaining 59/58/57; paginated Link	100/100/49 rows; 249 names, 247 tips
16:14:35.190013Z	GET repos/n8n-io/n8n	200; remaining 46	default master, wiki false, created 2019-06-22, SPDX NOASSERTION
16:14:35.409882Z	GET n8n-docs	200; remaining 45	default main, wiki true, created 2020-05-23, SPDX NOASSERTION
16:14:35.593560Z	GET docs commits page 1, one row	200; remaining 44; Link last=10913	ec418cd6..., 2026-08-14
16:14:35.798416Z	GET docs commits page 10913	200; remaining 43	root 7e853e24..., Initial commit
16:14:45.633962Z	GET docs commits page 10914	200; remaining 38	[]
16:14:36.035887Z	GET n8n releases page 1, 100 rows	200; remaining 42; response 1,008,199 bytes	stable 2.34.6, prerelease 2.35.3, stable 1.x 1.123.71
16:14:36.529914Z	GET docs Actions runs, main, 30 rows	200; remaining 41; Link last=135	4,022 total; latest 30: 6 success/9 skipped/15 cancelled
16:14:37.471146Z	GET pin check-runs	200; remaining 40	72: 21 success, 51 skipped
16:14:38.915168Z	GET pin combined status	200; remaining 39	success, 16 success statuses
16:26:28.344018Z	GET commits path=mkdocs.yml, page 1	200; remaining 28; Link last=616	last touch/delete 5db109aa...
16:26:28.641341Z	same, page 616	200; remaining 27	first path commit 9821b9c0..., 2022-04-13

43.3 GitHub compare and repeated branch-census requests

UTC	Branch/request	Status; quota/Link	Parsed body excerpt
16:14:14.106525Z	compare changelog-2x-gitbook, per_page=1	200; remaining 56; Link last=2	diverged, +2/-299, 2 files
16:14:14.594006Z	compare new_nav_exploration	200; remaining 55; Link last=3	diverged, +3/-1660, 5 files
16:14:15.077902Z	compare update-breaking-changes-21044125	200; remaining 54	behind, +0/-1469
16:14:15.452070Z	compare D0C-22200-preview	200; remaining 53	ahead, +1/-0, 27 files
16:14:15.939294Z	compare D0C-1093	200; remaining 52; Link last=17	diverged, +17/-1935, 0 net files

UTC	Branch/request	Status; quota/Link	Parsed body excerpt
16:14:16.513539Z	compare ligo-690-import-export-docs-preview	200; remaining 51; Link last=9	diverged, +9/-363, 8 files
16:14:17.119192Z	compare chore/openapi-sync-16167794424-1	200; remaining 50	diverged, +1/-1334, 1 file
16:14:17.579758Z	compare docs/remove-otel-self-hosted-only-hint	200; remaining 49; Link last=3	diverged, +3/-227, 0 files
16:14:17.980257Z	compare docs-discord-moderation-actions	200; remaining 48	diverged, +1/-210, 2 files
16:14:18.426468Z	compare update-variables-plans	200; remaining 47	behind, +0/-5518
16:26:10.805902–16:26:13.806390Z	detailed compare repeat for eight diverged/ ahead branches, per_page=100	all 200; served remaining values 37,36,59,35,34,33,32,31	commit subjects and net file patches summarized in 36.3
16:26:19.949837 , 20.211281 , 20.451696Z	repeat branches pages 1..3	all 200; remaining 30/29/58	duplicate-tip pairs and protected branches confirmed

[OBSERVED] Served quota values reset/non-monotonically changed during the sequence; the values are logged as returned and are not interpreted as repository state.

43.4 Documentation preflight/revalidation batches before the final canonical ledger

UTC span	GET resources	Status/body summary
16:24:18.048199–16:24:21.218405Z	root, robots, sitemap index, nine partitions, llms.txt	all 200; 1,334 sitemap URLs and 1,334 llms links
16:24:29.839220–16:24:31.032912Z	sitemap index/partitions and llms.txt repeated	all 200; normalized overlap 1,326; eight representation pairs differ
16:24:46.973121–16:24:52Z	16 rendered-route probes	canonical samples 200; old redirects final 200; synthetic nonexistent HTML 404
16:24:59.353887–16:25:00.597348Z	API auth/reference, save/publish, v3, terminology, and mistaken deploy/host-n8n/README.md Markdown forms	all HTTP 200; mistaken README form body was # Page Not Found
16:33:36.761096 , 16:33:43.492257Z	connect sitemap repeated	200; 23 generated resource groups plus Models identified
16:33:56.000482 , .310407Z	Role and nonexistent Role Mapping Rule Markdown	both HTTP 200; Role has Create only; missing form has Page Not Found body
16:34:16.977781–16:34:18.216554Z	architecture-overview mistake, queue, task runners, database, Code, CLI Markdown	all HTTP 200; mistaken overview.md body was Page Not Found; valid pages parsed
16:34:24.515401 , .625117 , 16:34:44.878397Z	architecture landing, task runners repeat, How n8n works	all 200; valid work-in-progress and runner statements confirmed

43.5 Final canonical documentation HTTP ledger

The following canonical request table contains the 40-row batch plus one final direct-Docker-page revalidation. Redirect chains, final URLs, relevant headers, byte counts, and bounded excerpts are retained per request.

UTC timestamp	Method	URL	Status/final/redirects	Relevant response headers	Bytes	Body excerpt (≤160 chars)
2026-08-15T16:38:29.152570+00:00	GET	https://docs.n8n.io/	200	{ "date": "Sat, 15 Aug 2026 16:38:29 GMT", "content-type": "text/html; charset=utf-8", "etag": "W/\\\"6ef3yfvrrbrdek\\\"", "cache-control": "public, max-age=0, must-revalidate", "server": "cloudflare" }	530654	<!DOCTYPE html><html data-dpl-id="p-3db14ad950af3a31c6bfe42c26e76d" lang="en" class="rounded-corners theme-clean no-tint sidebar-filled sidebar-list-default lin
2026-08-15T16:38:29.373451+00:00	GET	https://docs.n8n.io/robots.txt	200	{ "date": "Sat, 15 Aug 2026 16:38:29 GMT", "content-type": "text/plain", "content-length": "188", "cache-control": "public, max-age=0, must-revalidate", "server": "cloudflare" }	188	User-agent: * Content-Signal: ai-train=yes, search=yes, ai-input=yes Allow: /~gitbook/image? Allow: /~gitbook/icon? Allow: /favicon.ico Allow: / Sitemap: http
2026-08-15T16:38:29.488094+00:00	GET	https://docs.n8n.io/sitemap.xml	200	{ "date": "Sat, 15 Aug 2026 16:38:29 GMT", "content-type": "application/xml", "content-length": "921", "cache-control": "public, max-age=0, must-revalidate", "server": "cloudflare" }	921	<?xml version="1.0" encoding="utf-8"?><sitemapindex xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"><sitemap><loc>https://docs.n8n.io/sitemap-pages
2026-08-15T16:38:29.595912+00:00	GET	https://docs.n8n.io/sitemap-pages.xml	200	{ "date": "Sat, 15 Aug 2026 16:38:29 GMT", "content-type": "application/xml", "content-length": "830", "cache-control": "public, max-age=0, must-revalidate", "server": "cloudflare" }	830	<?xml version="1.0" encoding="utf-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"><url><loc>https://docs.n8n.io/</loc><priority>1<
2026-08-15T16:38:29.741845+00:00	GET	https://docs.n8n.io/deploy/sitemap-pages.xml	200	{ "date": "Sat, 15 Aug 2026 16:38:29 GMT", "content-type": "application/xml", "cache-control": "public, max-age=0, must-revalidate", "server": "cloudflare" }	23550	<?xml version="1.0" encoding="utf-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"><url><loc>https://docs.n8n.io/deploy/</loc><prio
2026-08-15T16:38:29.853655+00:00	GET	https://docs.n8n.io/build/sitemap-pages.xml	200	{ "date": "Sat, 15 Aug 2026 16:38:30 GMT", "content-type": "application/xml", "cache-control": "public, max-age=0, must-revalidate", "server": "cloudflare" }	24799	<?xml version="1.0" encoding="utf-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"><url><loc>https://docs.n8n.io/build/</loc><prior
2026-08-15T16:38:29.997177+00:00	GET	https://docs.n8n.io/integrations/sitemap-pages.xml	200	{ "date": "Sat, 15 Aug 2026 16:38:30 GMT", "content-type": "application/xml", "cache-control": "public, max-age=0, must-revalidate", "server": "cloudflare" }	175508	<?xml version="1.0" encoding="utf-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"><url><loc>https://docs.n8n.io/integrations/</loc>
2026-08-15T16:38:30.134246+00:00	GET	https://docs.n8n.io/connect/sitemap-pages.xml	200	{ "date": "Sat, 15 Aug 2026 16:38:30 GMT", "content-type": "application/xml", "cache-control": "public, max-age=0, must-revalidate", "server": "cloudflare" }	12454	<?xml version="1.0" encoding="utf-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"><url><loc>https://docs.n8n.io/connect/</loc><pri
2026-08-15T16:38:30.263141+00:00	GET	https://docs.n8n.io/administer/sitemap-pages.xml	200	{ "date": "Sat, 15 Aug 2026 16:38:30 GMT", "content-type": "application/xml", "cache-control": "public, max-age=0, must-revalidate", "server": "cloudflare" }	8397	<?xml version="1.0" encoding="utf-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"><url><loc>https://docs.n8n.io/administer/</loc><
2026-08-15T16:38:30.382699+00:00	GET	https://docs.n8n.io/contribute/sitemap-pages.xml	200	{ "date": "Sat, 15 Aug 2026 16:38:30 GMT", "content-type": "application/xml", "content-length": "888", "cache-control": "public, max-age=0, must-revalidate", "server": "cloudflare" }	888	<?xml version="1.0" encoding="utf-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"><url><loc>https://docs.n8n.io/contribute/</loc><
2026-08-15T16:38:30.514353+00:00	GET	https://docs.n8n.io/privacy-and-security/sitemap-pages.xml	200	{ "date": "Sat, 15 Aug 2026 16:38:30 GMT", "content-type": "application/xml", "content-length": "761", "cache-control": "public, max-age=0, must-revalidate", "server": "cloudflare" }	761	<?xml version="1.0" encoding="utf-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"><url><loc>https://docs.n8n.io/privacy-and-security/</
2026-08-15T16:38:30.657141+00:00	GET	https://docs.n8n.io/changelog/sitemap-pages.xml	200	{ "date": "Sat, 15 Aug 2026 16:38:30 GMT", "content-type": "application/xml", "cache-control": "public, max-age=0, must-revalidate", "server": "cloudflare" }	1493	<?xml version="1.0" encoding="utf-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"><url><loc>https://docs.n8n.io/changelog/</loc><p
2026-08-15T16:38:30.783059+00:00	GET	https://docs.n8n.io/llms.txt	200	{ "date": "Sat, 15 Aug 2026 16:38:31 GMT", "content-type": "text/markdown", "charset=utf-8", "cache-control": "public, max-age=0, must-revalidate", "server": "cloudflare" }	279594	# n8n Docs ## Get started - [n8n Docs] (https://docs.n8n.io/welcome.md) - [Choose how to use n8n] (https://docs.n8n.io/choose-how-to-use-n8n.md): Choose between

UTC timestamp	Method	URL	Status/final/redirects	Relevant response headers	Bytes	Body excerpt (≤160 chars)
2026-08-15T16:38:31.264536+00:00	GET	https://docs.n8n.io/choose-how-to-use-n8n	200	{"date":"Sat, 15 Aug 2026 16:38:31 GMT","content-type":"text/html; charset=utf-8","etag":"W/\\\"15r2gpzciulfhtr\\\"","cache-control":"public, max-age=0, must-revalidate","server":"cloudflare"}	723211	<!DOCTYPE html><html data-dpl-id="p-3db14ad950af3a31c6bfe42c26e76d" lang="en" class="rounded-corners theme-clean no-tint sidebar-filled sidebar-list-default lin
2026-08-15T16:38:31.423200+00:00	GET	https://docs.n8n.io/deploy/host-n8n/install-options/install-using-docker-compose	200	{"date":"Sat, 15 Aug 2026 16:38:31 GMT","content-type":"text/html; charset=utf-8","etag":"W/\\\"uwohg1377gjjl\\\"","cache-control":"public, max-age=0, must-revalidate","server":"cloudflare"}	771892	<!DOCTYPE html><html data-dpl-id="p-3db14ad950af3a31c6bfe42c26e76d" lang="en" class="rounded-corners theme-clean no-tint sidebar-filled sidebar-list-default lin
2026-08-15T16:38:31.597115+00:00	GET	https://docs.n8n.io/build/understand-workflows/save-and-publish-workflows	200	{"date":"Sat, 15 Aug 2026 16:38:31 GMT","content-type":"text/html; charset=utf-8","etag":"W/\\\"44z91c6yjd1fke\\\"","cache-control":"public, max-age=0, must-revalidate","server":"cloudflare"}	626784	<!DOCTYPE html><html data-dpl-id="p-3db14ad950af3a31c6bfe42c26e76d" lang="en" class="rounded-corners theme-clean no-tint sidebar-filled sidebar-list-default lin
2026-08-15T16:38:31.732857+00:00	GET	https://docs.n8n.io/integrations/built-in/core-nodes/n8n-nodes-base.webhook	200	{"date":"Sat, 15 Aug 2026 16:38:32 GMT","content-type":"text/html; charset=utf-8","etag":"W/\\\"rz7iio0ioliX2\\\"","cache-control":"public, max-age=0, must-revalidate","server":"cloudflare"}	1004476	<!DOCTYPE html><html data-dpl-id="p-3db14ad950af3a31c6bfe42c26e76d" lang="en" class="rounded-corners theme-clean no-tint sidebar-filled sidebar-list-default lin
2026-08-15T16:38:32.066879+00:00	GET	https://docs.n8n.io/connect/n8n-api/authentication	200	{"date":"Sat, 15 Aug 2026 16:38:32 GMT","content-type":"text/html; charset=utf-8","etag":"W/\\\"yq5a0vjays1ltn\\\"","cache-control":"public, max-age=0, must-revalidate","server":"cloudflare"}	982177	<!DOCTYPE html><html data-dpl-id="p-3db14ad950af3a31c6bfe42c26e76d" lang="en" class="rounded-corners theme-clean no-tint sidebar-filled sidebar-list-default lin
2026-08-15T16:38:32.230840+00:00	GET	https://docs.n8n.io/administer/manage-users-and-access/set-permissions-and-roles-rbac	200	{"date":"Sat, 15 Aug 2026 16:38:32 GMT","content-type":"text/html; charset=utf-8","etag":"W/\\\"6xou7qytiq937p\\\"","cache-control":"public, max-age=0, must-revalidate","server":"cloudflare"}	424115	<!DOCTYPE html><html data-dpl-id="p-3db14ad950af3a31c6bfe42c26e76d" lang="en" class="rounded-corners theme-clean no-tint sidebar-filled sidebar-list-default lin
2026-08-15T16:38:32.403642+00:00	GET	https://docs.n8n.io/contribute/terminology	200	{"date":"Sat, 15 Aug 2026 16:38:32 GMT","content-type":"text/html; charset=utf-8","etag":"W/\\\"a0690p2088bfxt\\\"","cache-control":"public, max-age=0, must-revalidate","server":"cloudflare"}	533907	<!DOCTYPE html><html data-dpl-id="p-3db14ad950af3a31c6bfe42c26e76d" lang="en" class="rounded-corners theme-clean no-tint sidebar-filled sidebar-list-default lin
2026-08-15T16:38:32.591130+00:00	GET	https://docs.n8n.io/privacy-and-security/sustainable-use-license	200	{"date":"Sat, 15 Aug 2026 16:38:32 GMT","content-type":"text/html; charset=utf-8","etag":"W/\\\"do06dow6sgcvtk\\\"","cache-control":"public, max-age=0, must-revalidate","server":"cloudflare"}	602498	<!DOCTYPE html><html data-dpl-id="p-3db14ad950af3a31c6bfe42c26e76d" lang="en" class="rounded-corners theme-clean no-tint sidebar-filled sidebar-list-default lin
2026-08-15T16:38:32.742912+00:00	GET	https://docs.n8n.io/changelog/release-notes	200	{"date":"Sat, 15 Aug 2026 16:38:32 GMT","content-type":"text/html; charset=utf-8","etag":"W/\\\"j3refp9yelh807\\\"","cache-control":"public, max-age=0, must-revalidate","server":"cloudflare"}	2484809	<!DOCTYPE html><html data-dpl-id="p-3db14ad950af3a31c6bfe42c26e76d" lang="en" class="rounded-corners theme-clean no-tint sidebar-filled sidebar-list-default lin
2026-08-15T16:38:32.933041+00:00	GET	https://docs.n8n.io/changelog/v30-breaking-changes	200	{"date":"Sat, 15 Aug 2026 16:38:33 GMT","content-type":"text/html; charset=utf-8","etag":"W/\\\"sn6kdgxbmaeq\\\"","cache-control":"public, max-age=0, must-revalidate","server":"cloudflare"}	485268	<!DOCTYPE html><html data-dpl-id="p-3db14ad950af3a31c6bfe42c26e76d" lang="en" class="rounded-corners theme-clean no-tint sidebar-filled sidebar-list-default lin
2026-08-15T16:38:33.092574+00:00	GET	https://docs.n8n.io/learning-paths	200	{"date":"Sat, 15 Aug 2026 16:38:33 GMT","content-type":"text/html; charset=utf-8","etag":"W/\\\"lsc3n5vek784jc\\\"","cache-control":"public, max-age=0, must-revalidate","server":"cloudflare"}	379166	<!DOCTYPE html><html data-dpl-id="p-3db14ad950af3a31c6bfe42c26e76d" lang="en" class="rounded-corners theme-clean no-tint sidebar-filled sidebar-list-default lin
2026-08-15T16:38:33.241537+00:00	GET	https://docs.n8n.io/hosting/	200 → https://docs.n8n.io/deploy/host-n8n redirects=[{"302","https://docs.n8n.io/hosting/","https://docs.n8n.io/hosting/"}], [{"307","https://docs.n8n.io/hosting/","https://docs.n8n.io/deploy/host-n8n"}]]	{"date":"Sat, 15 Aug 2026 16:38:33 GMT","content-type":"text/html; charset=utf-8","etag":"W/\\\"89w5h5ldgagjw\\\"","cache-control":"public, max-age=0, must-revalidate","server":"cloudflare"}	488074	<!DOCTYPE html><html data-dpl-id="p-3db14ad950af3a31c6bfe42c26e76d" lang="en" class="rounded-corners theme-clean no-tint sidebar-filled sidebar-list-default lin

UTC timestamp	Method	URL	Status/final/redirects	Relevant response headers	Bytes	Body excerpt (≤160 chars)
2026-08-15T16:38:33.676677+00:00	GET	https://docs.n8n.io/workflows/	200 → https://docs.n8n.io/build/understand-workflows redirects=[[{"302","https://docs.n8n.io/workflows/"},"https://docs.n8n.io/workflows"],[{"307","https://docs.n8n.io/workflows"},"https://docs.n8n.io/build/understand-workflows"]]	{"date":"Sat, 15 Aug 2026 16:38:33 GMT","content-type":"text/html; charset=utf-8","etag":"W/\\\"eqq7yoejo89iuz\\\"","cache-control\":\"public, max-age=0, must-revalidate\",\"server\":\"cloudflare\"}	444389	<!DOCTYPE html><html data-dpl-id=p-3db14ad950af3a31c6bfe42c26e76d\" lang=en\" class=rounded-corners theme-clean no-tint sidebar-filled sidebar-list-default lin
2026-08-15T16:38:33.988357+00:00	GET	https://docs.n8n.io/forensic-route-probe-not-real-20260815	404	{\"date\":\"Sat, 15 Aug 2026 16:38:34 GMT\", \"content-type\":\"text/html; charset=utf-8\", \"etag\":\"W/\\\"akaaJ0fc2741x8\\\"\", \"cache-control\":\"public, max-age=0, must-revalidate\", \"server\":\"cloudflare\"}	189133	<!DOCTYPE html><html data-dpl-id=p-3db14ad950af3a31c6bfe42c26e76d\" id= next_error \"><head><meta charSet=utf-8\"/><meta name=viewport\" content=width=device
2026-08-15T16:38:34.132571+00:00	GET	https://docs.n8n.io/connect/n8n-api/authentication.md	200	{\"date\":\"Sat, 15 Aug 2026 16:38:34 GMT\", \"content-type\":\"text/markdown; charset=utf-8\", \"cache-control\":\"public, max-age=0, must-revalidate\", \"server\":\"cloudflare\"}	13816	> For the complete documentation index, see [llms.txt](https://docs.n8n.io/llms.txt). Markdown versions of documentation pages are available by appending .md
2026-08-15T16:38:34.259607+00:00	GET	https://docs.n8n.io/connect/n8n-api/api-reference.md	200	{\"date\":\"Sat, 15 Aug 2026 16:38:34 GMT\", \"content-type\":\"text/markdown; charset=utf-8\", \"cache-control\":\"public, max-age=0, must-revalidate\", \"server\":\"cloudflare\"}	1553	> For the complete documentation index, see [llms.txt](https://docs.n8n.io/llms.txt). Markdown versions of documentation pages are available by appending .md
2026-08-15T16:38:34.375170+00:00	GET	https://docs.n8n.io/connect/n8n-api/role.md	200	{\"date\":\"Sat, 15 Aug 2026 16:38:34 GMT\", \"content-type\":\"text/markdown; charset=utf-8\", \"cache-control\":\"public, max-age=0, must-revalidate\", \"server\":\"cloudflare\"}	2532	> For the complete documentation index, see [llms.txt](https://docs.n8n.io/llms.txt). Markdown versions of documentation pages are available by appending .md
2026-08-15T16:38:34.500021+00:00	GET	https://docs.n8n.io/connect/n8n-api/role-mapping-rule.md	200	{\"date\":\"Sat, 15 Aug 2026 16:38:34 GMT\", \"content-type\":\"text/markdown; charset=utf-8\", \"cache-control\":\"public, max-age=0, must-revalidate\", \"server\":\"cloudflare\"}	1957	# Page Not Found The URL n8n-api/role-mapping-rule does not exist. This page may have been moved, renamed, or deleted. ## Suggested Pages You may be lookin
2026-08-15T16:38:34.623362+00:00	GET	https://docs.n8n.io/build/understand-workflows/save-and-publish-workflows.md	200	{\"date\":\"Sat, 15 Aug 2026 16:38:34 GMT\", \"content-type\":\"text/markdown; charset=utf-8\", \"cache-control\":\"public, max-age=0, must-revalidate\", \"server\":\"cloudflare\"}	5638	> For the complete documentation index, see [llms.txt](https://docs.n8n.io/llms.txt). Markdown versions of documentation pages are available by appending .md
2026-08-15T16:38:34.737792+00:00	GET	https://docs.n8n.io/changelog/v30-breaking-changes.md	200	{\"date\":\"Sat, 15 Aug 2026 16:38:34 GMT\", \"content-type\":\"text/markdown; charset=utf-8\", \"cache-control\":\"public, max-age=0, must-revalidate\", \"server\":\"cloudflare\"}	4170	> For the complete documentation index, see [llms.txt](https://docs.n8n.io/llms.txt). Markdown versions of documentation pages are available by appending .md
2026-08-15T16:38:34.851722+00:00	GET	https://docs.n8n.io/contribute/terminology.md	200	{\"date\":\"Sat, 15 Aug 2026 16:38:34 GMT\", \"content-type\":\"text/markdown; charset=utf-8\", \"cache-control\":\"public, max-age=0, must-revalidate\", \"server\":\"cloudflare\"}	4545	> For the complete documentation index, see [llms.txt](https://docs.n8n.io/llms.txt). Markdown versions of documentation pages are available by appending .md
2026-08-15T16:38:34.960011+00:00	GET	https://docs.n8n.io/deploy/host-n8n/understand-the-architecture/how-n8n-works.md	200	{\"date\":\"Sat, 15 Aug 2026 16:38:35 GMT\", \"content-type\":\"text/markdown; charset=utf-8\", \"content-length\":\"613\", \"cache-control\":\"public, max-age=0, must-revalidate\", \"server\":\"cloudflare\"}	613	> For the complete documentation index, see [llms.txt](https://docs.n8n.io/llms.txt). Markdown versions of documentation pages are available by appending .md
2026-08-15T16:38:35.100897+00:00	GET	https://docs.n8n.io/deploy/host-n8n/configure-n8n/scaling/enable-queue-mode.md	200	{\"date\":\"Sat, 15 Aug 2026 16:38:35 GMT\", \"content-type\":\"text/markdown; charset=utf-8\", \"cache-control\":\"public, max-age=0, must-revalidate\", \"server\":\"cloudflare\"}	21899	> For the complete documentation index, see [llms.txt](https://docs.n8n.io/llms.txt). Markdown versions of documentation pages are available by appending .md
2026-08-15T16:38:35.251820+00:00	GET	https://docs.n8n.io/deploy/host-n8n/configure-n8n/set-up-task-runners.md	200	{\"date\":\"Sat, 15 Aug 2026 16:38:35 GMT\", \"content-type\":\"text/markdown; charset=utf-8\", \"cache-control\":\"public, max-age=0, must-revalidate\", \"server\":\"cloudflare\"}	13035	> For the complete documentation index, see [llms.txt](https://docs.n8n.io/llms.txt). Markdown versions of documentation pages are available by appending .md
2026-08-15T16:38:35.383252+00:00	GET	https://docs.n8n.io/deploy/host-n8n/understand-the-architecture/understand-the-database.md	200	{\"date\":\"Sat, 15 Aug 2026 16:38:35 GMT\", \"content-type\":\"text/markdown; charset=utf-8\", \"cache-control\":\"public, max-age=0, must-revalidate\", \"server\":\"cloudflare\"}	5713	> For the complete documentation index, see [llms.txt](https://docs.n8n.io/llms.txt). Markdown versions of documentation pages are available by appending .md
2026-08-15T16:38:35.513746+00:00	GET	https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.code.md	200	{\"date\":\"Sat, 15 Aug 2026 16:38:35 GMT\", \"content-type\":\"text/markdown; charset=utf-8\", \"cache-control\":\"public, max-age=0, must-revalidate\", \"server\":\"cloudflare\"}	9865	> For the complete documentation index, see [llms.txt](https://docs.n8n.io/llms.txt). Markdown versions of documentation pages are available by appending .md

UTC timestamp	Method	URL	Status/final/redirects	Relevant response headers	Bytes	Body excerpt (≤160 chars)
2026-08-15T16:38:35.724972+00:00	GET	https://docs.n8n.io/connect/n8n-cli.md	200	{ "date": "Sat, 15 Aug 2026 16:38:35 GMT", "content-type": "text/markdown; charset=utf-8", "cache-control": "public, max-age=0, must-revalidate", "server": "cloudflare" }	8275	> For the complete documentation index, see [llms.txt](https://docs.n8n.io/llms.txt). Markdown versions of documentation pages are available by appending .md
2026-08-15T16:43:27.221248+00:00	GET	https://docs.n8n.io/deploy/host-n8n/install-options/install-with-docker	200	{ "date": "Sat, 15 Aug 2026 16:43:27 GMT", "content-type": "text/html; charset=utf-8", "etag": "W/\\u1371zib37evlr\\", "cache-control": "public, max-age=0, must-revalidate", "server": "cloudflare" }	694229	<!DOCTYPE html><html data-dpl-id="p-3db14ad950af3a31c6bfe42c26e76d" lang="en" class="rounded-corners theme-clean no-tint sidebar-filled sidebar-list-default lin

43.6 Filtered Git sessions

[OBSERVED] Read-only Git smart-HTTP sessions fetched: docs main ; all 29 sampled docs tips; wiki history; source pin; stable/prerelease tags; 13 consequential docs commits and seven required parents; and the 2022 MkDocs-introduction commit/parent. All repositories were bare/filtered and had no checkout. Direct fetch of revert-message SHA 8cecc29b... failed with upload-pack: not our ref ; it was not retried.

44. Evidence matrix, lifecycle status, unknowns, and audit

44.1 Evidence-matrix additions

Claim	Label	Status	Direct evidence
GitBook is current docs framework	[STATED]+[OBSERVED]	CURRENT	README; live GitBook HTML; current tree
Public docs route total is 1,334	[OBSERVED]	CURRENT	all nine sitemap partitions; exact source/generated reconciliation
llms.txt has 1,334 unique links but eight collection-root representations differ	[OBSERVED]	CURRENT	full in-memory set comparison
VuePress → MkDocs → GitBook forms	[OBSERVED]	HISTORICAL/CURRENT	root, 9821b9c..., 4f6df31..., current tree
Course content removed	[OBSERVED]	REMOVED	b924629... exact diff
Migration dropped then restored selected artifacts	[OBSERVED]	HISTORICAL/RESTORED PARTLY	migration and 44bb160... exact trees
Redirect cleanup attempted then fully reverted	[OBSERVED]	REVERTED	identical pre/revert tree SHA
Wiki exists with MkDocs-era guidance	[OBSERVED]	CURRENT ARTIFACT / HISTORICAL GUIDANCE	complete wiki history and current pages
Latest releases are stable 2.34.6, prerelease 2.35.3, maintained 1.123.71	[OBSERVED]	CURRENT AT REQUEST TIME	GitHub Releases API
v3 October 2026 transition	[STATED]	FUTURE	live/repository v3 page
16 v3 rule files exist but target constant is null	[OBSERVED]	FUTURE — CANDIDATE SOURCE / DISABLED	stable, prerelease, pin source
Current pin OpenAPI is 72 literal / 71 normalized / 113 operations	[OBSERVED]	CURRENT SOURCE PIN	complete ref resolution and build merge code
Live Role docs lag later pin's role/mapping routes	[OBSERVED]	CURRENT TEMPORAL DRIFT	live pages; three source states
Hosted DB page is incomplete/stale	[OBSERVED]	CURRENT DRIFT	24 headings versus 128/127 table pages; role wiring
Binary-data and task-runner path statements conflict	[STATED]+[OBSERVED]	CURRENT CONTRADICTIONS	current docs plus source config/Dockerfile
No local tests were run	[OBSERVED]	METHOD BOUNDARY	research log

44.2 Lifecycle/status summary

CURRENT

- GitBook-hosted docs with nine public collection partitions, 1,334 sitemap routes, 1,334 `llms.txt` links, `.md` page variants, collection `SUMMARY.md` navigation, and named built-in OpenAPI dependency.
- Docs `main@ec418cd6...`, 1,863 paths, 1,473 repository Markdown files, nine GitHub Actions workflow definitions, 16 workflow JSON artifacts, Vale/Lychee/script automation.
- Available four-page docs wiki at `d8b53a7...`, though platform-specific guidance is older than the GitBook migration.
- Stable 2.34.6, prerelease 2.35.3, and later implementation pin; current source still contains future-removal targets.
- Current source OpenAPI, generated database schemas, source-local architecture documents, and test/spec artifacts.

HISTORICAL

- VuePress at the 2020 root; MkDocs Material introduced in April 2022; GitBook migration in June 2026.
- Archived 0.x/1.x/2.x ledgers and migration pages.
- Pre-GitBook docs-repository `docs/api/v1/openapi.yml` and local MkDocs/Netlify build form.

REMOVED / REVERTED

- Course prose/images removed 2026-06-19.
- MkDocs navigation/config, snippets/overrides/archive tooling removed in migration cleanup; final Netlify/runtime/requirements residue removed 2026-07-23.
- Selected migration-dropped assets restored; 12 dropped JSON exports not restored by that commit.
- May 2026 redirect cleanup fully reverted to the exact prior tree.

FUTURE — STATED OR CANDIDATE

- n8n 3.0 scheduled for October 2026 in target-authored docs.
- Docker-only support/removals/security changes are future statements.
- Detailed v3 rules and one migration framework exist as candidate source, but target version is null in all three inspected current/release states.

UNKNOWN

- Exact GitBook OpenAPI synchronization source/version and cadence.
- Whether the enabled docs wiki remains intended contributor authority.
- Whether filesystem relay offload is a deliberate queue-mode exception.
- Which optional/module-generated tables exist in any particular live deployment.
- Deployment prevalence of Cloud/self-hosted, queue, multi-main, external-runner, and other forms.
- Final n8n 3.0 scope and whether every candidate rule ships unchanged.

- Runtime behavior of any documentation workflow artifact; none was executed.

44.3 Method incidents and test boundary

- [OBSERVED] No repository test suite was run. Test files establish intended checks, not observed pass results.
- [OBSERVED] Rename-aware migration diffing hydrated blobs and raised the workspace from about 3 MB to 43 MB, still below the 80 MB research ceiling. Later all-Markdown/schema scans peaked at 46 MB.
- [OBSERVED] No broad path-history scan, unreachable-object enumeration, checkout, or full-body persistence was attempted in the redo.
- [OBSERVED] Direct fetch of `8cecc29b...` failed because the remote did not advertise/serve that object. Reachable attempt/revert trees were sufficient to establish the exact reverted state.
- [OBSERVED] Compact parsing replaced unbounded console dumps. No source map, bundle, image, PDF body, or raw log was retained.

44.4 Cleanup and final resource audit

- [OBSERVED] Temporary filtered repositories, prior rejected Sprint 4 report, compact scratch ledger, and appendix scratch file were deleted after validation.
- [OBSERVED] No PDF was created; the final PDF remains reserved for Sprint 6.
- [OBSERVED] Final workspace: one file, `/home/user/working_sprint_4.md`.
- [OBSERVED] Final exact report size: 221321 bytes; final line count: 2177; `du -sh /home/user = 224K`; `find /home/user -type f | wc -l = 1`.

Sprint 5 will inspect the broad live public surface and infrastructure/routes using the predeclared endpoint sampling strategy; Sprint 6 remains the cross-source consolidation and final PDF sprint.

Sprint 4 of 6 completed. 2 research sprints remain.

Sprint 5 — live public surface, route generation, catalog APIs, and deployment signals

45. Sprint 5 method, evidence state, and sampling boundary

45.1 Current observation window and scope

- [OBSERVED] Direct HTTP collection ran from 2026-08-17T03:19:13.966782Z through 2026-08-17T03:27:36.589098Z (2026-08-17 Asia/Tehran). This sprint treats responses as a time-bounded public snapshot, not as a statement about every region, cache node, or later deployment.
- [OBSERVED] The live scope was the supplied marketing target n8n.io, its advertised sitemap families and machine-readable forms, and the public catalog/template service at api.n8n.io that the target’s HTML/configuration and current documentation explicitly identify. Selected exact source files were read at the existing implementation pin 2afc867f20d689891b516c0458d627bc52c979fd only to cross-check the live template boundary.
- [OBSERVED] No application instance, Cloud workspace, login page, creator portal, private workflow, user account, or administrative operation was intentionally inspected. Links to app.n8n.cloud and other authentication-oriented surfaces were inventoried from public HTML but not followed.
- [OBSERVED] No credential-bearing header was sent. No credential value was retained. No workflow was imported into an instance or executed. No POST, PUT, PATCH, or DELETE request was made.
- [OBSERVED] Full response bodies were never written to disk. Sitemap, HTML, JSON, source, and JavaScript responses were parsed serially in memory and reduced to counts, keys, hashes, bounded strings, and request excerpts of at most 160 characters. No source map or full bundle was retained.

45.2 Scale gates and deterministic sample

[OBSERVED] The current sitemap index still has 12 child maps and now contains 334,644 entries, so route bodies required sampling. The census itself was exhaustive: every sitemap body and every <loc> was parsed. The rendered-body sample was fixed before body interpretation:

- one lexical midpoint for each non-root first-path-segment stratum in the 235-entry general page sitemap: 38 routes; root was inspected separately;
- five evenly spaced lexical quantiles from each unique integration, workflow, creator, and native cross-integration set: 20 routes;
- the nearest lexical midpoint in each of six non-native cross-integration partitions: six routes;
- four evenly spaced lexical positions from the 60-entry AI benchmark set: four routes.

[OBSERVED] The resulting 68-route rendered sample had no duplicate URL. Additional architectural probes covered the two family hubs, template API forms, machine-readable report forms, canonical redirects, a synthetic missing route, static Nuxt chunks, and four pinned-source candidates.

Surface	Full current size	Rendered/body sample	Rationale
General page sitemap	235	38 plus root	Every non-root first-segment stratum represented.
Integration detail URLs	2,053 entries / 2,045 unique	5 plus hub	Lexical boundary/quartile sample; duplicates handled in census.
Workflow template URLs	11,462	5 plus hub	Lexical boundary/quartile sample, then matching catalog objects.

Surface	Full current size	Rendered/body sample	Rationale
Creator URLs	442	5 plus hub	Lexical boundary/quartile sample.
Native-labeled cross routes	46,963	5	Lexical boundary/quartile sample.
Six non-native-labeled cross partitions	273,429	6	One midpoint per partition.
AI benchmark model URLs	60	4 plus hub	Four lexical positions.
Public catalog/template API	endpoint-sized	21 direct requests plus target proxies	Documented forms, aliases, bounded search, and one OPTIONS preflight.

[OBSERVED] Three discovery searches were used to locate official current website/API references and a possible website source repository. The search tool did not expose its own request URL/status/timestamp. Architectural claims below rely on direct target, API, documentation, and pinned-source responses. No current public repository directly established itself as the production `n8n.io` marketing source; that remains [UNKNOWN].

45.3 Request accounting

- [OBSERVED] The compact ledger contains 146 recorded requests: 143 GET, two HEAD, and one OPTIONS.
- [OBSERVED] Final response statuses were 137 × 200, seven × 404, one × 204, and one × 503.
- [OBSERVED] Recorded host distribution: 115 `n8n.io`, 21 `api.n8n.io`, four `raw.githubusercontent.com`, three `api.github.com`, and one each to `docs.n8n.io`, `www.n8n.io`, and the Internet Archive CDX host.
- [OBSERVED] Total response bytes read serially were 83,283,797 (about 79.43 MiB); the largest single response was 4,787,250 bytes. Peak retained workspace remained below 1 MiB before report assembly because no body was persisted.

46. Live route census, growth, and generated graph

46.1 Exact current sitemap census

[OBSERVED] The index's `x-sitemap-prerendered` header says `2026-08-14T21:12:33.902Z`. None of the 334,644 `<url>` entries contained a `lastmod` value. Every location used host `n8n.io`.

Sitemap	Entries	Unique URLs	Change from Sprint 1 entries
General pages	235	235	+2
Integrations	2,053	2,045	+29 entries
Workflows	11,462	11,462	+10
Creators	442	442	0
Native-labeled cross integrations	46,963	46,963	0
Non-native cross 1	49,999	49,999	0
Non-native cross 2	49,999	49,999	0
Non-native cross 3	49,999	49,999	0
Non-native cross 4	49,999	49,999	0
Non-native cross 5	49,999	49,999	0
Non-native cross 6	23,434	23,434	0
AI benchmark	60	60	0
Aggregate	334,644	334,636	+41 entries

[OBSERVED] Sprint 1's 2026-08-13 snapshot had 334,603 entries. The exact +41 difference is accounted for by +2 general pages, +29 integration entries, and +10 workflow entries. Sprint 1 did not compute per-sitemap uniqueness, so the eight current duplicate entries are not classified as newly introduced.

46.2 Integration-sitemap duplicates

[OBSERVED] All eight duplicate entries are excess occurrences of four exact URLs:

Integration route	Occurrences	Excess entries
green-api-trigger	2	1
firstpromoter	4	3

Integration route	Occurrences	Excess entries
firstpromoter-trigger	4	3
bambuser	2	1
Total		8

[OBSERVED] Duplicate sitemap membership is not duplicate page content by itself; the URL set contains 2,045 distinct integration slugs.

46.3 Cross-integration route graph

The labels “native” and “non-native” below reproduce sitemap filenames. They are not independent runtime-support classifications.

Route set	Pair routes	Component slugs	Complete undirected pairs possible	Missing pairs	Self pairs	Reversed duplicate duos
Native-labeled map	46,963	307	46,971	8	0	0
Six non-native-labeled maps combined	273,429	801	320,400	not a standalone complete-graph claim	0	0
All cross maps combined	320,392	801	320,400	8	0	0

- [OBSERVED] The six non-native partitions contain no pair overlap with one another. All 307 native-set components also occur in the combined non-native-set component universe; another 494 occur only in the latter.
- [OBSERVED] The combined cross surface is a near-complete undirected graph: 320,392 of 320,400 possible two-component pairs. Pair order is not always lexical, but no reverse-order duplicate duo exists.
- [OBSERVED] The cross graph uses 801 component slugs. Eight hundred have an integration-sitemap page; one does not. That absent component appears in 800 pair routes, and 800 cross routes therefore reference at least one component without a corresponding current integration-sitemap entry.
- [OBSERVED] Conversely, 1,245 of the 2,045 unique integration-sitemap slugs do not appear in the cross-route component universe. URL membership alone does not establish why a component is eligible or ineligible for generated pair pages.

46.4 Rendered sample health and canonicalization

- [OBSERVED] All 68 sampled sitemap routes returned final HTTP 200 without a redirect. Every response contained Nuxt markers and a `__NUXT_DATA__` payload.
- [OBSERVED] Sixty-six of 68 canonical links exactly matched the requested sitemap URL.
- [OBSERVED] `/strapi-demo/` returned 200 with affiliate-program title/content and canonicalized to `/affiliates/`.
- [OBSERVED] `/subscription/thank-you/` returned 200 with plans/pricing content and canonicalized to `/pricing/`.
- [OBSERVED] `/embed/thank-you/` and `/oem/thank-you/` were in the sitemap, returned 200, and declared `noindex`.
- [OBSERVED] `/guide-hero-banner/` and `/guide-hero-basic/` were in the sitemap, returned 200, had exact canonicals and `index` robots values, but no HTML title text in the inspected responses.

- [OBSERVED] Five integration, five workflow, five creator, five native cross, six non-native cross, and four AI-benchmark samples all returned 200 with family-appropriate titles. This supports sampled route health only; it is not an exhaustive 334,636-route health test.
-

47. Current marketing-site rendering and deployment signals

47.1 Nuxt server-rendered form

[OBSERVED] The root response was approximately 351 KB of already rendered HTML. It contained:

- one executable entry script, `/_nuxt/C2mS7bxv.js`;
- 56 module-preload links and 14 stylesheet links;
- `__NUXT__`, `__NUXT_DATA__`, `data-nuxt-data`, and 750 scoped `data-v-*` attribute occurrences;
- an Organization JSON-LD object;
- no HTML form element in the root response;
- a public runtime configuration labeled `production`, with a build-seed field and named template/catalog, creator, workflow, Cloud registration, documentation, and benchmark endpoints.

[OBSERVED] Four bounded JavaScript chunks were fetched and parsed in memory: 742,416, 1,246, 775, and 285 bytes. The entry chunk imported 137 hashed modules and contained Nuxt/Vue runtime markers plus same-origin calls for workflow search, categories, node search filters, and website pages. None of the four sampled chunks contained a `sourceMappingURL` marker; no map was requested.

[OBSERVED] The root HTML publicly contains an internal-labeled website-backend hostname and the entry chunk contains additional host strings. Host strings and public configuration names are vocabulary/deployment leads, not proof that the browser can reach each host or that each service is active. The internal-labeled host was not requested and is not reproduced here.

47.2 Observed live data flow

The following is a bounded [INFERRED] reconstruction from rendered HTML, public configuration, successful same-origin proxies, direct public API responses, and static-resource links. It does not identify the undisclosed origin implementation or database.

MERMAID DIAGRAM SOURCE (RETAINED AS EVIDENCE)

```

flowchart LR
    B["Public browser"] --> CF["Cloudflare edge\nmarketing HTML/routes"]
    CF --> N["Nuxt SSR/hydration surface\n__NUXT_DATA__ + hashed chunks"]
    N --> P["Same-origin /api/product-api and /api/nodes proxies"]
    P --> T["api.n8n.io\ntemplate/catalog service"]
    N --> T
    T --> U["Catalog persistence/origin\nUNKNOWN"]
    N --> AZ["Azure Blob-hosted media\npublic asset references"]
    N -. public links only .-> APP["app.n8n.cloud\nauth-oriented surface not fetched"]
    N -. public links only .-> DOCS["docs/community/experts/creator surfaces"]
  
```

47.3 Edge, cache, and response-header forms

Resource class	Observed cache/edge form
Root/general marketing HTML	Cache-Control: public, max-age=0, must-revalidate; Cloudflare DYNAMIC.
Dynamic integration/workflow/catalog HTML	commonly stale-while-revalidate; Cloudflare DYNAMIC.
Hashed Nuxt JavaScript	public, max-age=31536000, immutable; sampled chunk was Cloudflare HIT; Access-Control-Allow-Origin: *.
Sitemap index	public, max-age=600, must-revalidate; prerender timestamp header.
Marketing robots.txt	max-age=14400, must-revalidate.
Template API GETs	generally max-age=14400; sampled responses were Cloudflare HIT or MISS; Vary: Origin, Accept-Encoding.
Synthetic missing marketing route	404, Cache-Control: no-cache, and Referrer-Policy: no-referrer.

[OBSERVED] Both `n8n.io` and `api.n8n.io` resolved through the sandbox resolver to the same three IPv4 and three IPv6 Cloudflare addresses at 2026-08-17T03:26:22.240970Z. DNS answers are time/region dependent and do not reveal the origin.

47.4 Security-header scope observed, without evaluation

- [OBSERVED] Root and all 68 rendered samples returned `Strict-Transport-Security: max-age=0; includeSubDomains`.
- [OBSERVED] Thirty-nine sampled responses used a common general-page header profile with `X-Frame-Options: DENY`, `X-Content-Type-Options: nosniff`, `Referrer-Policy: strict-origin-when-cross-origin`, `Access-Control-Allow-Origin: *`, and a **report-only** CSP. No enforced `Content-Security-Policy` header appeared in any of the 68 sample responses.
- [OBSERVED] The report-only policy named `frame-ancestors 'none'`, `object-src 'none'`, the public catalog API, analytics/reCAPTCHA/payment hosts, Azure-hosted media, and several frame hosts. The 29 catalog/detail responses did not carry that same general-page header set in the inspected responses, although all had the HSTS value above.
- [OBSERVED] HTTP root redirected 301 to HTTPS; `www.n8n.io` redirected 301 to the apex host. These redirect observations are separate from the `max-age=0` header fact.

47.5 Linked-surface boundary

[OBSERVED] Root HTML linked to the application login/registration host, docs, GitHub, community, experts, learning, blog, merchandise, creator, social, and other public surfaces. Direct links establish navigation relationships only. Authentication-oriented destinations and the internal-labeled host were not fetched. The deployed n8n application version behind any Cloud instance remains [UNKNOWN].

48. Public template/catalog API

48.1 Distinct API identity

[OBSERVED] `api.n8n.io` is the template/catalog service named by current marketing configuration, current template documentation, and pinned application source. It is distinct from an n8n instance's authenticated `/api/v1` Public API reconstructed in Sprint 4. The catalog endpoints inspected here were public GET surfaces and required no API key.

[STATED] Current template documentation says `N8N_TEMPLATES_HOST` may point an instance to a custom library and enumerates `/templates/workflows/{id}`, `/workflows/templates/{id}`, `/templates/search`, `/templates/collections`, `/templates/categories`, and `/health`. It explicitly says the two workflow endpoints require different response formats.

48.2 Current endpoint forms and schemas

GET endpoint	Status / bytes	Direct parsed structure
<code>/health</code>	200 / 15	<code>{"status": "OK"}</code> .
<code>/templates/categories</code>	200 / 3,966	31 categories; keys <code>id</code> , <code>name</code> , <code>displayName</code> , <code>icon</code> , <code>parent</code> .
<code>/templates/collections</code>	200 / 2,013	eight collections; collection keys include <code>id</code> , <code>rank</code> , <code>name</code> , <code>totalViews</code> , <code>createdAt</code> , <code>workflows</code> , <code>nodes</code> .
<code>/templates/search?rows=2&page=1</code>	200 / 86,280	<code>totalWorkflows=11,518</code> ; two workflow rows; three filter objects.
<code>/templates/search/filters</code>	200 / 1,126,106	31 category filter rows and 329 node filter rows.
<code>/api/creators?rows=2&page=1</code>	200 / 4,787,250	data contained 15,725 creator rows; supplied <code>rows/page</code> did not bound this response.
<code>/templates/workflows/11807</code>	200 / 200,440	outer <code>workflow</code> metadata wrapping an importable workflow definition.
<code>/workflows/templates/11807</code>	200 / 107,688	flat <code>id</code> , <code>name</code> , <code>workflow</code> import form.

- [OBSERVED] The same category/search/filter endpoints also returned 200 when prefixed with `/api`. The two workflow forms returned byte-identical bodies under their with- `/api` and without- `/api` aliases: SHA-256 `74377e...a68e8` for the wrapped metadata response and `607fdd...a6df` for the flat import response.
- [OBSERVED] Template 11807's nested workflow had 91 nodes, 67 connection-source keys, 31 distinct node-type strings, no node with a credential field, `active:false`, and settings keys `binaryMode`, `availableInMCP`, and `executionOrder`. These are stored template fields, not live execution state.
- [OBSERVED] No template response body was persisted. Workflow parameter, pin-data, metadata, and credential values were not retained.

48.3 Same-origin website proxies

Same-origin marketing endpoint	Status / parsed result
/api/product-api/categories?rootOnly=true	200; seven root categories.
/api/product-api/categories?skipEmpty=true	200; 28 nonempty categories.
/api/product-api/workflows/search?rows=2&page=1	200; same top-level schema and totalWorkflows=11,518.
/api/nodes/search-filters	200; 1,268 unique rows, all type:"app" ; 31 used logoIcon , 1,237 used logoUrl .

[OBSERVED] The Nuxt entry code caps each workflow-search request at 250 rows and, when a caller asks for more, partitions it into multiple page requests before concatenating responses. This is a client behavior observed in the sampled chunk, not a server-limit test.

[OBSERVED] An OPTIONS preflight to /templates/search with origin https://n8n.io returned 204, reflected that origin, listed GET/POST/PUT/PATCH/DELETE/OPTIONS/HEAD, listed allowed request headers, and set a one-year preflight max-age. Listing methods in CORS does not prove that this exact endpoint implements each method. No mutation method was called.

48.4 API-root boundary incident

[OBSERVED] A GET to the public API root unexpectedly followed a server redirect to an admin-labeled path before the client could stop it. The final HTML was 1,069 bytes. The path is redacted as [REDACTED_ADMIN_PATH] ; its body was not analyzed, no link or form was followed, and no further request was made to it. All API claims in this sprint rely on the documented public catalog endpoints, not that response.

49. Workflow, integration, creator, and benchmark forms

49.1 Workflow hub, sitemap, and API counts

[OBSERVED] Three simultaneous public counts differ:

- workflow sitemap: 11,462 detail URLs, prerendered 2026-08-14;
- catalog search API: `totalWorkflows=11,518` at both 03:20:52Z and a cache-busted 03:27:36Z request;
- `/workflows/` HTML title and H1: 11519 templates at 03:27Z.

[OBSERVED] The API exceeds the sitemap by 56; the hub text exceeds it by 57 and exceeds the API by one. The inspected surfaces do not disclose whether the differences reflect publication eligibility, synchronization cadence, filtering, counting convention, or another cause.

49.2 Five workflow-page/API pairs

All five page titles matched the catalog metadata names. Credential **values** were not inspected or retained; `credential-field nodes` counts only nodes with a nonempty `credentials` field.

Template ID	Definition nodes	Connection-source keys	Distinct node types	Credential-field nodes	Stored active field
1	3	2	3	1	absent
14,827	33	17	14	0	absent
2,541	30	13	12	5	true
6,287	11	6	8	5	false
9,999	21	9	12	6	absent

[OBSERVED] A template with `active:true` is a stored public artifact, not evidence that it is active in a live n8n instance. No sample was imported or executed.

49.3 Integration forms and count boundaries

- [OBSERVED] The integration sitemap has 2,045 unique page slugs; the website app-filter endpoint has 1,268 unique `type:"app"` rows; the template API's workflow-search filter has 329 node rows; and the pinned main source has the different code-path counts recorded in Sprint 4 (for example 556 `*.node.ts` implementations). These count different artifacts and must not be substituted for one another.
- [OBSERVED] Five sampled integration pages all returned 200. Two large responses (Magento 2 and Samsung SmartThings) had flattened Nuxt payloads with roughly 810 integration/app record key shapes, while the smaller 1Shot API, DriveLock API, and Zvid samples had different content structures. The sitemap filename, name, or page size is not converted into a runtime-support classification.

- [OBSERVED] Cross-integration pages are generated four-segment routes of the form `/integrations/<a>/and/<b>/` . Eleven sampled native/non-native routes all returned 200 and exact canonical links.

49.4 Creator surface

- [OBSERVED] Creator sitemap: 442 URLs. Five lexical samples all returned 200 and exact canonicals.
- [OBSERVED] The creator hub's flattened Nuxt payload contained 1,024 repeated creator-record key shapes, while the public creators API returned 15,725 records despite `rows=2&page=1` .
- [UNKNOWN] The selection/pagination rules that produce 442 sitemap creators, 1,024 hub-payload records, and 15,725 API rows. No identity or profile was treated as private; only already public creator page structure was sampled.

49.5 AI benchmark surface

- [OBSERVED] The AI benchmark sitemap has 60 model routes plus a general page in the general-page sitemap.
 - [OBSERVED] Four model samples—Claude Haiku 4.5, GPT-5 Mini, Llama 4 Scout, and Trinity Mini—returned 200, exact canonicals, model-detail/benchmark headings, and WebPage/SoftwareApplication JSON-LD.
 - [OBSERVED] Public configuration names a separate recommendation endpoint. It was not called because the exposed form was not established as a read-only GET interface and no request was needed for route reconstruction.
-

50. Machine-readable surfaces and route normalization

50.1 Marketing `llms.txt` is intentionally narrow in observed content

[OBSERVED] `https://n8n.io/llms.txt` returned 200, `text/plain`, 1,160 bytes, and ten lines. It contained four links representing two reports in HTML and Markdown forms:

- 2026 Workflow-based AI Agent Development Tools — HTML and `.md`;
- 2025 Enterprise AI Agent Development Tools — HTML and `.md`.

[OBSERVED] Unlike the documentation site's 1,334-link `llms.txt`, the marketing file is not a route inventory for the 334,636 unique sitemap URLs. This is a scope difference between two current machine-readable surfaces.

50.2 Markdown forms

- [OBSERVED] The 2026 report `.md` route returned 200, `text/markdown`, 6,875 bytes, and eight Markdown headings.
- [OBSERVED] A synthetic nonexistent marketing `.md` path returned 404 HTML, matching the marketing HTML-not-found status class.
- [OBSERVED] This differs from `docs.n8n.io`, where Sprint 4 observed synthetic `.md` not-found content under HTTP 200. Host-specific behavior is retained rather than generalized.

50.3 Redirect and missing-route forms

- [OBSERVED] `/sitemap.xml` redirected 307 to `/sitemap_index.xml`.
- [OBSERVED] Plain HTTP root and `www` each redirected 301 to canonical HTTPS apex root.
- [OBSERVED] A synthetic missing path without trailing slash redirected 301 to the slash form, which returned 404 HTML, 13,608 bytes, `no-cache`, and `Referrer-Policy: no-referrer`.
- [OBSERVED] `robots.txt` remained 76 bytes: `User-agent: *`, `disallow /cdn-cgi/`, and point to the sitemap index.

51. Source/document/live cross-check

51.1 Pinned application source points to the live catalog

[OBSERVED] At `2afc867f...`, `packages/@n8n/config/src/configs/templates.config.ts` is a 331-byte current source file with:

- `N8N_TEMPLATES_ENABLED` ;
- `N8N_TEMPLATES_HOST` ;
- default host `https://api.n8n.io/api/` .

[OBSERVED] Pinned `FrontendService` exposes `templates.enabled` and `templates.host` in frontend settings. This establishes the application's configurable catalog boundary; it does not make the marketing Nuxt code part of the application repository.

[OBSERVED] Current live documentation gives example links without `/api` , while pinned source defaults to the `/api/` base. Both observed URL forms returned matching schemas, and the two sampled workflow forms were byte-identical across aliases. The prefix difference is therefore a supported live alias in these observations, not a current break.

51.2 Marketing site versus application source

- [OBSERVED] The live marketing site is Nuxt-rendered and uses its own hashed asset graph and runtime configuration.
- [OBSERVED] The main product source reconstructed in Sprint 3 has a Vue/Vite editor and Node/Express CLI backend. The public marketing HTML does not identify itself as that editor build.
- [UNKNOWN] The exact production marketing source repository, commit, Nuxt version, origin runtime, content backend implementation, and deployment pipeline. Search found a separate `n8n-demo-website` repository, but no evidence identified it as the current production marketing source, so it is not used as implementation evidence.
- [UNKNOWN] The implementation, persistence technology, and deployment version of `api.n8n.io` . Its `X-Powered-By: n8n <n8n.io>` header is retained as an observed header string, not converted into a source-package claim.

51.3 Live version boundary

[OBSERVED] Marketing pages, template pages, and catalog responses did not establish the n8n version deployed to any Cloud editor instance. Stable/prerelease/source versions from Sprint 4 remain release/source evidence only. Public workflow definitions can contain versioned node fields, but they do not identify a serving application version.

52. Live contradiction and drift register

1. Sitemap/API/hub workflow totals

- [OBSERVED] 11,462 sitemap URLs, 11,518 API total, 11,519 hub title/H1.
- Status: **CURRENT MULTI-SURFACE COUNT DRIFT**. Cause [UNKNOWN].

2. Duplicate integration sitemap entries

- [OBSERVED] Eight excess entries across four URLs.
- Status: **CURRENT SITEMAP DUPLICATION**. Page duplication is not inferred.

3. Cross graph versus integration sitemap

- [OBSERVED] One cross-graph component lacks an integration-sitemap entry but participates in 800 pair URLs; 1,245 integration slugs do not participate in cross routes.
- Status: **CURRENT ELIGIBILITY/INDEXING ASYMMETRY**. Rule [UNKNOWN].

4. Sitemap URL versus canonical target

- [OBSERVED] `/strapi-demo/` canonicalizes to affiliates; `/subscription/thank-you/` canonicalizes to pricing while both remain sitemap entries.
- Status: **CURRENT ADDRESSABLE/CANONICAL DRIFT**. Removal or redirect intent is not inferred.

5. Sitemap versus robots metadata

- [OBSERVED] Two thank-you routes are sitemap entries and declare `noindex`.
- Status: **CURRENT INDEXING-SIGNAL TENSION**. Search-engine handling is outside this report.

6. Indexable routes without title

- [OBSERVED] Two `guide-hero-*` routes are sitemap-listed, exact-canonical, robots `index`, and have empty titles.
- Status: **CURRENT PUBLIC TITLELESS ROUTES**. Purpose [UNKNOWN]; the route names are not treated as implementation facts.

7. Template URL prefix

- [OBSERVED] Docs examples omit `/api`; source default includes it; both work for inspected endpoints.
- Status: **CURRENT ALIAS, NOT A LIVE CONTRADICTION**.

8. Machine-readable scope

- [OBSERVED] Marketing `llms.txt` has four links while sitemap has 334,636 unique URLs; docs `llms.txt` is a full 1,334-link documentation index.
- Status: **CURRENT HOST-SPECIFIC SCOPE DIFFERENCE**.

9. Catalog count vocabularies

- [OBSERVED] 2,045 integration slugs, 1,268 app-filter rows, 329 workflow-node filters, and different source implementation counts.
- Status: **DISTINCT ARTIFACT POPULATIONS**. No count is relabeled as “number of supported integrations” without target-defined equivalence.

10. Creator populations

- [OBSERVED] 442 sitemap pages, 1,024 hub payload record shapes, and 15,725 public API creator rows.
- Status: **CURRENT SELECTION/PAGINATION DRIFT OR DIFFERING POPULATIONS; EXACT RULE UNKNOWN**.

11. Header scope

- [OBSERVED] General pages and catalog/detail pages share Cloudflare/Nuxt/HSTS signals but differ in cache and security-header sets.
- Status: **CURRENT ROUTE-CLASS DIFFERENCE**, not assumed misconfiguration.

12. Historical archive gap

- [OBSERVED] The bounded Internet Archive CDX query returned 503.
 - Status: **UNKNOWN HISTORICAL MARKETING BUILD EPOCHS**. No retry or third-party reconstruction was used.
-

53. Sprint 5 compact HTTP request ledger

53.1 Header notation

Each row records the exact UTC client timestamp, method, requested URL, final status/redirect outcome, downloaded body bytes, relevant response-header summary, and an excerpt no longer than 160 characters. Compact header keys are: `ct` content type, `cache`, `srv`, `cf` Cloudflare cache status, `hsts`, `cspR0` report-only CSP present, `xfo`, `acao`, `powered`, `cors`, `rate`, and `prerender`. ETags and request IDs are retained only where they materially distinguish an immutable/static or hosted API response. Full CSP text is summarized here and described in Section 47.4 rather than repeated dozens of times.

53.2 Recorded requests

UTC timestamp	Method and requested URL	Final status / redirect outcome	Relevant headers	Bytes / body excerpt (≤160 chars)
2026-08-17T03:19:13.966782+00:00	GET https://n8n.io/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; hsts=max-age=0; includeSubDomains; xfo=DENY; ref=strict-origin-when-cross-origin; acao=*	350806; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:19:14.215171+00:00	GET https://n8n.io/robots.txt	200	ct=text/plain; charset=utf-8; cache=max-age=14400, must-revalidate; srv=cloudflare; hsts=max-age=0; includeSubDomains	76; User-agent: * Disallow: /cdn-cgi/ Sitemap: https://n8n.io/sitemap_index.xml
2026-08-17T03:19:14.300870+00:00	GET https://n8n.io/sitemap_index.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; hsts=max-age=0; includeSubDomains; xfo=DENY; ref=strict-origin-when-cross-origin; acao=*	1053; <?xml version="1.0" encoding="UTF-8"?><sitemapindex xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"><sitemap><loc>https://n8n.io/sitemap-pages.xml</loc></si
2026-08-17T03:19:14.487600+00:00	GET https://n8n.io/sitemap-pages.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; hsts=max-age=0; includeSubDomains; xfo=DENY; ref=strict-origin-when-cross-origin; acao=*	15427; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:19:14.587980+00:00	GET https://n8n.io/sitemap-integrations.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; hsts=max-age=0; includeSubDomains; xfo=DENY; ref=strict-origin-when-cross-origin; acao=*	128848; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:19:14.775852+00:00	GET https://n8n.io/sitemap-workflows.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; hsts=max-age=0; includeSubDomains; xfo=DENY; ref=strict-origin-when-cross-origin; acao=*	1388734; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:19:15.105977+00:00	GET https://n8n.io/sitemap-creators.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; hsts=max-age=0; includeSubDomains; xfo=DENY; ref=strict-origin-when-cross-origin; acao=*	25255; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:19:15.243667+00:00	GET https://n8n.io/sitemap-cross-integrations.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; hsts=max-age=0; includeSubDomains; xfo=DENY; ref=strict-origin-when-cross-origin; acao=*	3524575; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:19:16.137249+00:00	GET https://n8n.io/sitemap-cross-integrations-non-native-1.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; hsts=max-age=0; includeSubDomains; xfo=DENY; ref=strict-origin-when-cross-origin; acao=*	3724069; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:19:16.751355+00:00	GET https://n8n.io/sitemap-cross-integrations-non-native-2.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; hsts=max-age=0; includeSubDomains; xfo=DENY; ref=strict-origin-when-cross-origin; acao=*	3713184; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:19:17.322843+00:00	GET https://n8n.io/sitemap-cross-integrations-non-native-3.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; hsts=max-age=0; includeSubDomains; xfo=DENY; ref=strict-origin-when-cross-origin; acao=*	3771703; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:19:17.935725+00:00	GET https://n8n.io/sitemap-cross-integrations-non-native-4.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; hsts=max-age=0; includeSubDomains; xfo=DENY; ref=strict-origin-when-cross-origin; acao=*	3738242; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm

UTC timestamp	Method and requested URL	Final status / redirect outcome	Relevant headers	Bytes / body excerpt (≤160 chars)
2026-08-17T03:19:18.512389+00:00	GET https://n8n.io/sitemap-cross-integrations-non-native-5.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; hsts=max-age=0; includeSubDomains; xfo=DENY; ref=strict-origin-when-cross-origin; acao=*	3723654; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:19:19.133030+00:00	GET https://n8n.io/sitemap-cross-integrations-non-native-6.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; hsts=max-age=0; includeSubDomains; xfo=DENY; ref=strict-origin-when-cross-origin; acao=*	1730338; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:19:19.496057+00:00	GET https://n8n.io/sitemap-ai-benchmark.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; hsts=max-age=0; includeSubDomains; xfo=DENY; ref=strict-origin-when-cross-origin; acao=*	4332; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:19:43.969032+00:00	GET https://n8n.io/sitemap-integrations.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare	128848; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:19:59.353891+00:00	GET https://n8n.io/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	351159; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalabl
2026-08-17T03:20:28.575202+00:00	GET https://n8n.io/_nuxt/C2m57bxv.js	200	ct=text/javascript; cache=public, max-age=31536000, immutable; srv=cloudflare; cf=HIT; acao=*; etag=W/"acfc4c5802aa2a74a38fdb5d7c7d2831"	742416; const _vite_mapDeps=(i,m=_vite_mapDeps,d=(m.f\)\ (m.f=["./CBpc8LI5.js","./.dk8yEJm9.js","./.Dzv3zWH1.js","./.BKms0AG6.js","./.B6p1NHnW.js","./.h6Xfk3P1.js","./.ZABd
2026-08-17T03:20:28.695213+00:00	GET https://n8n.io/_nuxt/BRBWgaLQ.js	200	ct=text/javascript; cache=public, max-age=31536000, immutable; srv=cloudflare; cf=HIT; acao=*; etag="edabf387881ddabced6af56a8239f288"	1246; import(_ as o)from"/BxVYN8ec.js";import{h as r,c as t,o as m}from"/C2m57bxv.js";import"/. Bom8lh5Q.js";import"/.BXD_9xtS.js";import"/.BKms0AG6.js";import"/.X3R
2026-08-17T03:20:28.737854+00:00	GET https://n8n.io/_nuxt/Dt7hpg_D.js	200	ct=text/javascript; cache=public, max-age=31536000, immutable; srv=cloudflare; cf=HIT; acao=*; etag=W/"89d4b55c52dafc8e487834edf41b223d"	775; import{ _ as a,o as l,k as r,e}from"/.C2m57bxv.js";const s={},n={class:"pointer-events-none absolute aspect-1057/720 w-[130%] max-w-[1057px] origin-center -trans
2026-08-17T03:20:28.784328+00:00	GET https://n8n.io/_nuxt/xs0uZ0hN.js	200	ct=text/javascript; cache=public, max-age=31536000, immutable; srv=cloudflare; cf=HIT; acao=*; etag=W/"6621cf7d35364dd44494e9343ef5f0e"	285; import{h as s,o as a,c as r,s as n,w as o,r as c}from"/.C2m57bxv.js";const i=s({ _ name:"IconWrapper",props:{tag:{default:"div"}},setup(e){return(t,l)=>{a(),r(n(
2026-08-17T03:20:50.816282+00:00	GET https://api.n8n.io/	200; final=https://api.n8n.io/[REDACTED_ADMIN_PATH]	ct=text/html; charset=utf-8; srv=cloudflare; cf=DYNAMIC; powered=n8n <n8n.io>	1069; <!DOCTYPE html><html lang=en><head><meta charset=utf-8?><meta name=viewport content=width=device-width, initial-scale=1, viewport-fit=cover?><meta name=
2026-08-17T03:20:51.260159+00:00	GET https://api.n8n.io/health	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=HIT; powered=n8n <n8n.io>	15; {"status":"OK"}
2026-08-17T03:20:51.460957+00:00	GET https://api.n8n.io/templates/categories	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=HIT; powered=n8n <n8n.io>	3966; {"categories":[{"id":25,"name":"AI","displayName":null,"icon":"","parent":null},{id":47,"name":"AI Chatbot","displayName":"AI Chatbot, Assistant, Agent","icon
2026-08-17T03:20:51.696936+00:00	GET https://api.n8n.io/templates/collections	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=HIT; powered=n8n <n8n.io>	2013; {"collections":[{"id":7,"rank":0,"name":"n8n Key Concepts","totalViews":0,"createdAt":"2022-07-06T14:51:53.250Z","workflows":[{"id":1700},{id":1744},{id":1746
2026-08-17T03:20:51.929463+00:00	GET https://api.n8n.io/templates/search?rows=2&page=1	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=HIT; powered=n8n <n8n.io>	86280; {"totalWorkflows":11518,"workflows":[{"id":11807,"name":"Answer multi-channel support queries with OpenAI RAG and Supabase","totalViews":2462,"price":0,"purchas
2026-08-17T03:20:52.140045+00:00	GET https://api.n8n.io/templates/search/filters	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=EXPIRED; powered=n8n <n8n.io>	1126106; {"categories":[{"id":2,"name":"Sales","icon":"","hitCount":1674},{id":5,"name":"Engineering","icon":"⚙️","hitCount":538},{id":11,"name":"Other","icon":"","h
2026-08-17T03:20:52.699685+00:00	GET https://api.n8n.io/api/templates/categories	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=HIT; powered=n8n <n8n.io>	3966; {"categories":[{"id":25,"name":"AI","displayName":null,"icon":"","parent":null},{id":47,"name":"AI Chatbot","displayName":"AI Chatbot, Assistant, Agent","icon
2026-08-17T03:20:52.739300+00:00	GET https://api.n8n.io/api/templates/search?rows=2&page=1	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=HIT; powered=n8n <n8n.io>	86280; {"totalWorkflows":11518,"workflows":[{"id":11807,"name":"Answer multi-channel support queries with OpenAI RAG and Supabase","totalViews":2462,"price":0,"purchas
2026-08-17T03:20:52.787570+00:00	GET https://api.n8n.io/api/templates/search/filters	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=HIT; powered=n8n <n8n.io>	1126106; {"categories":[{"id":2,"name":"Sales","icon":"","hitCount":1674},{id":5,"name":"Engineering","icon":"⚙️","hitCount":538},{id":11,"name":"Other","icon":"","h

UTC timestamp	Method and requested URL	Final status / redirect outcome	Relevant headers	Bytes / body excerpt (≤160 chars)
2026-08-17T03:20:52.895374+00:00	GET https://api.n8n.io/api/creators?rows=2&page=1	200	ct=application/json; charset=utf-8; srv=cloudflare; cf=DYNAMIC; powered=n8n <n8n.io>	4787250; {"data":[{"id":"91416","name":"Sam Nesler","username":"badgr","bio":"","verified":false,"links":[{"partnerKey":null,"avatar":"https://gravatar.com/avatar/b5a5a0a
2026-08-17T03:21:23.805865+00:00	GET https://api.n8n.io/templates/workflows/11807	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=HIT	200440; {"workflow":{"id":"11807","name":"Answer multi-channel support queries with OpenAI RAG and Supabase","views":3218,"recentViews":392,"totalViews":3218,"createdAt":
2026-08-17T03:21:24.048084+00:00	GET https://api.n8n.io/api/templates/workflows/11807	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=HIT	200440; {"workflow":{"id":"11807","name":"Answer multi-channel support queries with OpenAI RAG and Supabase","views":3218,"recentViews":392,"totalViews":3218,"createdAt":
2026-08-17T03:21:24.119246+00:00	GET https://api.n8n.io/workflows/templates/11807	200	ct=application/json; charset=utf-8; srv=cloudflare; cf=DYNAMIC	107688; {"id":"11807","name":"Answer multi-channel support queries with OpenAI RAG and Supabase","workflow":{"id":"FcaGDP7u8LEV48Zy","meta":{"instanceId":"40d28ae80959439
2026-08-17T03:21:24.539841+00:00	GET https://api.n8n.io/api/workflows/templates/11807	200	ct=application/json; charset=utf-8; srv=cloudflare; cf=DYNAMIC	107688; {"id":"11807","name":"Answer multi-channel support queries with OpenAI RAG and Supabase","workflow":{"id":"FcaGDP7u8LEV48Zy","meta":{"instanceId":"40d28ae80959439
2026-08-17T03:21:43.537840+00:00	GET https://n8n.io/api/product-api/categories?rootOnly=true	200	ct=application/json; srv=cloudflare; cf=DYNAMIC	539; {"categories":[{"id":25,"name":"AI","displayName":null,"icon":"","parent":null},{id":30,"name":"Document Ops","displayName":null,"icon":"","parent":null},{
2026-08-17T03:21:43.659531+00:00	GET https://n8n.io/api/product-api/categories?skipEmpty=true	200	ct=application/json; srv=cloudflare; cf=DYNAMIC	3743; {"categories":[{"id":25,"name":"AI","displayName":null,"icon":"","parent":null},{id":47,"name":"AI Chatbot","displayName":"AI Chatbot, Assistant, Agent","icon
2026-08-17T03:21:43.740402+00:00	GET https://n8n.io/api/product-api/workflows/search?rows=2&page=1	200	ct=application/json; srv=cloudflare; cf=DYNAMIC	86280; {"totalWorkflows":11518,"workflows":[{"id":"11807","name":"Answer multi-channel support queries with OpenAI RAG and Supabase","totalViews":2462,"price":0,"purchas
2026-08-17T03:21:43.841980+00:00	GET https://n8n.io/api/nodes/search-filters	200	ct=application/json; srv=cloudflare; cf=DYNAMIC	265054; [{"id":"n8n-nodes-base.googleSheets","type":"app","value":"Google Sheets","label":"Google Sheets","logoIcon":"","logoUrl":"/nodes/google-sheets.svg"},{"id":"@n8
2026-08-17T03:21:43.934328+00:00	OPTIONS https://api.n8n.io/templates/search	204	srv=cloudflare; cf=DYNAMIC; acao=https://n8n.io; cors=GET,POST,PUT,PATCH,DELETE,OPTIONS,HEAD; cors-age=31536000	0; ``
2026-08-17T03:22:19.521563+00:00	GET https://n8n.io/affiliates/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	206575; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalabl
2026-08-17T03:22:19.676071+00:00	GET https://n8n.io/ai/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	292992; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalabl
2026-08-17T03:22:19.830724+00:00	GET https://n8n.io/ai-agents/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	398449; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalabl
2026-08-17T03:22:19.983442+00:00	GET https://n8n.io/ai-benchmark/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	378764; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalabl
2026-08-17T03:22:20.237739+00:00	GET https://n8n.io/automate-lead-management/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	299291; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalabl
2026-08-17T03:22:20.405733+00:00	GET https://n8n.io/careers/culture-code/multipliers/	200	ct=text/html; charset=utf-8; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	136347; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalabl
2026-08-17T03:22:21.192079+00:00	GET https://n8n.io/case-studies/kunai-part2/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	176847; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalabl

UTC timestamp	Method and requested URL	Final status / redirect outcome	Relevant headers	Bytes / body excerpt (≤160 chars)
2026-08-17T03:22:21.442789+00:00	GET https://n8n.io/contact/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	145378; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:21.582164+00:00	GET https://n8n.io/creators/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	749530; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:21.811600+00:00	GET https://n8n.io/embed/thank-you/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	122383; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:22.022253+00:00	GET https://n8n.io/enterprise/thank-you/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	123632; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:22.159766+00:00	GET https://n8n.io/expert-partners/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	177991; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:22.294076+00:00	GET https://n8n.io/features/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	299589; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:22.441458+00:00	GET https://n8n.io/guide-hero-banner/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	167665; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:22.646910+00:00	GET https://n8n.io/guide-hero-basic/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	169320; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:22.860719+00:00	GET https://n8n.io/imprint/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	128342; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:22.992418+00:00	GET https://n8n.io/integrations/categories/finance-and-accounting/2/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	360277; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:23.174335+00:00	GET https://n8n.io/itops/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	319047; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:23.345187+00:00	GET https://n8n.io/legal/master-enterprise-terms/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	242996; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:23.550180+00:00	GET https://n8n.io/limitless-integrations/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	296466; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:23.729977+00:00	GET https://n8n.io/newsletter/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	137622; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:23.921407+00:00	GET https://n8n.io/oem/thank-you/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	122484; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:24.106505+00:00	GET https://n8n.io/partners/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	171522; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:24.264828+00:00	GET https://n8n.io/press/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	126552; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:24.405390+00:00	GET https://n8n.io/pricing/startup-plan/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	179274; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable

UTC timestamp	Method and requested URL	Final status / redirect outcome	Relevant headers	Bytes / body excerpt (≤160 chars)
2026-08-17T03:22:24.588271+00:00	GET https://n8n.io/rag/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	394008; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:24.755805+00:00	GET https://n8n.io/report-a-vulnerability/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	122851; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:24.897720+00:00	GET https://n8n.io/reports/2026-ai-agent-development-tools/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	355404; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:25.089579+00:00	GET https://n8n.io/saas/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	296937; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:25.254548+00:00	GET https://n8n.io/sap-partnership/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	189930; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:25.397127+00:00	GET https://n8n.io/secops/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	299692; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:25.560392+00:00	GET https://n8n.io/strapi-demo/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	195886; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:25.775938+00:00	GET https://n8n.io/subscription/thank-you/	200	ct=text/html; charset=utf-8; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	124403; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:26.655405+00:00	GET https://n8n.io/supercharge-your-crm/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	308153; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:26.801136+00:00	GET https://n8n.io/support/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	139878; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:26.933599+00:00	GET https://n8n.io/tools/csv-to-json/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	212918; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:27.099814+00:00	GET https://n8n.io/vs/langgraph/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	248992; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:27.240162+00:00	GET https://n8n.io/workflows/categories/invoice-processing/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	767712; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:28.963497+00:00	GET https://n8n.io/integrations/1shot-api/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	229584; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:29.598687+00:00	GET https://n8n.io/integrations/drivelock-api/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	226797; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:30.729971+00:00	GET https://n8n.io/integrations/magento-2/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	1461646; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:32.855483+00:00	GET https://n8n.io/integrations/samsung-smarththings/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	1439453; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:35.494437+00:00	GET https://n8n.io/integrations/zvid/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	221514; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable

UTC timestamp	Method and requested URL	Final status / redirect outcome	Relevant headers	Bytes / body excerpt (≤160 chars)
2026-08-17T03:22:36.227854+00:00	GET https://n8n.io/workflows/1-insert-excel-data-to-postgres/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	173516; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:36.911003+00:00	GET https://n8n.io/workflows/14827-build-an-openai-rag-system-with-document-upload-semantic-search-and-caching/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	183399; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:37.641134+00:00	GET https://n8n.io/workflows/2541-ai-agent-to-chat-with-you-search-console-data-using-openai-and-postgres/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	179199; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:38.324732+00:00	GET https://n8n.io/workflows/6287-email-support-agent-w-gemini-and-gpt-fallback-using-gmail-google-sheets/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	179829; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:38.807572+00:00	GET https://n8n.io/workflows/9999-ai-powered-feedback-triage-jotform-to-trello-airtable-and-slack-with-gemini/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	194633; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:40.415962+00:00	GET https://n8n.io/creators/KlickTipp/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	265876; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:41.324735+00:00	GET https://n8n.io/creators/dudnyk/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	225814; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:41.774541+00:00	GET https://n8n.io/creators/lorenanda/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	251002; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:42.097769+00:00	GET https://n8n.io/creators/rckflr/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	231326; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:42.660664+00:00	GET https://n8n.io/creators/zospirlo21/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	236624; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:42.958300+00:00	GET https://n8n.io/integrations/action-network/and/activecampaign/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	409762; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:44.080949+00:00	GET https://n8n.io/integrations/brevo/and/gong/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	309451; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:45.081838+00:00	GET https://n8n.io/integrations/freshservice/and/harvest/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	470321; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:45.884005+00:00	GET https://n8n.io/integrations/lusha/and/mysql/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	322103; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:46.920208+00:00	GET https://n8n.io/integrations/zoom/and/zulip/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	300557; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:47.813142+00:00	GET https://n8n.io/integrations/gitlab/and/knack/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	297676; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:48.927563+00:00	GET https://n8n.io/integrations/float/and/questdb/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	270286; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:49.677993+00:00	GET https://n8n.io/integrations/deepseek-chat-model/and/giphy/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	304965; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable

UTC timestamp	Method and requested URL	Final status / redirect outcome	Relevant headers	Bytes / body excerpt (≤160 chars)
2026-08-17T03:22:51.556523+00:00	GET https://n8n.io/integrations/float/and/opencti/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	260226; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:52.493071+00:00	GET https://n8n.io/integrations/embedai/and/kadoa/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	258526; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:53.213483+00:00	GET https://n8n.io/integrations/domo/and/nyota/	200	ct=text/html; charset=utf-8; cache=stale-while-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	255862; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:53.718730+00:00	GET https://n8n.io/ai-benchmark/claude-haiku-4.5/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	202192; <!DOCTYPE html><html lang=a charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:53.864153+00:00	GET https://n8n.io/ai-benchmark/gpt-5-mini/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	201885; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:54.019836+00:00	GET https://n8n.io/ai-benchmark/llama-4-scout/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	202525; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:22:54.175313+00:00	GET https://n8n.io/ai-benchmark/trinity-mini/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	201050; <!DOCTYPE html><html lang=en><head><meta charset=utf-8><meta name=viewport content=width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:23:23.810588+00:00	GET https://api.n8n.io/templates/workflows/1	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=HIT	14829; {"workflow":{"id":1,"name":"Insert Excel data to Postgres","views":13545,"recentViews":0,"totalViews":13545,"createdAt":"2019-08-31T00:05:02.587Z","description"
2026-08-17T03:23:24.022396+00:00	GET https://api.n8n.io/templates/workflows/14827	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=HIT	64583; {"workflow":{"id":14827,"name":"Build an OpenAI RAG system with document upload, semantic search and caching","views":710,"recentViews":0,"totalViews":710,"c rea
2026-08-17T03:23:24.220276+00:00	GET https://api.n8n.io/templates/workflows/2541	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=HIT	61858; {"workflow":{"id":2541,"name":"AI agent to chat with you Search Console data, using OpenAI and Postgres","views":9517,"recentViews":0,"totalViews":9517,"created
2026-08-17T03:23:24.441196+00:00	GET https://api.n8n.io/templates/workflows/6287	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=HIT	30722; {"workflow":{"id":6287,"name":"Email support agent w/ Gemini & GPT fallback using Gmail + Google Sheets","views":36073,"recentViews":2,"totalViews":36073,"creat
2026-08-17T03:23:24.637467+00:00	GET https://api.n8n.io/templates/workflows/9999	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=HIT	57399; {"workflow":{"id":9999,"name":"AI-powered feedback triage: Jotform to Trello, Airtable & Slack with Gemini","views":660,"recentViews":0,"totalViews":660,"cr eate
2026-08-17T03:23:34.745783+00:00	GET https://n8n.io/api/nodes/search-filters	200	ct=application/json; srv=cloudflare; cf=DYNAMIC	265054; [{"id":"n8n-nodes-base.googleSheets","type":"app","value":"Google Sheets","label":"Google Sheets","logoIcon":"","logoUrl":"/nodes/google-sheets.svg"},{"id":"@n8
2026-08-17T03:23:53.104459+00:00	GET https://n8n.io/sitemap-integrations.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; cf=DYNAMIC	128848; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:23:53.189159+00:00	GET https://n8n.io/sitemap-cross-integrations.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; cf=DYNAMIC	3524575; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:23:53.466963+00:00	GET https://n8n.io/sitemap-cross-integrations-non-native-1.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; cf=DYNAMIC	3724069; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:23:53.725881+00:00	GET https://n8n.io/sitemap-cross-integrations-non-native-2.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; cf=DYNAMIC	3713184; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:23:53.980755+00:00	GET https://n8n.io/sitemap-cross-integrations-non-native-3.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; cf=DYNAMIC	3771703; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm

UTC timestamp	Method and requested URL	Final status / redirect outcome	Relevant headers	Bytes / body excerpt (≤160 chars)
2026-08-17T03:23:54.230679+00:00	GET https://n8n.io/sitemap-cross-integrations-non-native-4.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; cf=DYNAMIC	3738242; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:23:54.547843+00:00	GET https://n8n.io/sitemap-cross-integrations-non-native-5.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; cf=DYNAMIC	3723654; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:23:54.817688+00:00	GET https://n8n.io/sitemap-cross-integrations-non-native-6.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; cf=DYNAMIC	1730338; <?xml version="1.0" encoding="UTF-8"?><urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xm
2026-08-17T03:24:36.329271+00:00	GET https://n8n.io/_nuxt/C2m57bxv.js	200	ct=text/javascript; cache=public, max-age=31536000, immutable; srv=cloudflare; cf=HIT; etag=W/"acfc4c5802aa2a74a38fdb5d7c7d2831"	742416; const _vite_mapDeps=(i,m=_vite_mapDeps,d=(m.f\\ \\)(m.f=["../CBpc8LI5.js","./dk8yEJm9.js","./Dzv3zWH1.js","./BKms0AG6.js","./B6p1NHw.js","./h6xfk3P1.js","./ZABd
2026-08-17T03:25:03.545203+00:00	GET https://n8n.io/api/pages?filters%5B%24and%5D%5B0%5D%5Burl%5D%5B%24eq%5D=%2F&pagination%5Blimit%5D=1	404; 301 https://n8n.io/api/pages?filters%5B%24and%5D%5B0%5D%5Burl%5D%5B%24eq%5D=%2F&pagination%5Blimit%5D=1 → https://n8n.io/api/pages/?filters[&and][0][url][\$eq]=&pagination[limit]=1	ct=application/json; cache=no-cache; srv=cloudflare; cf=DYNAMIC	193; { "error": true, "url": "https://n8n.io/api/pages/?filters[&and][0][url][\$eq]=&pagination[limit]=1", "statusCode": 404, "statusMessage": "Page not found", "mes
2026-08-17T03:25:32.838181+00:00	GET https://api.github.com/repos/n8n-io/n8n/contents/packages/%40n8n/config/src/configs?ref=2afc867f20d689891b516c0458d627bc52c979fd	200	ct=application/json; charset=utf-8; rate=58/60	64715; [{ "name": "__tests__", "path": "packages/@n8n/config/src/configs/__tests__", "sha": "25fef4f81e888306f91674efc510bd052dbd1a96", "size": 0, "url": "https://api.github.com
2026-08-17T03:25:33.029406+00:00	GET https://api.github.com/repos/n8n-io/n8n/contents/packages/frontend/editor-ui/src/api?ref=2afc867f20d689891b516c0458d627bc52c979fd	404	ct=application/json; charset=utf-8; rate=57/60	127; { "message": "Not Found", "documentation_url": "https://docs.github.com/rest/repos/contents#get-repository-content", "status": "404" }
2026-08-17T03:25:33.181374+00:00	GET https://api.github.com/repos/n8n-io/n8n/contents/packages/frontend/editor-ui/src/stores?ref=2afc867f20d689891b516c0458d627bc52c979fd	404	ct=application/json; charset=utf-8; rate=56/60	127; { "message": "Not Found", "documentation_url": "https://docs.github.com/rest/repos/contents#get-repository-content", "status": "404" }
2026-08-17T03:25:42.256899+00:00	GET https://raw.githubusercontent.com/n8n-io/n8n/2afc867f20d689891b516c0458d627bc52c979fd/packages/@n8n/config/src/configs/templates.config.ts	200	ct=text/plain; charset=utf-8; cache=max-age=300; acao=*	331; import { Config, Env } from './decorators'; @Config export class TemplatesConfig { /* Whether to enable loading and showing workflow templates. */ @Env('N8N_T
2026-08-17T03:25:42.450998+00:00	GET https://raw.githubusercontent.com/n8n-io/n8n/2afc867f20d689891b516c0458d627bc52c979fd/packages/@n8n/config/src/configs/__tests__/templates.config.test.ts	404	ct=text/plain; charset=utf-8; acao=*	14; 404: Not Found
2026-08-17T03:25:42.595296+00:00	GET https://raw.githubusercontent.com/n8n-io/n8n/2afc867f20d689891b516c0458d627bc52c979fd/packages/cli/src/services/frontend.service.ts	200	ct=text/plain; charset=utf-8; cache=max-age=300; acao=*	29726; import type { FrontendSettings, ITelemetrySettings, N8nEnvFeatFlags } from '@n8n/api-types'; import { LicenseState, Logger, ModuleRegistry } from '@n8n/backend-
2026-08-17T03:25:42.755686+00:00	GET https://raw.githubusercontent.com/n8n-io/n8n/2afc867f20d689891b516c0458d627bc52c979fd/packages/cli/src/services/frontend.service.test.ts	404	ct=text/plain; charset=utf-8; acao=*	14; 404: Not Found
2026-08-17T03:25:52.521379+00:00	GET https://docs.n8n.io/build/ways-of-building-workflows/use-templates.md	200	ct=text/markdown; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare	13203; > For the complete documentation index, see [lms.txt](https://docs.n8n.io/lms.txt). Markdown versions of documentation pages are available by appending \.md
2026-08-17T03:26:20.767888+00:00	HEAD https://n8n.io/lms.txt	200	ct=text/plain; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	0; ``
2026-08-17T03:26:20.920045+00:00	GET https://n8n.io/lms.txt	200	ct=text/plain; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	1160; # n8n > n8n is a workflow automation platform that uniquely combines AI capabilities with business process automation, giving technical teams the flexibility of
2026-08-17T03:26:21.001261+00:00	GET https://n8n.io/sitemap.xml	200; 307 https://n8n.io/sitemap.xml → https://n8n.io/sitemap_index.xml	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	1053; <?xml version="1.0" encoding="UTF-8"?><sitemapindex xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"><sitemap><loc>https://n8n.io/sitemap-pages.xml</loc></si
2026-08-17T03:26:21.133651+00:00	GET http://n8n.io/	200; 301 http://n8n.io/ → https://n8n.io/	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	350806; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:26:21.290405+00:00	GET https://www.n8n.io/	200; 301 https://www.n8n.io/ → https://n8n.io/	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; cspR0=yes; xfo=DENY; nosniff; ref=strict-origin-when-cross-origin; acao=*	350806; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable

UTC timestamp	Method and requested URL	Final status / redirect outcome	Relevant headers	Bytes / body excerpt (≤160 chars)
2026-08-17T03:26:21.498196+00:00	GET https://n8n.io/forensic-route-probe-not-real-20260817	404; 301 https://n8n.io/forensic-route-probe-not-real-20260817 → https://n8n.io/forensic-route-probe-not-real-20260817/	ct=text/html; charset=utf-8; cache=no-cache; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains; xfo=DENY; nosniff; ref=no-referrer	13608; <!DOCTYPE html><html><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1"><style>@layer properties{@supports ((-webkit
2026-08-17T03:26:28.239485+00:00	GET https://n8n.io/llms.txt	200	ct=text/plain; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC	1160; # n8n > n8n is a workflow automation platform that uniquely combines AI capabilities with business process automation, giving technical teams the flexibility of
2026-08-17T03:26:41.929915+00:00	GET https://n8n.io/reports/2026-ai-agent-development-tools.md	200	ct=text/markdown; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC	6875; # Workflow-based AI Agent Development Tools (2026) > Independent technical evaluation of enterprise-grade, workflow-based AI agent development tools. > This is
2026-08-17T03:26:42.079614+00:00	GET https://n8n.io/forensic-route-probe-not-real-20260817.md	404	ct=text/html; charset=utf-8; cache=no-cache; srv=cloudflare; cf=DYNAMIC	13605; <!DOCTYPE html><html><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1"><style>@layer properties{@supports ((-webkit
2026-08-17T03:26:56.277821+00:00	GET https://web.archive.org/cdx/search/cdx?url=n8n.io%2F%output=json&fl=timestamp%2Coriginal%2Cstatuscode%2Cmime%2Cdigest&filter=statuscode%3A200&filter=mime%3Atext%2Fhtml&collapse=digest&from=2019&to=2026	503	ct=text/html; cache=no-cache; srv=nginx	107; <html><body> 503 Service Unavailable No server is available to handle this request. </body></html>
2026-08-17T03:27:13.557660+00:00	HEAD https://n8n.io/sitemap_index.xml	200	ct=text/xml; charset=UTF-8; cache=public, max-age=600, must-revalidate; srv=cloudflare; cf=DYNAMIC; prerender=2026-08-14T21:12:33.902Z	0; ``
2026-08-17T03:27:28.984556+00:00	GET https://n8n.io/workflows/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	496024; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:27:29.144753+00:00	GET https://n8n.io/integrations/	200	ct=text/html; charset=utf-8; cache=public, max-age=0, must-revalidate; srv=cloudflare; cf=DYNAMIC; hsts=max-age=0; includeSubDomains	1771060; <!DOCTYPE html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=2.0, user-scalable
2026-08-17T03:27:36.589098+00:00	GET https://api.n8n.io/templates/search?rows=16page=1&u=202608179327	200	ct=application/json; charset=utf-8; cache=max-age=14400; srv=cloudflare; cf=MISS	63963; {"totalWorkflows":11518,"workflows":[{"id":11807,"name":"Answer multi-channel support queries with OpenAI RAG and Supabase","totalViews":2462,"price":0,"purchas

53.3 Non-HTTP discovery and correction notes

- [OBSERVED] DNS resolution at 2026-08-17T03:26:22.240970Z used the sandbox resolver through socket.getaddrinfo ; it is not an HTTP request and therefore has no method/status row.
- [OBSERVED] One first attempt to the bounded same-origin /api/pages query followed its normalization redirect to a 404 but raised before the compact logger appended the row. A corrected repeat immediately recorded the exact 301→404 chain. No claim relies on the unlogged attempt.
- [OBSERVED] The public API-root request’s unexpected admin-labeled final path is redacted and was not followed further.
- [OBSERVED] Discovery-search transport details remain [UNKNOWN], as noted in Section 45.2.

54. Evidence matrix, lifecycle status, unknowns, and Sprint 5 audit

54.1 Evidence-matrix additions

Claim	Label	Status	Direct evidence
Marketing deployment is Nuxt-rendered behind Cloudflare	[OBSERVED]	CURRENT	root HTML, payload/asset markers, sampled chunks, headers
Current sitemap has 334,644 entries / 334,636 unique	[OBSERVED]	CURRENT	complete 12-map parse
Sitemap grew by 41 entries since Sprint 1	[OBSERVED]	HISTORICAL DELTA	two exact dated censuses
Integration sitemap has eight excess duplicate entries	[OBSERVED]	CURRENT	complete URL multiset
Cross surface is 320,392 of 320,400 pairs over 801 components	[OBSERVED]	CURRENT GENERATED ROUTE FORM	complete pair parse
68-route body sample all returned 200	[OBSERVED]	CURRENT SAMPLED HEALTH	deterministic rendered sample
Two sitemap routes canonicalize elsewhere; two noindex; two empty-title	[OBSERVED]	CURRENT DRIFT	rendered HTML/canonical/robots/title parse
Public template API backs marketing and configurable in-app templates	[OBSERVED]+[STATED]	CURRENT	live config/proxies/API, docs, pinned config source
Two documented template workflow response shapes differ	[OBSERVED]+[STATED]	CURRENT	live endpoint bodies and current docs
Public workflow totals disagree across sitemap/API/hub	[OBSERVED]	CURRENT DRIFT	near-simultaneous responses
Marketing <code>llms.txt</code> exposes only two reports in two forms	[OBSERVED]	CURRENT MACHINE SURFACE	HEAD/GET and exact link parse
Marketing production source/deployment version	[UNKNOWN]	UNKNOWN	no authoritative source mapping found
No workflow/API/source test was executed	[OBSERVED]	METHOD BOUNDARY	research log

54.2 Lifecycle/status summary

CURRENT

- Cloudflare-fronted Nuxt marketing surface with server-rendered HTML, hydration payloads, hashed immutable assets, route-class cache policies, sitemap generation, and same-origin catalog proxies.

- Twelve sitemap families, 334,644 entries, 334,636 unique URLs, 2,045 unique integration slugs, 11,462 workflow detail routes, 442 creator routes, 320,392 cross routes, and 60 benchmark model routes.
- Public template/catalog API with health, categories, collections, search, search filters, creator data, wrapped template metadata, and flat workflow-import forms.
- Public links to Cloud/app, docs, community, creator, expert, learning, and other ecosystem surfaces; link targets are not collapsed into one deployment.
- Narrow marketing `llms.txt` plus two report Markdown forms.

HISTORICAL DELTA

- From the 2026-08-13 Sprint 1 snapshot to the 2026-08-17 Sprint 5 snapshot: +2 general-page entries, +29 integration entries, +10 workflow entries, and no entry-count change in the other nine maps.

ADDRESSABLE / DRIFTED, NOT CLASSIFIED REMOVED

- `strapi-demo` and subscription thank-you URLs remain 200/sitemap-listed but canonicalize to other pages.
- Embed/OEM thank-you URLs remain sitemap-listed and `noindex`.
- The two `guide-hero-*` URLs remain indexable and titleless; their names do not establish purpose.

UNKNOWN

- Current marketing source repository/commit, exact Nuxt version, origin runtime, content-backend implementation, and deployment pipeline.
- Catalog-service implementation/persistence/deployed version.
- Cause and intended synchronization semantics of 11,462/11,518/11,519 workflow totals.
- Rules for integration cross-route eligibility, the eight absent pairs, and the one component missing from the integration sitemap.
- Rules selecting sitemap/hub/API creator populations.
- Whether current duplicate integration entries or canonical/noindex/title anomalies persist after the observed prerender.
- Historical marketing frontend/build epochs; the archive query failed with 503.
- Deployed version and topology of any Cloud editor instance; auth-oriented surfaces were not fetched.

54.3 Method incidents and bounded limits

- [OBSERVED] The API root's public redirect unexpectedly reached an admin-labeled path. Collection stopped for that route, the path was redacted, and no admin content or action was inspected.
- [OBSERVED] `/api/creators?rows=2&page=1` ignored the supplied bound and returned 15,725 rows / 4,787,250 bytes. It was parsed in memory once and not repeated.
- [OBSERVED] The first `/api/pages` correction incident and the Internet Archive 503 are documented in Section 53.3 and the ledger.
- [OBSERVED] No external JavaScript dependency, source map, media asset, workflow export file, raw body, or binary was retained. Four first-party hashed chunks were parsed in memory only.
- [OBSERVED] No local repository test, workflow execution, API mutation, authentication attempt, or deployment probe was performed.

54.4 Cleanup and final resource audit

- [OBSERVED] The compact temporary ledger and appendix-generation file were deleted after consolidation.
- [OBSERVED] No PDF was created; the consolidated PDF remains reserved for Sprint 6.
- [OBSERVED] Final workspace: one file, `/home/user/working_sprint_5.md`.
- [OBSERVED] Final exact report size: `317466` bytes; final line count: `2781`; `du -sh /home/user = 312K`; `find /home/user -type f | wc -l = 1`.

Sprint 6 will perform the cross-source audit, consolidate the full reconstruction, preserve evidence/status boundaries, and produce the final PDF.

Sprint 5 of 6 completed. 1 research sprint remains.

Sprint 6 — final cross-source consolidation and provenance audit

55. Final-sprint method and consolidation boundary

55.1 No new target collection

- **[OBSERVED]** Sprint 6 made no HTTP request to the target or its ecosystem, performed no Git fetch, and executed no repository test, build, workflow, migration, container, or application process. It consolidates the dated evidence already collected in Sections 1–54.
- **[OBSERVED]** The final audit used the cumulative Markdown as its source of truth, checked internal arithmetic/status boundaries, added unified matrices and diagrams, and generated the final PDF. PDF rendering and PDF text/page validation are deliverable-generation checks, not tests of n8n.
- **[OBSERVED]** The evidence remains time-bounded: source/ref pins are from 2026-08-15 or earlier; documentation responses are from 2026-08-15; marketing/catalog responses are from 2026-08-17; Sprint 6 consolidation date is 2026-08-18 Asia/Tehran.

55.2 Final evidence hierarchy

Evidence class	Use in this report	Boundary
Direct source object/tree/diff	Implementation, exact path state, ancestry, deletion, and configuration at a named ref	Source capability is not deployment proof.
Direct public HTTP response	Current route/status/header/body structure at the request timestamp	One edge/time/route sample is not universal deployment state.
Target-authored docs/README/commit text	Project terminology, declared status, migration guidance, and historical statements	A statement is not independently observed runtime behavior.
Test source and hosted checks	Intended assertions/coverage and hosted check metadata	No local pass result was produced.
Bounded inference	Minimal relationship required to connect direct observations	Marked [INFERRED] ; no intention, psychology, or quality inference.
Unresolved point	Facts not established under the passive/public boundary	Marked [UNKNOWN] ; not silently filled from names or common practice.

55.3 Lifecycle discipline retained in the final synthesis

- **CURRENT**: directly present at a current pin or returned by a current public response.
- **CURRENT — CONDITIONAL**: current source wiring gated by process role, configuration, module, license, or instance role.
- **CURRENT — EXPERIMENTAL/NOT READY**: current tracked source with explicit target-authored incompleteness.
- **HISTORICAL**: directly observed in a dated reachable state.
- **REMOVED**: exact later tree deletes the form.
- **REVERTED**: a later tree restores the exact earlier state.

- **BRANCH-ONLY / EXACT TIP UNMERGED**: present on a sampled branch but not established in the pinned default branch.
 - **FUTURE — STATED / CANDIDATE SOURCE**: explicitly scheduled/documented or present as disabled transition code.
 - **ABANDONED**: used only if direct evidence supports that label. No branch is called abandoned merely because it is old, unmerged, renamed, or removed.
 - **UNKNOWN**: lifecycle or deployment status not established.
-

56. Forensic identity and evidence topology

56.1 What the supplied target resolves to

[OBSERVED] `https://n8n.io/` is a public marketing/product surface that directly links a public source repository. The target is therefore **TYPE 3 — HYBRID**, but the evidence does not collapse every target-owned surface into one application or deployment.

Surface	Current observed form	Primary evidence state
Marketing/public catalog	Cloudflare-fronted Nuxt-rendered <code>n8n.io</code> , 12 sitemap families, same-origin catalog proxies	Live 2026-08-17 responses
Template/catalog service	Public <code>api.n8n.io</code> health/categories/collections/search/template forms	Live 2026-08-17 responses and current template docs
Product documentation	GitBook-rendered <code>docs.n8n.io</code> , source in <code>n8n-io/n8n-docs</code>	Docs pin <code>ec418cd6...</code> , live 2026-08-15
Documentation wiki	Four-page GitHub wiki with complete 48-commit history	Wiki tip <code>d8b53a7...</code>
Product source	Public <code>n8n-io/n8n</code> monorepo	Main pin <code>2afc867f...</code> ; stable/prerelease refs below
Stable release	GitHub release <code>n8n@2.34.6</code>	<code>d743b0ad...</code> , published 2026-08-14
Prerelease	GitHub prerelease <code>n8n@2.35.3</code>	<code>0dccd14a...</code> , published 2026-08-14
Hosted editor/application instances	Authentication-oriented surfaces linked but not inspected	Deployment/version/topology [UNKNOWN]

56.2 Evidence-topology diagram

The diagram is a compact [INFERRED] map of directly observed links, source configuration, and successful public calls. It does not assert common deployment ownership behind each edge.

MERMAID DIAGRAM SOURCE (RETAINED AS EVIDENCE)

```
flowchart TB
    MKT["n8n.io marketing\nNuxt + Cloudflare"] --> CAT["api.n8n.io\ntemplate/catalog API"]
    MKT --> DOCS["docs.n8n.io\nGitBook"]
    MKT --> SRC["github.com/n8n-io/n8n\nproduct source"]
    DOCS --> DSR["github.com/n8n-io/n8n-docs\ndocs source"]
    DSR --> WIKI["n8n-docs GitHub wiki\nseparate 48-commit history"]
    APP["n8n application source\nCLI + editor + execution engine"] -->|default template host| CAT
    APP --> PUB["Per-instance Public API\n/api/v1 + API key scopes"]
    MKT -. links only .-> CLOUD["Cloud editor/auth surfaces\nnot inspected"]
    SRC --> REL["Stable/prerelease tags\nand release branches"]
    ENG["@n8n/engine / Engine 2.0\nseparate current attempt"] -. not wired to primary CLI .-> APP
```

56.3 Versions are evidence-state labels, not one deployed version

- [OBSERVED] The pinned default branch package field says 2.35.0 ; the stable release is 2.34.6 ; the prerelease is 2.35.3 . Package text, publication channel, and commit chronology are recorded separately.
 - [OBSERVED] Live GitBook OpenAPI group structure most closely aligns with the prerelease's 23 root tags, while the later source pin has 24 and adds Role Mapping Rule operations.
 - [UNKNOWN] Which n8n version runs in any Cloud editor instance. Marketing, documentation, release, and source states do not establish it.
-

57. Unified historical timeline

Date/epoch	Directly observed form or transition	Final lifecycle classification
2019-06-23 root 9cb9804e...	Seven-package Lerna workspace; combined Express REST/webhook/static UI server; SQLite and MongoDB; 40 node/trigger candidates; Docker/Node 10	HISTORICAL reachable root
2019-06 to 2019-09	run → execute ; Convict configuration; Vorpal→oclif; Read File From URL removed into HTTP Request functionality	HISTORICAL transitions; node path REMOVED
2020-01	OAuth-support merge explicitly reverted by 70e972ff...	REVERTED attempt
2020-07	Main-repository docs removed by 62612d0a... with move to n8n-docs	HISTORICAL docs relocation / old tree REMOVED
2021-01	MongoDB application persistence removed by b33a5fcd...	REMOVED persistence adapter
2021-02	Queue worker and separate webhook command/server introduced	HISTORICAL transition; evolved into CURRENT conditional forms
2022-03	User management and credential sharing expansion; license text changed to Sustainable Use License	HISTORICAL transitions with CURRENT descendants
2022-04	Documentation moved from VuePress to MkDocs Material at 9821b9c...	HISTORICAL docs form
2022-06 to 2022-11	Public API added; Lerna→Turborepo; npm→pnpm	HISTORICAL transitions with CURRENT descendants
2023-07	Source-control/environments release and Vue 3 editor/design-system migration	HISTORICAL transitions with CURRENT descendants
2023-11	Multiple-main workflow-activation coordination	HISTORICAL introduction; CURRENT conditional descendant
2024-10	Initial Code task-runner support	HISTORICAL introduction; CURRENT JS/Python runner form
2024-12	Execution-engine code moved from n8n-workflow to n8n-core	HISTORICAL structural move / old path REMOVED
2025-03 and 2025-07	Backend module runtime/decorators, then module-generation/tooling expansion	CURRENT module system descendants
2025-08	Latest reachable docs-wiki update; content still describes MkDocs syntax	CURRENT artifact / HISTORICAL platform guidance
2026-05	Docs redirect cleanup attempted and next-commit reverted to identical prior tree	REVERTED exact-tree attempt
2026-06-19	Course prose/images removed	REMOVED docs content
2026-06-24	MkDocs→GitBook release migration	CURRENT docs hosting form
2026-06-26 to 2026-07-23	Migration cleanup; selected dropped workflows/PDFs restored; final MkDocs/Netlify residue removed	REMOVED and PARTLY RESTORED artifacts
2026-07	Descriptor-v2 frontend SDK current form; public v3 change page and candidate rule source appear	CURRENT SDK form; FUTURE v3 stated/candidate
2026-08-14/15	Stable 2.34.6, prerelease 2.35.3, source pin 2afc867f..., docs pin ec418cd6...	Current evidence refs
2026-08-17	Marketing sitemap 334,644 entries; Nuxt/catalog API current snapshot	CURRENT public-surface snapshot
2026-10 (scheduled)	Public documentation schedules n8n 3.0	FUTURE — STATED, not observed shipped

[OBSERVED] The timeline establishes multiple architectural and publishing forms. It does not imply a single linear replacement for every subsystem: release branches, prototypes, compatibility paths, and parallel memory/agent branches coexist in the evidence.

58. Product architecture forms over time and current composition

58.1 Structural evolution

Form	Direct evidence	Status
Combined CLI server + editor static files + REST/webhooks	Root <code>Server.ts</code> , route registration, static editor serving	HISTORICAL 2019
Dual SQLite/MongoDB persistence	Root entities/adapters and Docker docs	HISTORICAL; MongoDB application adapter REMOVED 2021
SQL primary persistence with SQLite/PostgreSQL migrations	@n8n/db, TypeORM options, generated schemas	CURRENT
Queue worker and dedicated webhook process	worker, webhook, Bull/Redis, command wiring	CURRENT — CONDITIONAL
Multi-main lease/pub-sub coordination	Redis lease, leader events, 40 typed scaling events	CURRENT — CONDITIONAL
Task-broker + JS/Python runners	requester/broker/runner protocol, internal/external launch, Code-node adapters	CURRENT
Decorator/DI controllers and hybrid Public API	109 bounded controller files; handwritten and decorator OpenAPI merge	CURRENT
Backend modules and module-contributed entities	32 module declarations, eligibility/license/role filtering	CURRENT — CONDITIONAL
Extension SDK example	tracked manifest/setup, no primary loading path established	CURRENT — ACTIVATION UNKNOWN
Engine 2.0 separate server/database/API	package, Dockerfile, Postgres entities, tests, explicit not-ready statement	CURRENT — EXPERIMENTAL/NOT READY
Marketing Nuxt site and catalog service	live public HTML/config/API	CURRENT, separate from editor source
GitBook documentation source/rendering	docs README/tree/live responses	CURRENT, separate publishing system

58.2 Current primary execution composition

MERMAID DIAGRAM SOURCE (RETAINED AS EVIDENCE)

```

flowchart LR
  UI["Vue editor UI"] --> REST["Main/start process\ninternal REST + Public API"]
  HOOK["Webhook HTTP"] --> ROLE{"regular or queue mode"}
  REST --> ROLE
  ROLE -->|regular/local| WX["WorkflowRunner → WorkflowExecute"]
  ROLE -->|queue| BULL["Bull jobs / Redis"]
  BULL --> WORKER["Worker JobProcessor"]
  WORKER --> WX
  WEB["Dedicated webhook process\nqueue mode"] --> BULL
  WX --> LIFE["Execution lifecycle hooks"]
  LIFE --> SQL["SQLite or PostgreSQL\nexecution metadata/data"]
  LIFE --> BLOB["DB/filesystem/S3/Azure\nconfigured execution data"]
  CODE["Code node requester"] --> BROKER["Task broker"]
  BROKER --> JS["JavaScript runner"]
  BROKER --> PY["Python runner"]
  REDIS["Redis pub/sub + leases"] --> MAIN["Leader/follower mains"]
  MAIN --> REST
  MOD["Eligible backend modules"] --> REST
  E2["Engine 2.0 server"] -. separate/not wired .-> WX

```

- [OBSERVED] Regular execution is the pinned default. Queue mode uses Bull over Redis; worker processing reloads persisted execution state and invokes the same primary `WorkflowExecute` path.
- [OBSERVED] Redis pub/sub/leases are distinct from Bull job storage. Multi-main services start/stop on leader changes.
- [OBSERVED] Lifecycle composition differs across regular main, scaling worker, scaling main, and subexecution roles to divide node events, data saves, metadata updates, deletion, push, and module hooks.
- [OBSERVED] Current Code-node JavaScript and native Python dispatch through task runners. Retained `vm2` `JavaScriptSandbox.ts` is not imported by the current Code node; reachability elsewhere remains [UNKNOWN].

58.3 Current persistence and data forms

- [OBSERVED] Primary runtime database choices are SQLite and PostgreSQL; `DB_TYPE` defaults to SQLite. Runtime entity composition combines core and eligible module entities.
- [OBSERVED] Execution persistence records a relational entity plus serialized data. `database`, `filesystem`, `s3`, and `azure` storage forms map to row tags and reads follow the row's stored location.
- [OBSERVED] Generated schema documentation has 128 PostgreSQL table pages and 127 SQLite table pages. `workflow_statistics_delta` is the sole PostgreSQL-only page after filename normalization.
- [OBSERVED] Engine 2.0 has separate PostgreSQL-only `workflow_execution` and `workflow_step_execution` tables absent from the primary generated schemas.

59. Current interface and public-surface matrix

59.1 Interface forms must remain separate

Interface	Current observed role	Authentication/deployment boundary
Marketing HTML/routes	Product, integration, workflow, creator, report, and benchmark pages	Public; not the workflow editor
Marketing machine forms	12 sitemaps, narrow <code>llms.txt</code> , two report <code>.md</code> forms	Public
Template/catalog API	Template metadata/search/categories/collections/import forms	Public GETs inspected; distinct from instance Public API
Instance Public API	<code>/api/v1</code> operations, API-key middleware/scopes, OpenAPI	Source/docs; actual hosted instance not probed
Internal REST controllers	Editor/backend controller surface	Source capability; runtime count varies by module/license/role
Webhook/test-webhook paths	Workflow-trigger HTTP ingress	Source/docs; no webhook invoked
Server CLI	Same-host commands with database-adjacent administration	Source/docs; not run
n8n CLI remote client	API-key client wrapping Public API	Docs page says beta; not run
Instance MCP server	Distinct connect surface	Docs/source; not probed
Documentation MCP server	Documentation-oriented surface	Distinct from instance MCP; not probed

59.2 Current public-scale reconciliation

- [OBSERVED] Marketing: 334,644 sitemap entries / 334,636 unique; 2,045 unique integration detail slugs; 11,462 workflow detail routes; 442 creator routes; 320,392 pair routes; 60 model benchmark routes.
- [OBSERVED] Template API: search total 11,518; workflow hub text 11,519. These are not silently substituted for the 11,462 sitemap routes.
- [OBSERVED] Documentation: 1,334 unique routes = 1,310 repository-backed public pages + 24 generated OpenAPI pages. Documentation `llms.txt` also has 1,334 links after its host-specific representation rules.
- [OBSERVED] Product source at the pin: 26,727 tracked paths, 87 manifests, 423 unique internal dependency edges, 110 bounded controller files, 537 method-decorator calls, 329 core migration files, 32 backend module declarations, and 5,790 conventionally named test/spec files under Sprint 3's test-file heuristic.
- [OBSERVED] Counts have different units. Integration pages, app filters, node filters, node implementation paths, credential paths, OpenAPI operations, tests, and templates are not interchangeable measures.

59.3 Target-native terminology

[STATED] Current target-authored prose prefers **workflow**, **node**, **execution**, **publish/unpublish**, **app node**, **core node**, **cluster/root/sub-node**, **community node**, **canvas**, **sticky note**, **sub-workflow**, **self-hosted**, **n8n Cloud**, **instance**, **project**, **folder**, **environment variable**, and **data table**.

[OBSERVED] Historical and identifier forms still include `active`, activate/deactivate paths/scopes, worker `job` concepts, `WorkflowExecute`, `runData`, and legacy node names. Identifiers and historical source are retained verbatim rather than rewritten to current prose.

60. Consolidated lifecycle registry

60.1 CURRENT

- Primary `WorkflowRunner` + `WorkflowExecute` runtime; regular mode default.
- Conditional Bull/Redis worker, dedicated webhook, and multi-main forms.
- JS/Python task runners with internal/external modes and Code-node integration.
- SQLite/PostgreSQL primary persistence and DB/filesystem/S3/Azure execution-data forms.
- Decorator/DI controller system, hybrid Public API specification, scoped API-key authorization, and current role operations.
- Conditional backend modules and module-contributed entities.
- GitBook docs, separate current docs wiki artifact, generated schema docs, and source-local architecture documents.
- Nuxt marketing surface, public template/catalog service, generated catalog routes, sitemap and machine-readable report forms.
- Stable 2.34.6, prerelease 2.35.3, maintained 1.x release, and later pinned default-branch source as separate evidence states.

60.2 HISTORICAL

- 2019 Lerna/Vorpal-era combined server/UI and SQLite/MongoDB form.
- Convict/oclif transitions, queue/webhook introduction, user-management expansion, Public API introduction, Turbo/pnpm migrations, Vue 3 migration, source-control environments, and task-runner introduction.
- VuePress and MkDocs documentation systems, archived release ledgers, and pre-GitBook OpenAPI file.
- Old active-workflow terminology and pre-publish lifecycle prose.

60.3 REMOVED

Exact form	Removal evidence
<code>ReadFileFromUrl.node.ts</code>	Deleted in <code>c1e753e9...</code> ; HTTP Request modified same commit
MongoDB application persistence adapter	Nine entities/migrations/index files deleted in <code>b33a5fcd...</code>
Main-repository 2020 docs tree	Deleted in <code>62612d0a...</code> , message points to <code>n8n-docs</code>
Committed SQLite DB file	Deleted in <code>48a1e0c4...</code>
<code>RoutingNode</code> under workflow package	Deleted there and added under core in <code>5a055ed5...</code>
MkDocs/Netlify documentation build/navigation residue	Removed across GitBook migration cleanup and <code>4998c228...</code>
Course prose/images	134 deletions in <code>b924629...</code>

Exact form	Removal evidence
Twelve migration-dropped JSON exports not restored by restoration commit	Exact migration/restoration tree comparison

60.4 REVERTED

- [OBSERVED] OAuth-support merge/code was explicitly reverted by 70e972ff... in 2020.
- [OBSERVED] The May 2026 docs redirect-cleanup attempt was reverted in the next commit; pre-attempt and post-revert tree SHA are identical.

60.5 BRANCH-ONLY / EXACT TIP UNMERGED

- In-memory webhook-registration/cache experiment.
- CheckProcessed node/manager form.
- Git-connection DB→module→CLI/public-API stack.
- Publish-state switcher/composable, named-version prototype, insights prototype, Instance AI setup rewrite, and A2A agentic-users form.
- Earlier @n8n/module-sdk package-name/form.
- Multiple cross-thread, episodic, and observational memory branch chains/parallel lines.
- Two newest Sprint 1 refs that disappeared before Sprint 2; exact commits remained public and unmerged at collection.

[UNKNOWN] Patch-level derivation from these branches into same-term current code unless exact ancestry was established. Current same-name or same-domain implementation does not retroactively classify an old exact tip as merged.

60.6 CURRENT EXPERIMENTAL / FUTURE

- [OBSERVED] Engine 2.0 is **CURRENT — EXPERIMENTAL/NOT READY**: separate package/server/database/image/tests, explicit “not ready” status, missing primary CLI dependency, in-memory queue limitations, no standalone v1 executor, and acceptance TODOs.
- [STATED] n8n 3.0 is scheduled for October 2026 with named removals/changes.
- [OBSERVED] Sixteen v3 rule files and a migration framework are **FUTURE — CANDIDATE SOURCE** because MIGRATION_REPORT_TARGET_VERSION is null in stable, prerelease, and pin states.

60.7 ABANDONED status boundary

[OBSERVED] No item is labeled **ABANDONED** solely from branch age, ref deletion, unmerged status, missing current path, or a prototype name. The report uses directly supportable labels—REMOVED, REVERTED, BRANCH-ONLY, EXACT TIP UNMERGED, or UNKNOWN—rather than inferring intent.

61. Attempts, parallel branches, and commit-scale evidence

61.1 Source-branch scale and sample

- [OBSERVED] Sprint 2 graph-scanned all 1,645 advertised source heads; 504 had more than ten commits not reachable from current `master` at that snapshot.
- [OBSERVED] Sixty current refs received independent merge-base/tree-diff inspection, plus two disappeared refs through public commit/compare APIs. All 60 exact sampled tips were unmerged relative to the Sprint 2 pin.
- [OBSERVED] The sample covered fixed oldest/newest refs, all strict keyword matches under the prescribed rule, release representatives, prototypes, rewrites, docs, v2, core, and memory families.
- [OBSERVED] A zero-file merge-base-to-tip delta for one 13-ahead memory UI branch demonstrates that unique commits do not necessarily imply a unique final tree.

61.2 Documentation-branch scale and sample

- [OBSERVED] All 249 documentation branch names / 247 unique tips were enumerated. Twenty-nine deterministic tips across seven strata plus `main` were tree/metadata inspected; ten received GitHub comparisons and eight detailed parses.
- [OBSERVED] Sample statuses included one ahead-only, two behind-only, and seven diverged comparisons. Ahead commits with zero net files were not treated as surviving content differences.
- [OBSERVED] The current docs branch name `update-variables-plans` points to a merge about execution timeout, directly demonstrating why branch names are not implementation facts.

61.3 Consequential commit chain

[OBSERVED] Twenty-four source commits received bounded direct tree diffs in Sprint 2, including configuration/CLI replacements, node removal, OAuth revert, database/docs removal, queue/webhook introduction, MongoDB removal, user management, licensing, Public API, build/package-manager transitions, source control, Vue 3, multi-main, task runners, execution-engine movement, backend modules, and descriptor-v2 SDK work.

[OBSERVED] Thirteen consequential docs commits plus required parents established VuePress→MkDocs→GitBook, course removal, migration cleanup, link repair, artifact restoration, residual deletion, release-feed rename, and v3-page addition.

62. Documentation, OpenAPI, schema, diagram, and test artifacts

62.1 Documentation system

- [OBSERVED] Current docs repository: 1,863 paths, 1,473 Markdown files, 1,437 Markdown files under docs/ , 16 workflow-shaped JSON artifacts, two PDFs, and GitBook collection navigation.
- [OBSERVED] Public route arithmetic is exact: nine collection partitions contain 1,310 repository-backed pages; GitBook contributes 24 generated OpenAPI routes; total 1,334.
- [OBSERVED] Complete wiki history has 48 commits and four paths. Current wiki text still teaches MkDocs syntax while current publishing uses GitBook. Intended authority remains [UNKNOWN].
- [OBSERVED] Migration frontmatter, include/hint/tab syntax, workflow/PDF restoration, and generated machine-readable Markdown forms are retained as distinct artifact classes.

62.2 OpenAPI and scope states

State	Merged literal path keys	Normalized path shapes	Operations	Root tags	Declared/consumed scopes
Stable 2.34.6	68	68	109	22	86 / 86
Prerelease 2.35.3	70	70	110	23	88 / 88
Pin 2afc867f...	72	71	113	24	91 / 91

- [OBSERVED] Every source state has exact equality between declarable API-key resources and operation-consumed scopes.
- [OBSERVED] Current authentication prose lists 67 unique scope rows. Relative to the pin, 25 consumed scopes are omitted and one explicit reserved/non-consumed scope, user:enforceMfa , is present.
- [OBSERVED] Live Role content exposes create only and no Role Mapping Rule group, aligning with the bounded prerelease tag structure for the newly changing role area rather than the later pin.

62.3 Schemas and diagrams

- [OBSERVED] Current generated primary schemas: 128 PostgreSQL tables and 127 SQLite tables across 257 one-diagram Markdown files.
- [OBSERVED] Main source has 298 Mermaid fences in 271 of 592 Markdown files at the pin; 257 are generated schema diagrams and 41 occur across 14 other source files.
- [OBSERVED] Docs source has three Mermaid fences in two pages and references binary queue/task-runner/database diagrams. Binary bodies were not retained.
- [OBSERVED] Documentation workflow artifacts parse to 168 nodes, 88 connection-map entries, 45 distinct node-type strings, and nine credential-field nodes. Public template API samples were parsed structurally without retaining credential values.

62.4 Tests and checks

- [OBSERVED] The pinned tree has 5,790 conventionally named test/spec files under Sprint 3's file rule, spanning CLI, frontend, nodes, runners, scaling, Public API, Engine 2.0, compatibility, deployment, and Playwright areas.
 - [OBSERVED] Documentation hosted metadata showed successful/skipped/cancelled checks and statuses at the docs pin. Those observations are not local execution.
 - [OBSERVED] No product/docs test suite was run in any sprint. Test code establishes intended assertions, not passing behavior.
-

63. Consolidated contradiction and drift register

ID	Evidence tension	Direct status	Unresolved boundary
C-01	Stable 2.34.6, prerelease 2.35.3, pin package 2.35.0	Distinct release/source states	Deployed Cloud version UNKNOWN
C-02	Sprint 3 provisional 70 current paths versus corrected 72 literal / 71 normalized	Corrected by complete OpenAPI merge resolution	None for pin count
C-03	Live Role docs expose prerelease-level create; pin adds list/read/mapping	Current synchronization drift	Exact GitBook spec source/cadence UNKNOWN
C-04	Auth docs 67 scope rows versus 88 prerelease / 91 pin consumed scopes	Current prose incomplete relative to source	Publication update cadence UNKNOWN
C-05	Hosted DB page 24 headings and “role unused” versus 127-128 generated tables and wired role APIs	Current hosted page stale/incomplete	Per-deployment optional tables UNKNOWN
C-06	v2 page says binary default /available-modes removed; env page/source retain parts with automatic default rules	Current documentation contradiction; source resolves pin behavior	Future v3 final behavior UNKNOWN
C-07	Task-runner docs use /etc/task-runners.json once and /etc/n8n-task-runners.json elsewhere/source	Current documentation path contradiction	Source image uses latter
C-08	Queue docs say filesystem unsupported, then allow shared-filesystem large-webhook relay	Current scope ambiguity	Intended exception boundary UNKNOWN
C-09	Docs README says Sustainable Use License; docs LICENSE.md contains Commons Clause/Apache text; hosted SPDX NOASSERTION	Current repository-internal text contradiction	No legal conclusion made
C-10	Enabled current docs wiki teaches MkDocs while source/live use GitBook	Current artifact with historical guidance	Intended wiki authority UNKNOWN
C-11	Migration frontmatter URLs/path fields versus live collection-root mappings	Migration residue, not route truth	None inferred
C-12	Hidden/outdated Docker page remains directly 200	Current addressable but hidden/outdated	Removal intent UNKNOWN
C-13	Style guide says Preview; six current pages use beta	Current terminology drift	Runtime maturity not inferred
C-14	Existing docs .md and missing docs .md can both return 200; missing body says Page Not Found	Current status/content mismatch	Body classification required
C-15	Marketing workflow sitemap 11,462, API 11,518, hub 11,519	Current count drift	Eligibility/sync/count rule UNKNOWN
C-16	Integration sitemap has eight excess duplicate entries	Current sitemap duplication	Whether later prerender fixes it UNKNOWN
C-17	One cross component absent from integration sitemap; 1,245 integration slugs absent from cross graph	Current indexing/eligibility asymmetry	Generation rule UNKNOWN
C-18	Two sitemap pages canonicalize elsewhere; two are noindex; two titleless	Current route/indexing metadata drift	Purpose/retention UNKNOWN
C-19	Creator sitemap 442, hub payload 1,024 shapes, API 15,725 rows	Current population/pagination drift	Selection rules UNKNOWN
C-20	Code hint says sandbox has no network; JS runner exposes HTTP helper RPC	Current wording/capability tension	“Direct primitives” interpretation UNKNOWN
C-21	Runner comments forecast 5/60 defaults in v2; TS defaults remain 10/300 while Python uses 5/60	Current source-comment/default drift	Intended scope UNKNOWN
C-22	Engine 2.0 has package/API/image/tests but explicit not-ready/unwired limitations	Current maturity tension, not contradiction	Future integration/shipping UNKNOWN
C-23	Runners image README says Preview; live docs prescribe external runners for production	Current artifact/status scope tension	Packaging versus operating-guidance scope

ID	Evidence tension	Direct status	Unresolved boundary
C-24	v3 public summary names selected removals; disabled candidate source covers 35 node types and more rules	Future summary versus candidate detail	Final v3 scope UNKNOWN
C-25	Marketing <code>llms.txt</code> has four links; marketing sitemap has 334,636 unique URLs; docs <code>llms.txt</code> is exhaustive	Host-specific machine-surface scope difference	No common <code>llms.txt</code> contract inferred

[OBSERVED] The register records tensions without rating severity, quality, or correctness beyond exact source/body reconciliation. Where current source resolves a documented value, the report says so; where deployment or intent is absent, status remains UNKNOWN.

64. Final unknowns, non-conclusions, provenance, and deliverable audit

64.1 Consolidated unknowns

Deployment and live behavior

- [UNKNOWN] Version/topology of any hosted n8n editor instance; prevalence of regular, queue, multi-main, external-runner, module, storage, and Engine 2.0 forms.
- [UNKNOWN] Runtime controller/endpoint count after all license, feature, module, instance-role, and configuration filters.
- [UNKNOWN] Production marketing source repository/commit, exact Nuxt version, origin runtime, content backend, and deployment pipeline.
- [UNKNOWN] Template/catalog service implementation, persistence, deployment version, creator selection, workflow-count synchronization, and cross-route generation rule.
- [UNKNOWN] Which optional/module tables exist in a specific live database.

Source and branch lineage

- [UNKNOWN] Patch-level lineage from most branch experiments to same-term current code; exact cherry-pick/rewrite relationships where hashes/ancestry do not establish them.
- [UNKNOWN] Lifecycle intent for most extant unmerged refs and reasons for the two observed ref removals.
- [UNKNOWN] Unreachable/orphaned/deleted remote refs not advertised by public Git.
- [UNKNOWN] Reachability of retained `vm2` sandbox outside the current Code node and loading of the extension SDK Insights example.
- [UNKNOWN] Reliable repository LOC; deliberately not hydrated/computed.

Documentation and future state

- [UNKNOWN] Exact GitBook OpenAPI synchronization source/version/cadence and intended authority of the enabled MkDocs-era wiki.
- [UNKNOWN] Final n8n 3.0 rule set and whether every candidate rule ships unchanged.
- [UNKNOWN] Final Engine 2.0 integration path, persistence/queue changes, and resolution of explicit TODOs.
- [UNKNOWN] Historical marketing frontend epochs; the bounded archive query failed.

64.2 Explicit non-conclusions

- [OBSERVED] This report is not a security assessment, vulnerability test, legal interpretation, product evaluation, recommendation, redesign, competitor comparison, or operational benchmark.
- [OBSERVED] Source presence is not deployment proof; documentation wording is not runtime proof; a test file is not a passing test; a route name is not an implementation fact.

- [OBSERVED] Branch age/name/unmerged status is not abandonment intent. A stored workflow `active` field is not evidence that the public artifact is running.
- [OBSERVED] Page, node, credential, operation, scope, test, table, and template counts have separate units and are not converted into a single feature count.

64.3 Provenance and reproducibility summary

Dimension	Final accounting
Source branch census	1,645 heads graph-scanned; 60 current refs tree-sampled; two disappeared refs API-inspected
Source consequential commits	24 bounded direct tree diffs, plus branch-specific commits/ancestry
Documentation branches	249 names / 247 tips enumerated; 29 deterministic tips inspected; ten compares
Wiki	All 48 reachable commits; all four paths
Current source pin	<code>2afc867f20d689891b516c0458d627bc52c979fd</code>
Stable/prerelease refs	<code>d743b0ad...</code> / <code>0dccd14a...</code>
Docs/wiki pins	<code>ec418cd6...</code> / <code>d8b53a7...</code>
Documentation routes	All 1,334 names reconciled; bounded rendered/request sample
Marketing routes	All 334,644 sitemap entries parsed; 68 deterministic bodies plus hubs/probes
Sprint 5 request ledger	146 recorded requests: 143 GET, two HEAD, one OPTIONS
Local product/docs tests	None run
Credential handling	No credential supplied; credential-like values not retained
Historical checkout	None; filtered bare Git and object streaming only

64.4 Research-chain incidents retained

- [OBSERVED] Sprint 1 briefly exceeded the preferred 80 MB working target during broad history hydration and immediately deleted the clone.
- [OBSERVED] Sprint 2 had a severe partial-clone hydration incident reaching about 3.3 GB/14,902 files; the process and clone were immediately removed and the method was replaced with bounded API/object calls.
- [OBSERVED] Sprint 3 stopped two broad `JavaScriptSandbox` reference searches after timeouts.
- [OBSERVED] Sprint 4 redo remained under the 80 MB workspace research target; direct fetch of one revert-message SHA failed but reachable trees established the state.
- [OBSERVED] Sprint 5 kept retained workspace below 1 MiB before report assembly, but one bounded creators query ignored row parameters and returned 4.79 MB; an API-root redirect unexpectedly reached an admin-labeled path and collection stopped for that route.

64.5 Final deliverables and validation

- [OBSERVED] Final cumulative Markdown: `/home/user/working_sprint_6.md`.
 - [OBSERVED] Final consolidated PDF: `/home/user/n8n_forensic_reconstruction_final.pdf`.
 - [OBSERVED] PDF rendering used the cumulative Markdown with local CSS/fonts and blocked external-resource retrieval. Mermaid blocks remain as bounded source diagrams in the PDF; no external JavaScript renderer or diagram file was retained.
 - [OBSERVED] PDF validation: 151 pages; text extraction located the report title, final synthesis, evidence labels, and completion line.
 - [OBSERVED] Final Markdown: 353822 bytes and 3212 lines.
 - [OBSERVED] Final workspace: 2 files; `du -sh /home/user` = 1.3M; no temporary clone, HTML, script, cache, or separate diagram remains.
-

Sprint 6 of 6 completed. 0 research sprints remain.